Communicating with Models and Data

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Abstract

Many real-world decisions rely on models and data supplied by experts, such as consultants, financial advisors, and even AI chatbots. I study how model simplification changes experts' incentives to collect data when their goal is to make decisionmakers optimistic. A sender samples data and reports a model of how a receiver's choices map to outcomes. He is either "accurate," and correctly estimates the functional form mapping choices to outcomes; or "simple," and calibrates the *curvature* of this relationship to a predetermined value. I show that simple senders only sample data about extreme choices. Because the receiver knows a sender's model may be simple, she entertains every model consistent with these data. By keeping the receiver in the dark about the effects of the interior choices she is likely to make, a simple sender ensures that these alternative models promise her higher expected utility. Strategic sampling thus makes model simplicity valuable: accurate senders benefit when receivers perceive their models as simple. When an accurate sender can choose the composition of simple senders in the population, he surrounds himself with a polarized group of simple senders who estimate extremely low or extremely high curvature models. In the limit, these correspond to linear regressions and "worst-case scenario" models.

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1 Introduction

Consider a policymaker tasked with designing a carbon tax. She understands the basic nonlinear relationship: low taxes do not curb emissions, while high taxes generate economic costs. The welfare-maximizing tax lies somewhere in the middle. To help her figure out where it is, she hires a consultant, who collects data and estimates a *model* mapping taxes to welfare. While his model is potentially valuable, it also presents several issues. First, it may be too simple. Imagine he runs a linear regression of welfare on taxes, totally neglecting the nonlinear relationship the policymaker entertains. The policymaker suspects his model is simplistic, and thus considers other models that could have produced the same data. This leads to a second issue: the consultant chose which data to collect. They may mostly describe the effects of low or high taxes, and say little about the moderate taxes she is considering. Third, the consultant knows that if the policymaker is optimistic about how much she can improve welfare, she will launch a tax reform for which he can take credit. He is thus incentivized to collect data and present models that make her optimistic.

This example reflects a broader phenomenon: decisionmakers rely on experts to model the world for them. Investors hire financial advisors to model stock returns, voters use political narratives to understand laws, and, increasingly, any of these decisionmakers may ask an AI chatbot for a recommendation. Models and narratives shape how decisionmakers *perceive* the *impact* of their choices, generating optimistic hiring (Flynn and Sastry, 2025), political polarization (Levy et al., 2022), and media engagement (Eliaz and Spiegler, 2026). A growing literature documents that we often rely on *simple* models, which may lead to inefficient investing or price-setting (Shiller, 2017; Han et al., 2026). At the same time, experts choose the data underlying these models (Di Tillio et al., 2021) and, in cases like AI, the mapping from data to a model is opaque (Jin et al., 2024).

This paper studies how *model simplicity and strategic data collection* shape what decisionmakers learn from experts who want to make them optimistic. A receiver (she) must make a choice to maximize an outcome. She has a prior over the functional form mapping choices to outcomes. She hires an expert sender (he) to provide a model of this mapping. The sender collects data and provides the best-fitting model to that data. The sender is either "accurate" or "simple." An accurate sender considers the same models as the receiver and estimates the mapping correctly. A simple sender estimates misspecified models whose curvature is calibrated to a fixed value. Since the receiver knows the sender might be simple, she entertains not only the sender's original model but also alternative models that could have produced his data.

The paper has two key sets of results. First, I show that a simple sender strategically samples data about extreme choices, so that these alternative models promise the receiver higher expected utility than his original model. If he assumes low curvature (such as a linear regression), he samples bimodal data about extreme choices. If he assumes high curvature, he samples local data about a single extreme choice. Either way, these data say little about choices the receiver is likely to make. Moreover, with multiple choice variables, he also samples data that are highly collinear, so that the effects of different choices cannot be separately learned. Second, I show that an accurate sender thus benefits if the receiver suspects his model might be simple, since she entertains

alternative models that deliver higher expected utility than his accurate model. In particular, I show that this benefit is maximized when accurate senders only surround themselves with extreme senders who calibrate their models to very low or high curvature values.

Model Overview A receiver — such as a policymaker, CEO, investor, or voter — must choose an action x to maximize an outcome y — such as profits or the value of a public good. She believes the true functional form mapping x to y is given by a set of data-generating processes (DGPs) that capture a tradeoff between a linear benefit and a convex cost:

$$y = \alpha_0 + \alpha_1 x - \gamma x^2.$$

The parameter $\gamma \in [0, \overline{\gamma}]$ represents the *curvature* of tradeoffs or magnitude of costs.

The receiver hires a sender to acquire information for her. This information takes the form of a *model*: a functional form mapping choices x to outcomes y, which is in turn estimated on data. The sender first samples that data. He chooses a distribution of choices x to collect information about. He then observes the corresponding y values, producing data (y, x) which are unobservable to the receiver. The sender then fits a model to the data. He has a private type that determines the set of models he can fit. He is either "accurate" or "simple." An accurate sender considers the same models as the receiver. A simple sender considers similar models, but he is *misspecified* and calibrates the curvature parameter γ to a predetermined, *fixed value*. If he assumes γ is low, he *neglects* tradeoffs the receiver considers. For example, if he assumes $\gamma = 0$, he runs a linear regression of y on x. If he assumes γ is very high, he *exaggerates* tradeoffs between choices and outcomes. For example, if he assumes $\gamma = \overline{\gamma}$, he assumes the convexity of costs is as large as possible, which I refer to as a "worst-case scenario" model. Given the data he collects, he provides the receiver with the best-fitting model to that data. Although the receiver cannot observe these raw data, the interpretation is that the sender's model must be defensible and can be audited either by the receiver's organization or the scientific community.

The receiver is Bayesian: she observes the sender's model, updates over the set of models she considers, and makes the choice that maximizes her expected utility. The sender wants to make the receiver optimistic: formally, he maximizes her *perceived* expected utility. This reflects the reality that a receiver's decision is only implemented if others believe it will be effective. Policymakers must convince politicians to pass tax reforms, and CEOs must convince their boards to adopt workforce initiatives. The more optimistic a sender can make the receiver, the more effectively she can justify her choice to others, and the more likely a new initiative is to be approved. The sender gains credit and reputation for helping launch the initiative, regardless of its eventual outcome.

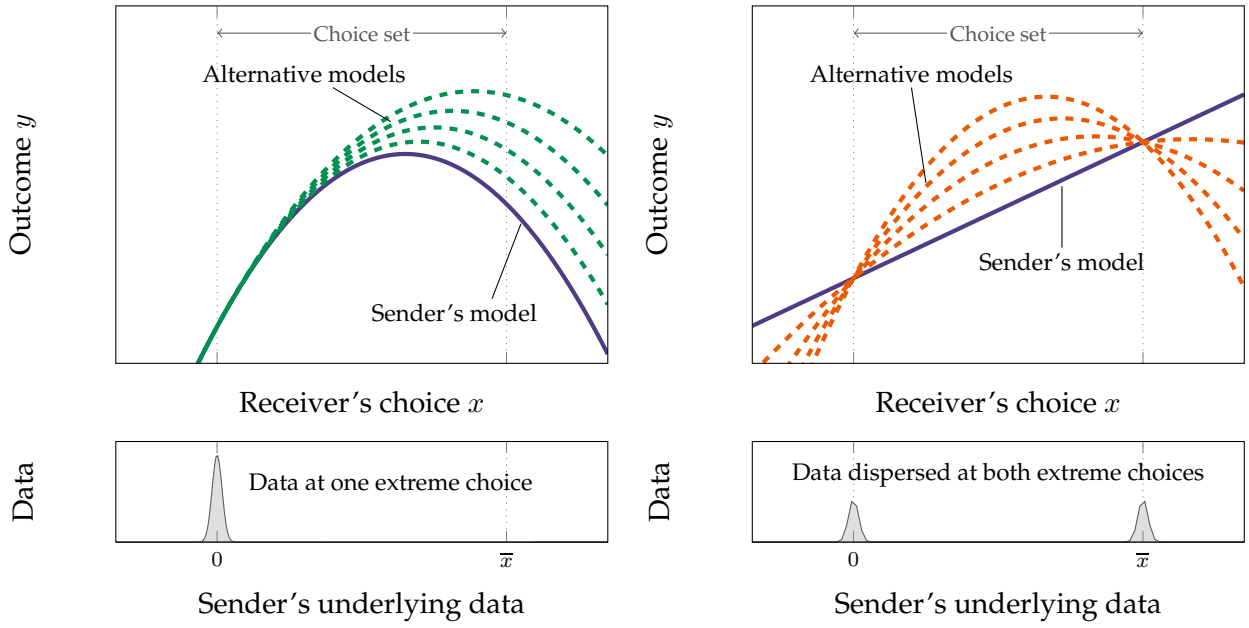
Receiver Learning and Sender Sampling I show that the receiver's learning depends on the interaction between the sender's model simplicity and how he collects data. If the receiver knew the model came from an accurate sender, she would take it as the true mapping between x and y. But because it may have come from a simple, misspecified sender, she also entertains alternative

models that could have rationalized his data. Since a simple sender calibrates curvature to a fixed γ, these alternatives differ in their curvature and, as a result, in their intercept and slope.

The key is that every alternative model the receiver finds plausible satisfies two constraints: its *mean* and its average *first derivative* over the sender's data must match the sender's model. Suppose the sender's underlying data sampled choices x from a distribution with mean x_δ and variance σ_δ^2. All plausible models must have the same derivative as the sender's original model at x_δ, and also coincide with the original model at $x_\delta - \sigma_\delta$ and $x_\delta + \sigma_\delta$. The spread of the underlying data determines where these alternative models lie relative to the original model, shown in Figure 1. The solid curve indicates the sender's original model and the dashed curves alternative models.

When the sender samples data in a small neighborhood of a single point, every plausible model is *tangent* to the sender's model at x_δ (left panel). I refer to this as *local* sampling: the data describe the relationship between x and y only in a narrow range of choices. When the sender instead samples data with a large variance, the plausible models are *secants* that cross the sender's model at $x_\delta \pm \sigma_\delta$ (right panel). When data is sampled with the largest possible variance, I refer to this as *dispersed* sampling, in which case the data are bimodal and mostly describe extreme choices.

Figure 1: Sender's model (solid) vs. alternative models receiver entertains (dashed) and data underlying sender's model



(a) Alternative models under "local" data

(b) Alternative models under "dispersed" data

The receiver's beliefs thus depend on the alternative models she entertains, which the sender can control through how he samples data. I show that a simple sender whose model has *low curvature*, such as a linear regression, samples *dispersed* data, as in the right panel. Because his model assumes a low value of γ, the alternative models the receiver entertains are mostly *steeper* than the sender's, shown by the orange dashed lines. Under local sampling (left panel), these

steeper models lie *below* his model. Under dispersed sampling (right panel), they lie *above* his model for a large set of interior values. So, for the same model, the receiver's *perceived* expected utility rises the more dispersed his data are. This requires placing substantial mass on extreme choices, some of which lie outside the receiver's choice set.

A simple sender who estimates a *high curvature* model, such as a worst-case scenario, samples *locally*. Upon seeing a high curvature model, the receiver mostly entertains alternative models with *flatter* curvature. Under local sampling (left panel), these flatter models lie above his model. Under dispersed sampling (right panel), they lie below it. Local sampling therefore makes the receiver more optimistic. Moreover, he samples in a small neighborhood of either the receiver's lowest or highest feasible choice, where he expects his model to be steepest.

This combination of model simplicity and strategic sampling benefits the accurate sender. Because the receiver cannot tell whether a model came from a simple or an accurate sender, she entertains alternative models even when the model is in fact accurate. Since simple senders sample strategically, these alternatives deliver higher expected utility on average than the original model itself. The receiver thus weighs not only the true model, which the accurate sender knows to be correct, but also alternatives more optimistic than the truth, which benefits the accurate sender.

I then extend the baseline model to multidimensional choices and show that simple senders collect data that maximize collinearity between variables. If a sender assumes a low *interaction* effect between two variables, he samples data so that they are as correlated as possible; if he assumes a high interaction effect, he makes them as anticorrelated as possible. The intuition is that if the sender's model assumes away (negative) interactions, the sample mean in the receiver's underlying data is in part driven by a negative interaction of two highly correlated variables. If the receiver were to unwind this correlation, the sample mean would be higher than what she initially observed. The opposite logic holds when the sender's model assumes a high interaction effect. This means that senders who estimate simple models collect datasets that are heavily unbalanced.

Moreover, I show that a receiver's choices are more accurate under an intermediate degree of sampling, rather than local or dispersed sampling. Thus, conditional on model simplification, simple senders do not even sample data in a way that is second-best for the receiver, and include either precise or disperse data on the edges of the choice space. This low quality of data emerges precisely as a consequence of model simplification, strategic data collection, and optimism.

The baseline model assumes the sender always estimates a misspecified model with *fixed* curvature. However, I also show that the results generalize if the sender can estimate models with a range of curvatures that are still a subset of the receiver's. Moreover, while most of the paper assumes the receiver's payoff y is quadratic in x, I also show how the results are robust to the receiver's payoffs being given by a linear benefit less a generic convex cost.

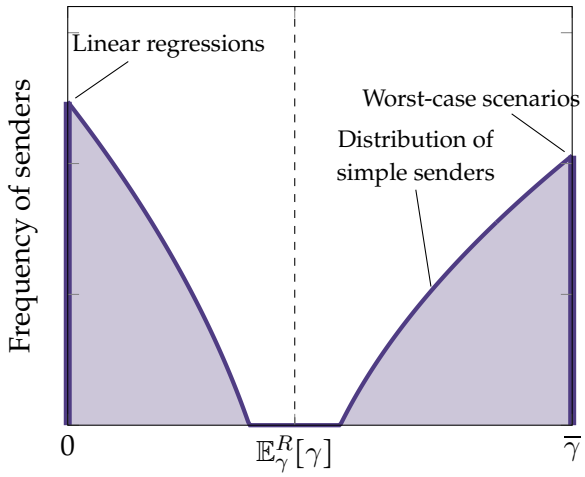
The Supply of Simple Senders The baseline model shows that the accurate sender gains from being mistaken for a simple sender. However, I show that this gain varies with a simple sender's type, i.e., the value of γ the simple sender uses to calibrate his model's curvature. If the accurate

sender could *choose* which simple senders operate alongside him, which would he choose? Formally, what *distribution* of simple sender types would maximize the accurate sender's utility? I use the answer to this question to characterize what sorts of models simple senders will supply in equilibrium and how frequently the receiver will see different simple models.

In practice, the closest analogue of this problem is one where an accurate sender is able to hire additional simple senders to work alongside him. A senior consultant, for instance, may build a team by hiring junior consultants who analyze problems in simple ways. In turn, his clients may be unsure as to how closely he supervised the model the team provides them. The same logic applies more broadly: a politician may propagate simple narratives in the media through extremist co-partisans, or an AI platform may deploy simpler LLMs alongside more complex ones.

I show that the accurate sender surrounds himself entirely with simple senders who assume very low curvature values (close to $\gamma = 0$) or very high curvature values (close to $\overline{\gamma}$), and no senders who assume intermediate curvature values. This is illustrated graphically in Figure 2. The intuition is that the accurate sender's gain from being misperceived as simple is higher the farther his model's curvature departs from the receiver's prior expectation $\mathbb{E}_R^{\gamma}[\gamma]$. Thus, an accurate sender will prefer to place mass on simple senders who calibrate their curvature to very extreme γ values, and no mass on those who calibrate their models to intermediate values. The simple models a receiver encounters are therefore systematically extreme and *polarized*. In the limit, as the receiver becomes confident she faces the accurate sender, the only remaining simple models are linear regressions and worst-case scenarios. The prevalence of these extreme models need not reflect experts' ignorance or lack of data. Instead, it can arise because extreme models are the ones that make accurate experts look best.

Figure 2: Distribution of simple sender types chosen by the accurate sender



Overview of Paper Section 2 lays out the basic model. Section 3 characterizes learning, presents the results on strategic data sampling, and then the results with multivariate choices. Section 4

extends the model with forecast error and general functional forms. Section 5 studies the accurate sender's design problem and the supply of simple models. Section 6 concludes. The remainder of this section reviews the relevant literature.

Literature This paper broadly adds to our understanding of how we learn from externally-provided models and data, and how incentives to make us optimistic distort the design of information through strategic data collection and model simplification. Its key contribution is to characterize what *sorts* of data model senders may strategically sample, how incentives to strategically sample data are informed by the simplicity of their models' curvature, and how the interaction of both can generate gains from systematically pursuing highly simplistic data-generating processes.

Many of the examples in the paper concern *delegation* to experts. There is a long line of literature — particularly in organizations and finance — studying this process, including mistrust in experts (Cheng and Hsiaw, 2022), uncertainty about expertise (Kishishita, 2020), and its interaction with persuasion (Kolotilin and Zapechelnyuk, 2025). My contribution to this literature is to study how delegation and communication operate when an expert's recommendation comes in the form of a *model* organizing endogenously collected *data*. This conceptually contrasts my paper with work like Deimen and Szalay (2019), who also study delegation to experts, but are concerned with the information that can be gleaned from experts' reports when their preferences may be misaligned with decisionmakers'. The most closely related paper in the finance literature is Li and Wang (2018), who study delegation when an investment advisor provides a model of returns to a client.

While there are many papers that study learning from models, much of this literature takes as given the past history of data that a model is fit on — whether those data are exogenous or endogenously generated by a principal's or agent's choices (Eliaz and Spiegler, 2020; Schwartzstein and Sunderam, 2021; Levy et al., 2022). My contribution is to study a sender who first chooses a data collection rule and then provides the best-fitting model to those data. Thus, although data collection is endogenous and unobserved by the receiver, the sender's hands are still tied by his initial data collection. This assumption contrasts with other papers where data are endogenously generated either through follow-on evidence generation (Dai et al., 2026), selective disclosure (Felgenhauer and Schulte, 2014), nonrandom sampling (Di Tillio et al., 2021), or sequential data acquisition with endogenous stopping (Henry and Ottaviani, 2019). These papers show that the ability of a sender to persuade a receiver often depends on the sorts of signals about data a sender can transmit. In my setting, I show that a sender's desire to make the receiver optimistic coupled with model simplicity creates a tractable characterization of data sampling: the sender either locally samples data in the smallest possible neighborhood of an extreme receiver choice, or as distantly as possible across a wide range of receiver choices.

The use of strategic *sampling* interacts with a sender's model simplicity to make the receiver believe her expected utility is *higher* than under the model provided by the sender. This channel

through which a receiver can be made more optimistic is thus related to model simplicity and data sampling, as opposed to departures from a status quo or the inclusion of confounding variables, such as in Eliaz and Spiegler (2020). This also distinguishes the paper from work where a receiver selects between different models with different sample sizes and more or less complex models (Montiel Olea et al., 2022).

My paper shows that gains from model simplicity or strategic sampling can arise without interpreting the role of noise in driving data-generating processes. This finding contrasts with papers that investigate gains from model-based persuasion that involve trading off model fit and belief manipulability — such as Schwartzstein and Sunderam (2021), Aina (2023), Barron and Fries (2024), and Aina and Schneider (2025). The present paper does, however, show that senders who make receivers more optimistic often create greater *forecast* error for a receiver, suggesting that receiver optimism may come at the cost of uncertainty over models.

The particular angle the present paper takes is closely related to omitted variable bias — in that a sender cannot (or refuses to) fully model a tradeoff in the way a receiver would like. Thus, this paper is related to an older literature on omitted variable bias in econometrics (see Theil, 1957; Altonji et al., 2005; Oster, 2019, and references therein). It also connects the paper to theoretical work like Levy et al. (2022), Gleyze and Pernoud (2022), and Eliaz et al. (2025) — who show that model misspecification may interact with competitive pressures or uncertainty about data estimation to generate a range of outcomes from improved communication and engagement to political cycles and the prevalence of false models. More broadly, this paper studies a model where a receiver entertains a well-specified model and a sender estimates a misspecified model. It thus contributes to a deeper theoretical literature on Berk-Nash equilibrium and learning with misspecified beliefs (Alonso and Câmara, 2016; Esponda and Pouzo, 2016; Spiegler, 2016; Heidhues et al., 2018; Mailath and Samuelson, 2020; Bohren and Hauser, 2021; Fudenberg et al., 2021; De Clippel and Zhang, 2022; Levy et al., 2022; Bohren and Hauser, 2025).

A key finding of this paper is that senders may generally benefit from providing *simpler* models to receivers. The idea that senders can benefit from simple models is also captured by Perez-Richet and Prady (2012), Bloedel and Segal (2018), Lipnowski et al. (2020), Eliaz et al. (2021), and Burkovskaya and Starkov (2026). While many of these papers are driven by behavioral assumptions like bounded rationality or rational inattention, the findings in this paper are related to sampling and simplicity: given a sender's model, a receiver will entertain additional models. The set of these additional models is determined by the *simplicity* of the sender's model, and the expected utility they generate for the receiver increases or decreases with the spread of the sender's data sampling. In particular, the present paper shows that a sender informed of the true model of the world can benefit by making the receiver think his model was estimated by a sender who had an artificially simple model *and* strategically sampled. The most closely related paper is Gleyze and Pernoud (2022), who show that an informed sender can often gain by delegating model provision to a (potentially misspecified) sender.

Finally, because of its reliance on models as information structures, the present paper also

adds to a longer literature on (Bayesian) persuasion (Kamenica and Gentzkow, 2011), as well as persuasion and competition (Gentzkow and Kamenica, 2017; Wang, 2013). In particular, receivers in the present paper are uncertain as to the source of data or the broader class of models a sender entertained. This connects the paper to models of persuasion and cheap talk with *uncertainty* about information structures (Huang, 2024; Min, 2026) and ambiguity aversion (Colo, 2024).

2 Model Setup

I begin by laying out a game between a single receiver R (she) and an expert sender S (he). R must make a choice to maximize an outcome, and enlists the services of S. S collects data and provides R a (potentially misspecified) model describing those data. R sees S's model, updates, and then makes the choice that maximizes the expected outcome. Throughout, I use the running example from the introduction where R is a policymaker tasked with designing a carbon tax, and S is a consultant who models the effects of different taxes.

2.1 Receiver and Models

R faces a set of choices $X = [0, \overline{x}] \subseteq \mathbb{R}$ for $\overline{x} > 0$. Each of these choices maps to an outcome $y \in \mathbb{R}$. Ex-ante, R does not know the relationship between y and x. However, she believes that the value of y given x can be described by a data-generating process (DGP) $y = f_R(x)$ for a function $f_R : X \to \mathbb{R}$. We refer to f_R as a **model**. We denote F_R as the set of *models* f_R that R entertains.

I assume that for each $f_R \in F_R$, we have

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma x^2,$$

where $\alpha_0 \sim p_0^R$ on $\mathbb{R}$, $\alpha_1 \sim p_1^R$ on $\mathbb{R}$, and $\gamma \sim p_\gamma^R$ on $[0, \overline{\gamma}] := \Gamma$ for continuous priors p_0^R, p_1^R, p_γ^R. Her prior over each of the three components is independent. p_0^R is uniform on $[\underline{\alpha}_0, \overline{\alpha}_0]$ for $\underline{\alpha}_0$ arbitrarily negative and $\overline{\alpha}_0$ arbitrarily large. p_1^R is uniform on $[\underline{\alpha}_1, \overline{\alpha}_1]$ and p_γ^R is symmetric. I assume that $\underline{\alpha}_1 \leq 0$ and $\overline{\alpha}_1 \geq 2\overline{\gamma x}$.

Each of R's models is concave, and can be represented as an affine function less a convex, quadratic loss.[1] The magnitude of this quadratic loss is governed by the parameter γ. γ represents the *curvature* of the data-generating process, and hence how sharp the receiver's *tradeoffs* are.

The receiver's goal is to maximize the expected value of y. Let q denote her belief over F_R, i.e. her (joint) belief over these three parameters. Her expected utility is given by:

$$U_q(x) = \mathbb{E}_q[f_R(x)].$$

Her goal is to choose x to maximize $U_q(x)$.

[1] I assume a quadratic form for tractability but show that the central intuitions generalize in Appendix B.3.

In the running example, R is a policymaker who designs a carbon tax x. y represents the welfare of a representative household or voter. Higher taxes reduce emissions and improve health outcomes, but may also increase energy prices and reduce private sector activity. The single choice of a tax rate x may have different consequences for welfare, depending on the *aggregate* of these different effects. The policymaker would like to balance these effects and choose the tax rate that maximizes expected welfare. However, there are many other interpretations of the receiver and her preferences. If R is an investor, she cares about how much money she can make by investing in a choice x. If R is a CEO restructuring her workforce, she cares about the effect of layoffs on profits. If R is a voter, she cares about the impact of a policy x on her wellbeing.[2]

2.2 Sender and Data

The receiver R contracts a sender S to provide her with information about the relationship between x and y. This sender is an *expert* — such as a scientist or experienced consultant — who provides domain expertise and access to data. Of particular interest is also the case where S is an AI program, such as a large language model (LLM) that R can interface with. The key idea is that, as an expert, the sender is able to gather and interpret data for the receiver, which she may find difficult to do on her own.

The sender has a private type θ determining the sorts of models he can provide the sender. The order of events is as follows.

1. Nature draws S's type and the receiver's true model f_R.
2. S collects data and calculates the model of best fit, given his type.
3. S provides the best-fitting model to R. R updates about the true data-generating process.
4. R makes a choice. S receives utility. Uncertainty resolves, and R receives her utility based on the realized outcome of her choice.

I now provide more detail about each of these components.

Sender's Type A sender's private type θ affects the *simplicity* of the models he is capable of providing to R. We denote the type space by Θ. There are two varieties of sender. The first is an **accurate** sender. We denote his type a. This sender provides a model in the set

$$F_a := \{f_a \in F_R : f_a(x) = \beta_0 + \beta_1 x - \delta x^2, (\beta_0, \beta_1) \in \mathbb{R}^2, \delta \in \Gamma\}.$$

We note that $F_a = F_R$, i.e. the accurate sender is estimating *exactly* the same models as R. Additionally, I assume he has the same prior over models as the receiver. The ex-ante probability of seeing a type a sender is p_a.

[2] I utilize these simple preferences for higher y in the baseline model, but Section 4.2 extends them to evaluate settings where R also values *forecast accuracy*, and thus knowing the correct model of the world.

The second variety of sender is a **simple** sender. This sender has a type δ, where $\delta \in [0, \overline{\gamma}]$. A simple sender is limited in how he can estimate the *curvature* of models. [3] A type δ sender is able to provide R a model in the set F_δ.

$$F_\delta := \{f_\delta \in F_R : f_\delta(x) = \beta_0 + \beta_1 x - \delta x^2, (\beta_0, \beta_1) \in \mathbb{R}^2\}.$$

The interpretation is that a simple, type δ sender **calibrates** the curvature of the convex loss to δ, and does not estimate it on the data. For example, the consultant may calibrate the effect of carbon taxes on energy prices using a fixed elasticity from the literature, rather than estimating it on data that may be more or less relevant to the receiver's setting. Or, he may believe that the cost of actions is always given by δx^2, based on his own idiosyncratic experiences. The key is that the simple sender believes the *curvature* of the relationship between x and y takes a fixed value.

We denote by $P^\delta_{f_S}$ a type δ sender's prior over models $f_S \in F_\delta$. Effectively, his prior is over β_0 and β_1. We assume his prior over β_1 is identical to the receiver's prior over α_1. The simple sender thus estimates *misspecified models*; he truthfully believes that his way of modeling the problem (in simple terms) is the right way.

The receiver sees a simple sender with ex-ante probability $1 - p_a$. A simple sender's type is in turn drawn from a distribution T over $[0, \overline{\gamma}]$ known to R. If T is continuous with pdf t, the ex-ante density of a type δ simple sender is thus $(1 - p_a)t(\delta)$. In turn, the full type space Θ is given by $\Theta = \Gamma \cup \{a\}$.

Two Simple Types There are two simple sender types that will be of particular interest to our analysis. The first is a type $\delta = 0$ sender. He estimates models of the form $y = \beta_0 + \beta_1 x$. This sender is effectively running a linear regression of y on x, even though the receiver believes the true relationship is given by a nonlinear DGP. In the example, the policymaker may believe small carbon taxes generate meaningful reductions in pollution without adverse effects, but very high taxes lead to higher energy prices or lower business activity. These nonlinear effects would be neglected if a consultant presented a linear regression of welfare on tax rates.

The second is a type $\delta = \overline{\gamma}$ sender. He estimates models of the form $y = \beta_0 + \beta_1 x - \overline{\gamma} x^2$, where $\overline{\gamma}$ is fixed at its highest possible value. This sender estimates a "worst-case" model: he always assumes the magnitude of the convex loss is as large as possible and, hence, assumes the *curvature* of the tradeoff between choices and outcomes is as sharp as possible. In our example, the consultant may assume that higher taxes always have maximally deleterious effects on the business sector, which the policymaker may believe is unlikely.

Sender's Data Before providing a model to the receiver, the sender first collects data about certain choices. He selects a distribution of choices x to collect information about. He then observes

[3] Appendix B.1 relaxes this assumption and allows types to be intervals of curvature values, so that a sender may neglect or exaggerate some tradeoffs while correctly estimating others. The main results are robust to this richer type space. However, the design problem studied in Section 5 shows that if a receiver saw a simple sender in equilibrium, she would only see one of these *singleton* types.

the y value associated with that distribution of choices. This gives him a dataset he can use to fit his model. While the receiver cannot observe or interpret this dataset on her own, the model-fitting process is auditable.

Prior to estimating a model, S chooses a data sampling strategy μ, which is a distribution over $\mathbb{R}$. Writing $x_\mu := \mathbb{E}_\mu[x]$, the set of distributions he can sample from is given by the set M.

$$M := \Big\{\mu \in \Delta(\mathbb{R}) : \ x_\mu \in X, \ \ x - x_\mu \overset{d}{=} x_\mu - x, \ \ \text{supp}(\mu) \subseteq [x_\mu - \overline{\sigma}, x_\mu + \overline{\sigma}], \ \ \text{Var}_\mu(x) \geq \underline{\sigma}^2\Big\},$$

where $\underline{\sigma} > 0$ is arbitrarily small and $\overline{\sigma}$ is large. That is, μ is symmetric about a mean in the receiver's choice set X, lies within $\overline{\sigma}$ of that mean, and has variance bounded away from zero, but can otherwise take any shape. [4]

We will refer to a type θ sender's choice of μ as μ_θ. We denote its mean $x_\theta := \mathbb{E}_{\mu_\theta}[x]$ and variance $\sigma_\theta^2 := \text{Var}_{\mu_\theta}(x)$. I note that μ does allow the sender to sample information about x values outside of the receiver's choice set X. This is consistent with the interpretation that the receiver's choices are "bounded" in a broader space of potential choices, and that choices outside of X are concretely "irrelevant" to the receiver's eventual choice.

Given this sampling strategy over *choices*, S then directly observes the true joint distribution of $(y|x, x)$, subject to observational noise. Formally, if the true data-generating process is f_R^*, he observes a joint distribution $D(y|x, x)$, where the marginal distribution of x is given by $D_x = \mu_\theta$ and the conditional distribution of y given x is given by $D_y(y|x) = f_R^*(x) + \eta$, where η is a mean-zero noise term with unknown variance. This is equivalent to S observing an arbitrarily large (or infinite) number of i.i.d. draws from the true data-generating process. [5]

Sender's Information With the sender's type and sampling strategy in hand, we define the model S provides to R. We say that a model f_S **best-fits the data** with respect to θ and a data-sampling strategy μ_θ if:

$$f_S = \arg\min_{f \in F_\theta} \mathbb{E}_{D(\mu_\theta)}\big[\big(y - f(x)\big)^2\big].$$

For example, when $\theta = \delta = 0$, f_S represents the *best-fitting linear model* to the data.[6]

I assume S always provides R the best-fitting model to the data. There are several interpretations of this assumption. The first is that the sender makes an honest effort to provide the receiver with a best-fitting model, conditional on the data he collected. The second is that, although the receiver may be unable to observe or interpret the raw data herself, the sender's model will be audited at some point. That audit could be performed by the receiver's organization, or externally

[4] The assumption of symmetry is used for tractability to discipline the set of models the receiver will entertain, but the results generalize. Relatedly, the constraint that $\sigma > 0$ is a regularity assumption made to ensure learning is structured in equilibrium.

[5] I allow S to observe the distribution of data to shut down the role of idiosyncratic data draws in shaping posterior beliefs.

[6] The existence of a best-fitting model is given by the Hilbert Projection Theorem.

by a broader scientific community.

In the running example, the assumption that the consultant provides the best-fitting model given his type can be justified in three ways. First, imagine that he is a trained economist who relies on well-known methods such as regressions or structural estimation. Even if the policymaker cannot interpret the raw data herself, her organization can request a replication package to verify his methodology. Second, his work is subject to scrutiny from the broader scientific community — in this caseww, the economics profession — which would expose fabricated data or a poorly fitting model. Third, the policymaker can work back and forth with the consultant until he justifies why his model is the best way to describe the data. In each case, the receiver learns about the data only through the sender's model, which allows me to isolate how model *simplicity* and strategic *data collection* shape learning and sender behavior.

Sender's Preferences The sender's goal is to maximize the receiver's *perceived expected utility*, after she has made her choice but prior to the resolution of any uncertainty. Suppose S provides a model $f_S \in F_\theta$. R has a posterior belief over the true DGP $f_R \in F_R$ as a function of f_S, which are also affected by her beliefs about the underlying data via the sender's choice of μ and her beliefs about the sender's type. Let $x^*(f_S)$ be the receiver's choice of $x \in X$ conditional on her posterior beliefs. A type θ sender's utility is then:

$$U^\theta(f_S) = \mathbb{E}^R_{f_R|f_S}[f_R(x^*(f_S))].$$

Conditional on providing a model f_S, S's utility is given by the receiver's *perceived* expected utility at her consequent choice.

The interpretation is that the sender would like to make the receiver *optimistic* about the expected impact of her choices. In our running example, imagine that the policymaker is trying to ascertain whether she should launch a new carbon tax program. If she is sufficiently optimistic about the transformative impact that such a program will have, she is more easily able to justify launching that program to politicians and bureaucrats. In the event the program is launched, the consultant can take credit for helping out with the program, which can make it easier for her to receive future business, regardless of whether that eventual program succeeds or fails.[7]

These preferences apply more broadly across a variety of settings. For example, a management consulting firm may inform the choice of a CEO to engage in mass layoffs. The CEO is better able to justify his choice to the board if he is more optimistic that mass layoffs will increase profits. If the sender is a financial advisor, he gains from making his clients think their investment returns will be high. If the sender is a politician and can make voters optimistic about the effects of a new law once reelected, they may be more willing to turn out to vote for him. Finally, if the sender is an AI chatbot, optimistic users are more likely to engage with the chatbot and, as a result, may be

[7] One of the key insights of the paper is that the receiver will entertain not only the sender's original model but also alternative models that, on average, deliver strictly higher expected utility. Thus, in equilibrium, the expected outcomes the receiver realizes will tend to be higher than the sender's original model.

more likely to purchase a premium subscription.

Equilibrium I formally define equilibrium for the sender and receiver. Given the probability p_a of an accurate sender and the distribution T of simple sender types, an **equilibrium** consists of a sampling strategy $\mu_\theta^* \in M$ for each type $\theta \in \Theta$, a belief $\hat{\mu}_\theta \in M$ about each type's sampling strategy, a receiver posterior belief $q(f_R|f_S)$ over F_R for each model f_S, and a choice $x^*(f_S) \in X$ for each model f_S, such that the following are satisfied.

1. A type θ sender who samples according to μ_θ^* provides the model $f_S \in F_\theta$ that best-fits the data with respect to F_θ and μ_θ^*.
2. For each f_S, $q(f_R|f_S)$ is derived by Bayes' rule from R's prior over f_R, the probability p_a, the type distribution T, and her belief $\hat{\mu}_\theta$ about the sampling strategy of each type $\theta \in \Theta$.
3. For each f_S, $x^*(f_S) \in \arg\max_{x \in X} \mathbb{E}_{q(f_R|f_S)}[f_R(x)]$.
4. For each $\theta \in \Theta$,

$$\mu_\theta^* \in \arg\max_{\mu \in M} \mathbb{E}^\theta_{f_S}\Big[\max_{x \in X} \mathbb{E}_{\hat{q}(f_R|f_S;\mu)}[f_R(x)]\Big],$$

   where $\hat{q}(f_R|f_S;\mu)$ is S's second-order belief about R's posterior given f_S, when R believes S sampled according to μ.
5. For each $\theta \in \Theta$, $\hat{\mu}_\theta = \mu_\theta^*$, i.e. the receiver's beliefs over μ_θ are equal to their actual distribution.

2.3 Additional Comments

The receiver's learning is affected by two key frictions. The first is that the sender's models are *simple*: S may be unable to think about problems exactly the way R would like. In the two-simple-type example, S either estimates a linear model that *totally* neglects tradeoffs, or a worst-case model that totally exaggerates them. This assumption is consistent with several interpretations.

1. **Infeasibility**: the sender's model may represent the culmination of a highly complex data synthesis process that makes it difficult to directly and fully estimate more complicated models. In the running example, the consultant may perform complex climate simulations which in turn feed into an intricate structural model before delivering a recommendation. Simple models may emerge as a necessity to make data synthesis tractable.
2. **Calibration**: instead of estimating parameters organically on data, S may calibrate his models' curvature to a fixed value, such as an elasticity from the literature. Relatedly, the sender may believe that, based on his own experience, the true magnitude of the sender's cost is always given by a fixed value.
3. **Strategic behavior**: the sender may be able to make the receiver *more optimistic* if she thinks his model was a simplified model. This will be a central result of the paper (Section 5).

The second friction is that the receiver does not simply request data from the sender and fit the model herself. The interpretation is that the receiver cannot process raw data on her own, and precisely hires an expert to interpret that data for her. That is, the only way for her to process information is through an expert's model. Additionally:

1. The sender may have access to a proprietary dataset, which cannot be directly disclosed. Commonly used datasets like those from Nielsen provide proprietary information on "audience segments," for instance, to help advertisers run targeted campaigns (Nielsen, 2024).

2. The sender's model may also embed *expertise*. The strategy literature suggests that firms often develop internal ways of approaching a problem based on analogous situations they have experienced in the past (Carroll and Sørensen, 2024). Accordingly, a consulting firm's recommendations may be informed by experience working on cases similar to each other or drawn from many different cases and past contexts .

3. The sender is an AI program. It is often difficult for users of these programs (or even their developers) to understand the mapping from internal data to recommendations (Zuccarelli, 2024). In this sense, the mapping from the data to a model is a *black box*.

What matters is that R knows S has access to useful data, but must infer how those data may have been strategically sampled when estimating a model.[8] The case where S provides linear regression models is also of particular interest, given their common usage in the business world, economics, and data-driven social sciences.

Additionally, R may also be unable (or unwilling) to expend organizational time and resources in collecting, understanding, and synthesizing complex data. Instead, she delegates the collection of data and its synthesis into a model to the sender, expediting her decision-making process in the interest of convenience, but potentially at the cost of information quality.

I focus largely on settings where the nonlinear component of f_R is quadratic. While this is done for tractability, Appendix B.3 discusses the assumptions necessary to generalize the logic.

3 Analysis

This section first characterizes the set of models that the receiver entertains, and then shows how senders strategically sample data. I show that simple senders who assume low-curvature models — such as linear regressions — collect *dispersed data* that place mass bimodally on extreme choices, which may lie outside of the receiver's choice set. Simple senders who assume high-curvature models — like worst-case scenarios — collect *local data* in a small neighborhood of either the receiver's lowest choice ($x = 0$) or highest choice ($x = \overline{x}$). In either case, the data underlying these models disproportionately contain information about choices that will be *irrelevant* to the receiver's eventual choice.

[8] Accordingly, I will be interested in exactly *what sorts of data* the sender will collect in equilibrium, and *how* that data collection interacts with the simplicity of his models.

Consequently, I show that accurate senders gain from being misperceived as simple senders, in that the existence of simple senders allows their models to make the receiver more *optimistic* than they would otherwise. The reason is that strategic sampling coupled with the model simplicity of simple senders forces the receiver to entertain not just the sender's original model, but also alternative models that deliver higher expected utility. This means that even when the accurate sender provides the true DGP, the receiver rationally considers these alternative DGPs. From the perspective of the accurate sender, this makes her more optimistic than the truth.

I then further generalize the baseline model with *multidimensional choices*, and show that simple senders maximize *collinearity* between different variables. In particular, if a sender *neglects* interaction effects between two variables, he samples data so that they are as *correlated* as possible. If he *exaggerates* interaction effects, he samples data so that they are as *anticorrelated* as possible. Thus, although receivers correctly infer the sort of data that senders collect in equilibrium, those data may be *low quality*, in that they suffer from a high degree of collinearity and embed information about choices that will (in expectation) be irrelevant.

3.1 Receiver's Learning and Behavior

I begin by characterizing receiver learning as a function of data-sampling strategies and S's type.

Receiver's Updating Suppose S provided a model of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ which best fits the data. If the sender is an accurate sender, this is the true model. However, if the sender is a simple sender, this may not be the true model. We define $\phi(f_S; \mu_\delta, \mu_a)$ as the set of functions that could rationalize the data for R, given f_S and the sampling strategies of a type δ and a type a sender.

$$\phi(f_S; \mu_\delta, \mu_a) = \left\{ f_R \in F_R : f_S = \arg\min_{f \in F_\theta} \mathbb{E}_{\mu_\theta}\left[\left(f_R(x) - f(x)\right)^2\right], \theta \in \{a, \delta\} \right\}.$$

If $\theta = a$, then because $F_a = F_R$, the best-fitting model is f_R itself, so that $f_R = f_S$. If $\theta = \delta$, we necessarily have that for each such f_R:

$$\frac{\partial}{\partial \beta_i} \mathbb{E}_{\mu_\delta}\left[\left(f_R(x) - f(x)\right)^2\right] = 0,$$

which yields the following moment conditions familiar from ordinary least squares (OLS):

$$\mathbb{E}_{\mu_\delta}\left[f_R(x) - \beta_0 - \beta_1 x + \delta x^2\right] = 0 \tag{1}$$

$$\mathbb{E}_{\mu_\delta}\left[x\left(f_R(x) - \beta_0 - \beta_1 x + \delta x^2\right)\right] = 0 \tag{2}$$

Equation (1) is the moment condition with respect to the intercept, and states that the difference between f_R and f_S must average to zero. We note that this is the same as the constraint that a model must "fit the data." Equation (2) can be combined with equation (1) to read that the

covariance between x and $f_R(x) - \beta_0 - \beta_1 x + \delta x^2$ is zero. Both expectations are taken with respect to μ_δ, which is the type δ sender's sampling strategy. The following proposition characterizes the set of functions that R entertains given f_S, δ, and μ_δ: i.e., assuming the receiver *knows* the simple type δ sender's underlying data sampling strategy.

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ best-fits the data and the simple sender's sampling strategy μ_δ has mean and variance x_δ and σ_δ^2. Then, for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$, we have:*

$$f_R(x) = f_S(x) + (\gamma - \delta)\big(\sigma_\delta^2 - (x - x_\delta)^2\big).$$

The proof of this result is contained in Appendix A. Consider the policymaker and consultant. The consultant's model may be accurate. In this case, $f_R(x) = f_S(x)$, and the consultant's model is the true model. However, it is also possible that the sender's model was simple. Thus, she considers both the consultant's original model and other models that could rationalize the tax-welfare data the consultant saw.

In particular, note that the set of models she considers depends *only* on the sampling strategy of the type δ sender, not the type a sender: the accurate sender's model f_S is itself the member of this set with $\gamma = \delta$. Mathematically, the set of models she considers, given a best-fitting model f_S, takes a simple form: they are all the sender's original model plus a constant, less a quadratic term centered at x_δ. In the case where $\delta = 0$, we have

$$f_R(x) = \beta_0 + \beta_1 x + \gamma\big(\sigma_\delta^2 - (x - x_\delta)^2\big)$$

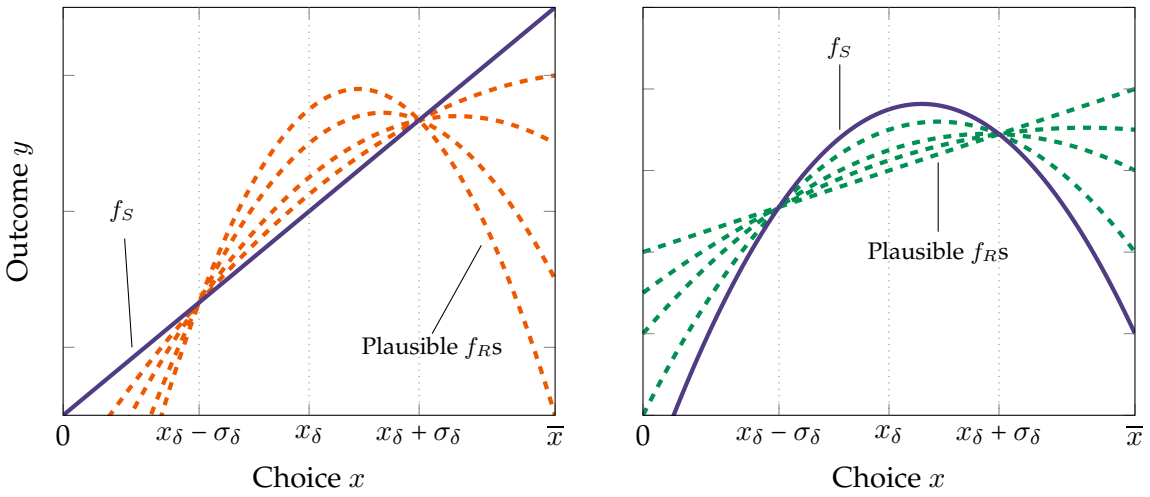
for each $\gamma \in [0, \overline{\gamma}]$. In the case where $\delta = \overline{\gamma}$, we have

$$f_R(x) = \beta_0 + \beta_1 x - \overline{\gamma} x^2 + (\gamma - \overline{\gamma})\big(\sigma_\delta^2 - (x - x_\delta)^2\big)$$

for each $\gamma \in [0, \overline{\gamma}]$. In both cases, the set of models R considers involves a fixed "affine component" plus a constant and quadratic term, which may be positive or negative. Moreover, all plausible f_Rs have the same slope at x_δ.

This intuition is shown for the type $\delta = 0$ in the left panel of Figure 3 below. The purple line shows the sender's linear regression model. The orange dashed lines indicate all the potential alternative models f_R that the receiver entertains. When the type $\delta = 0$ sender estimates a linear regression model, he neglects any tradeoffs R thinks are important. Accordingly, all the additional models that R entertains have a steeper curvature than the regression model. However, across the sampling range, they have the same sample mean and average slope.

Figure 3: Plausible receiver models f_R as a function of sender model f_S and sampling strategy μ_δ



(a) Plausible models when the sender provides a low-curvature "linear" model

(b) Plausible models when the sender provides a high-curvature "worst-case" model

The right panel shows the intuition for the type $\delta = \overline{\gamma}$ sender, whose high-curvature model exaggerates tradeoffs. Here, all the additional models that R entertains have shallower curvature than f_S. Just like the case with the linear regression, however, they all have the same average slope and sample mean with respect to μ_δ.

A key insight of Proposition 1 is that the set of alternative models the receiver entertains is solely a function of the first and second moments of the *simple* sender's sampling, i.e. x_δ and σ^2_δ. This observation will play a key role in qualitatively characterizing equilibrium sender sampling.

Receiver's Decisionmaking The receiver's learning takes a relatively simple form: regardless of the limitation on S's ability, R always learns what the data-generating process looks like "up to the first derivative at x_δ and a constant." Moreover, as σ changes — and thus S's sampling strategy changes — the set of plausible f_Rs remains identical, except that they shift up or down by a constant. This means that R's optimal *action* depends only on the probability $\Pr(\theta = a|f_S)$ that the model is accurate, her expectation of γ given that the model is simple, and x_δ. Given f_S, this choice has the following expression.

Corollary 1. *Given $f_S = \beta_0 + \beta_1 x - \delta x^2$ and a type δ sender's sampling strategy μ_δ, R's optimal action is given by $x^*(f_S)$, where:*

$$x^*(f_S) = \max\{0, \min\{\tilde{x}, \overline{x}\}\}$$
$$\tilde{x} = x_\delta + \frac{f_S'(x_\delta)}{2\Big(\Pr(\theta = a|f_S)\,\delta + \Pr(\theta = \delta|f_S)\,\mathbb{E}^R_\gamma[\gamma|f_S, x_\delta, \theta = \delta]\Big)}.$$

We note that $\tilde{x}$ is simply R's optimal interior action, and that $x^*(f_S)$ is this interior choice truncated between 0 and $\overline{x}$.

3.2 Sender's Data Sampling and Model Provision

I use these insights to show how S chooses the *spread* of his data as a function of his type. We say that a type δ sender is a **low-curvature** sender if $\delta < \mathbb{E}^R_\gamma[\gamma]$, i.e. the simple sender's assumed value of curvature is lower than the receiver's prior expectation. The type $\delta = 0$ linear regression sender is low-curvature. We say that a type δ sender is a **high-curvature** sender if $\delta > \mathbb{E}^R_\gamma[\gamma]$, i.e. the simple sender's assumed value of curvature is *higher* than the receiver's prior expectation. The type $\delta = \overline{\gamma}$ worst-case scenario sender is high-curvature.

These definitions allow me to state my main result on strategic sender sampling.

Theorem 1. *Suppose S is a type δ simple sender.*

1. *If S is low-curvature, he collects* dispersed *data about extreme choices. μ^*_δ is a bimodal distribution that places mass $\frac{1}{2}$ on each of $x^*_\delta - \overline{\sigma}$ and $x^*_\delta + \overline{\sigma}$, where $x^*_\delta \in \left[0, \mathbb{E}^\delta_{f_S}[x^*(f_S)]\right)$.*

2. *If S is high-curvature, he collects* local *data about a single extreme choice. μ^*_δ is any distribution in M with variance $\underline{\sigma}^2$, and has mean x^*_δ equal to either 0 or $\overline{x}$.*

Theorem 1 can be interpreted as follows. If the sender is low-curvature — such as when he only provides linear regressions — he collects data about choices as far away from the center of the receiver's choice space as possible. In fact, the dataset underlying his model is so dispersed that his data are collected from a two-point distribution, which may even draw only on data outside of the receiver's choice set.

However, if the sender is high-curvature, such as the worst-case scenario sender, the opposite happens. This sender collects data in a small local neighborhood of a single extreme choice. In particular, the theorem shows that this sender will sample data locally near either $\overline{x}$ or 0; his equilibrium choice of μ^*_δ is such that x^*_δ is either $\overline{x}$ or 0 and $\sigma^*_\delta = \underline{\sigma}$ is as low as possible.

The intuition is driven by the interaction of the curvature of the receiver's additional models with the sender's method of *data sampling*. The logic is shown visually in Figure 4 below. The top panels (a) and (b) look at the case where f_S is a linear regression of y on x, and illustrate the logic for low-curvature senders. The bottom panels (c) and (d) look at the case where f_S is the worst-case model fitting the data, and illustrate the logic for high-curvature senders. The smaller graphs below the graphs of f_S and f_R show what the distribution of *data* underlying different choices is.

Recall that when the receiver sees a model f_S, she entertains alternative models, which are characterized by the set ϕ. The relative location of these alternative models depends on the simple sender's sampling strategy μ_δ and, in particular, only on its mean x_δ and variance σ^2_δ.

Suppose a low-curvature S collects data in a small, local neighborhood of a point x_δ. This is shown in panel (a), where the underlying data of the simple sender μ_δ are concentrated near x_δ. In this case, the variance σ^2_δ of those underlying data is small. As a result, the alternative plausible models f_R are tangent to f_S at x_δ. These alternative models have steeper curvature (i.e. a higher γ value) than f_S, and lie *below* f_S: i.e., $f_R(x) \leq f_S(x)$ for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$. Thus, when the

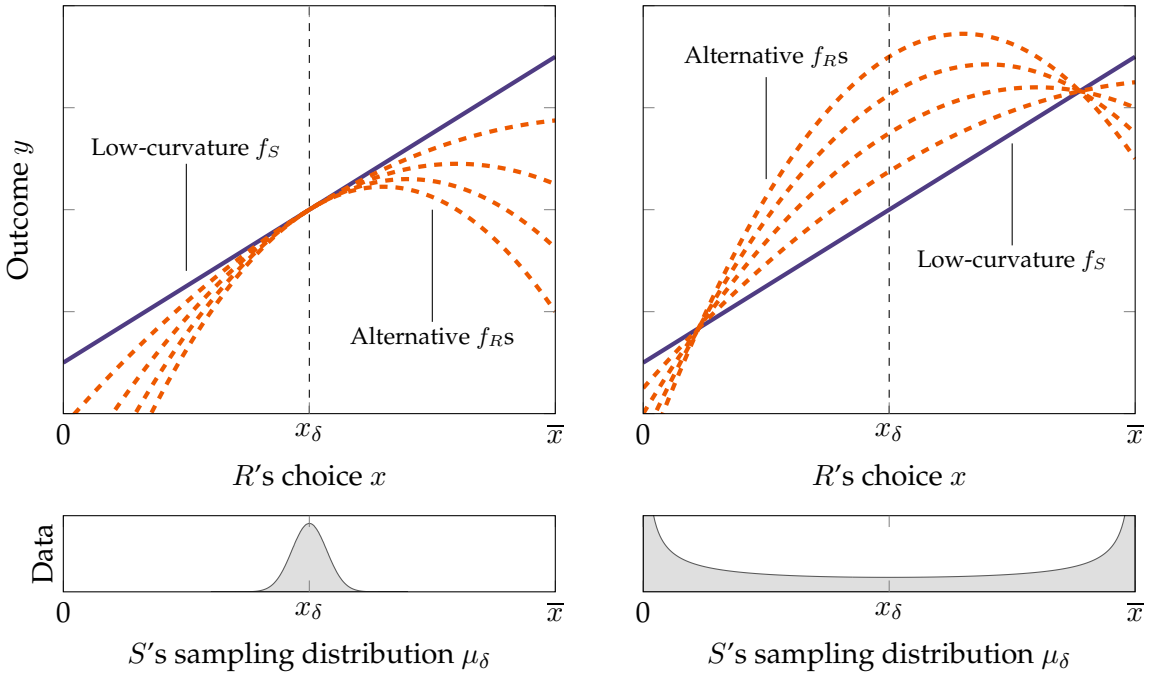
variance of the underlying data is *small*, conditional on seeing f_S the receiver's maximal expected utility is *lower* than under f_S.

However, in panel (b), the plausible alternative models are the same as in panel (a), except that each is shifted up. For values of x close to x_δ, they each deliver strictly higher utility than f_S: $f_R(x_\delta) > f_S(x_\delta)$ for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$. In fact, as σ_δ^2 grows higher and higher, the receiver's expected utility conditional on seeing f_S grows higher and higher.

In this sense, by sampling data with a wider distribution, the sender can make the receiver conditionally more optimistic. Fixing x_δ, the sender would thus sample from the distribution in M with the highest possible variance. The distribution in M with the highest variance is the two-point distribution that places equal mass on $x_\delta - \overline{\sigma}$ and $x_\delta + \overline{\sigma}$. Hence, the low-curvature simple sender samples from this two-point distribution. I refer to this sort of distribution as corresponding to *dispersed* data, and note that this is a bimodal distribution concentrated on the two most extreme possible points. The variance of this distribution is given by $\overline{\sigma}^2$.

Figure 4: Alternative plausible models that R entertains when S estimates low- vs. high-curvature models, as function of underlying data

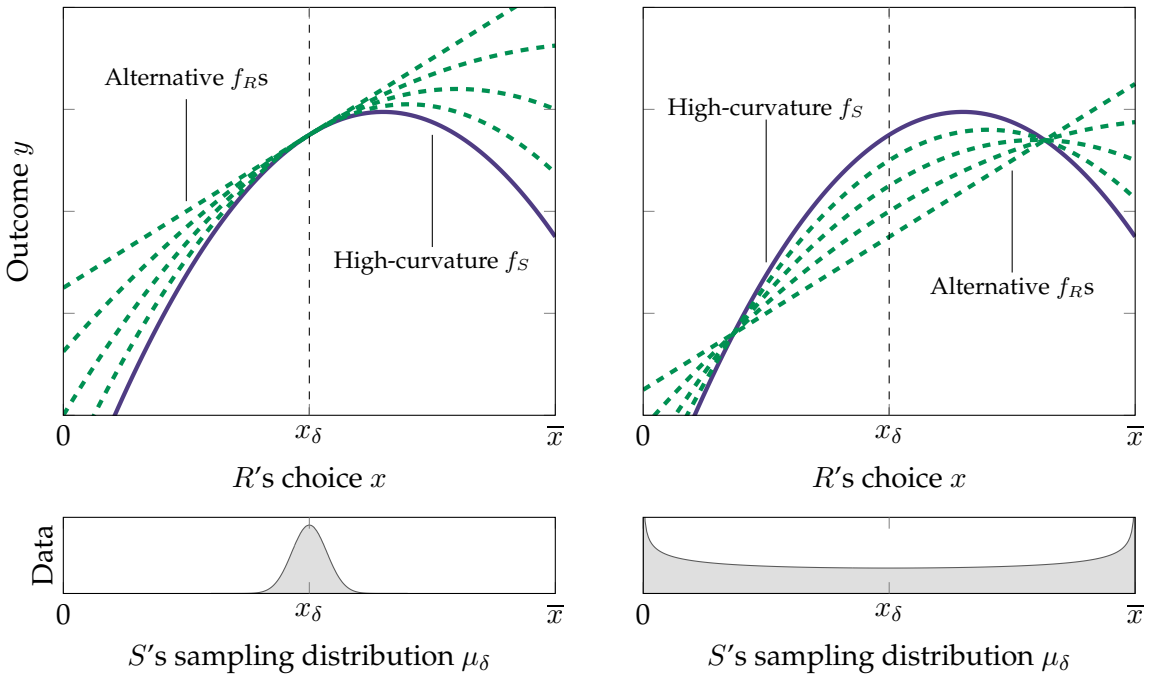
(i) S estimates a low-curvature "linear" model



(a) Underlying data local near x_δ

(b) Underlying data dispersed across X

(ii) S estimates a high-curvature "worst-case" model



(c) Underlying data local near x_δ

(d) Underlying data dispersed across X

The logic is flipped for a high-curvature sender. Panels (c) and (d) look at what happens for a sender who provides a worst-case model. On the left in panel (c), when the sender's underlying data are sampled in a tiny neighborhood of x_δ, all the alternative plausible models R considers have shallower curvature (i.e. lower γ) than the sender's model. However, they all lie *above* the sender's original model, and thus deliver higher expected utility, conditional on f_S. By contrast, for distributions that are more spread out, as on the right in panel (d), these alternative models deliver lower expected utility, conditional on the same model f_S. Thus, a high-curvature sender has an incentive to collect data *locally* in the small neighborhood of a choice.

Finally, Theorem 1 additionally characterizes the *mean* x^*_δ of the distribution μ^*_δ that a type δ sender samples from in equilibrium. The mean of a low-curvature sender's data lies between 0 and the receiver's expected choice, roughly in the center of the choice space X; what matters, however, is that the data around this mean are spread out as widely as possible. The mean of a high-curvature sender's data is at an edge of X. To see why, it helps to write out the receiver's expected utility, conditional on seeing f_S. We define $\Delta(f_S, x_\delta)$ as:

$$\Delta(f_S, x_\delta) = \mathbb{E}^R_\gamma[\gamma | f_S, x_\delta, \theta = \delta] - \delta.$$

Fixing f_S, Δ represents the *difference* between R's expectation of the quadratic parameter γ, conditional on the model being simple, and the sender's provided parameter δ. For example, Δ is always positive for the linear regression sender $\delta = 0$ and always negative for the worst-case scenario sender $\delta = \overline{\gamma}$. In turn, R's maximal expected utility can be expressed as:

$$\max_x f_S(x) + \Pr(\theta = \delta | f_S)\Delta(f_S, x_\delta)\big(\sigma^2_\delta - (x - x_\delta)^2\big),$$

which she attains at $x = x^*(f_S)$. The sender's choice of x_δ is then shaped by two forces.

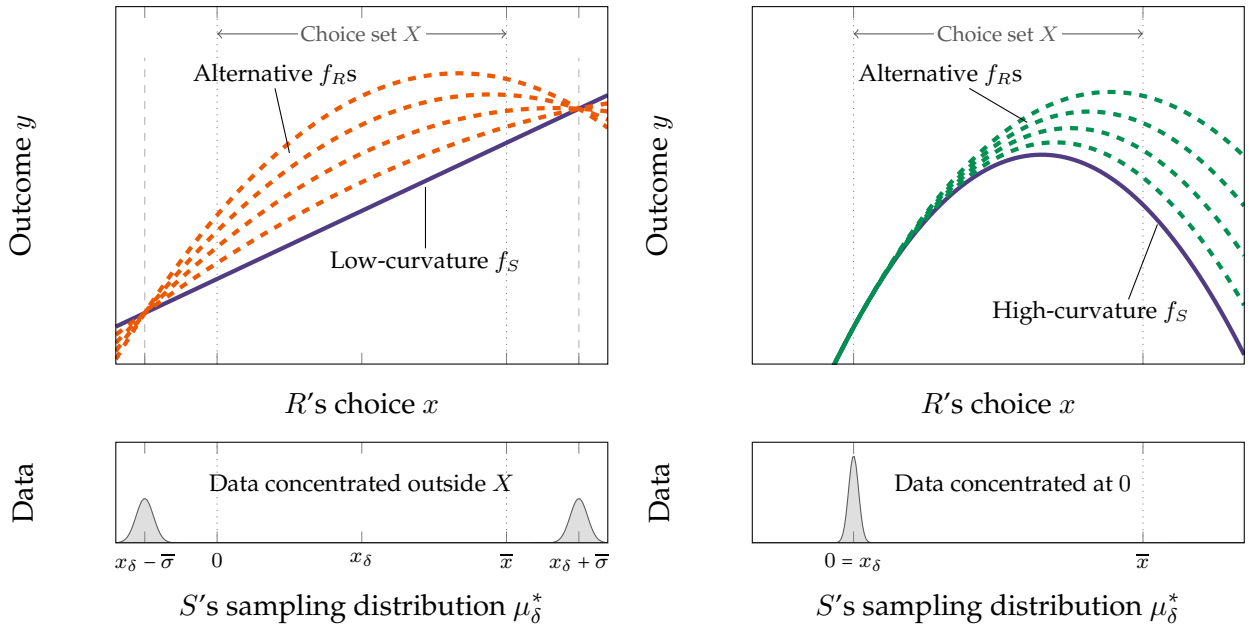
The first is a *learning* force: x_δ affects $\Delta(f_S, x_\delta)$ itself. When the sender samples at x_δ, the receiver learns $f'_R(x_\delta) = \alpha_1 - 2\gamma x_\delta$. When $x_\delta = 0$, she receives a perfect signal about α_1 and no signal about γ; as x_δ grows, she learns more about γ. Because the sender believes his own curvature δ is correct, he expects this information to pull the receiver's posterior curvature towards δ. The assumption of a uniform distribution on α_1 and a symmetric distribution on γ allows a tractable characterization of this effect: the proof shows that $\mathbb{E}^\delta_{f_S}\big[\mathbb{E}^R_\gamma[\gamma | f_S, x_\delta, \theta = \delta]\big]$ is decreasing in x_δ for $\delta < \mathbb{E}^R_\gamma[\gamma]$ and increasing in x_δ for $\delta > \mathbb{E}^R_\gamma[\gamma]$. A low-curvature sender wants the receiver's posterior curvature to be high, while a high-curvature sender wants it to be low, so for both types this force pushes the sender towards sampling at 0, where his data are uninformative about curvature.

The second is a *location* force. When $\Delta(f_S, x_\delta) \geq 0$, the receiver's expected utility above is concave in x_δ. This means that, fixing $\Delta(f_S, x_\delta) \geq 0$, the sender would like to choose $x_\delta = x^*(f_S)$. S does not know ex-ante where this maximum will be, so this force pushes x_δ towards where he *expects* the receiver's choice to be, $\mathbb{E}^\delta_{f_S}[x^*(f_S)]$. Combined with the learning force, a low-curvature sender's choice of x_δ is thus somewhere between 0 and $\mathbb{E}^\delta_{f_S}[x^*(f_S)]$. However, when $\Delta(f_S, x_\delta) < 0$, the receiver's expected utility is convex in x_δ, so the sender would prefer to sample at a corner.

This pushes his preferred sampling mean towards $x_\delta = 0$ or $x_\delta = \overline{x}$, and the learning force tilts this choice towards 0.

Figure 5 thus visualizes the sender's equilibrium choices. The left shows a low-curvature (linear regression) sender collecting dispersed data, with mean x_δ roughly in the center of the choice space. The right shows a high-curvature (worst-case) sender collecting local data near 0.

Figure 5: All plausible models $\phi(f_S; \mu_a, \mu_\delta)$ receiver considers and equilibrium data collection μ_δ^* of low vs. high curvature senders



(a) Low-curvature S samples dispersed data

(b) High-curvature S samples local data

In this sense, simple senders sample data that are *poorly suited* to the receiver's eventual choice. Even though the low-curvature sender's data are centered roughly between 0 and $\overline{x}$, the actual choices he samples about are highly dispersed. In fact, those choices are so dispersed that they may not even lie inside the receiver's choice set. This is shown in the left panel (a).

The high-curvature sender's data may also be poorly suited to the receiver's eventual choice. He samples in the smallest possible neighborhood of 0 or $\overline{x}$. In this sense, he accurately describes the expected value of y given x at just one of these extreme points, but nowhere else. In the example of the policymaker and consultant, this means that a consultant would mostly collect data where a carbon tax is close to 0, or where a carbon tax is excessively high, neither of which is likely to represent the policymaker's optimum.

The coupling of *model simplicity* with *strategic sampling* thus creates incentives to collect data that are poorly suited to assessing the effects of choices on outcomes. Intuitively, by sampling information on the edges of the receiver's choice space, senders create leeway for receivers to optimistically "fill in the blanks" about the values of outcomes at intermediate choices. Indeed, Section 4.2 formally shows that, if the receiver values accuracy, the *second-best* degree of sampling

would be an interior amount between $\underline{\sigma}$ and $\overline{\sigma}$.

The linear regression model (type $\delta = 0$ sender) is also of note. The result on strategic sampling says that this sender always estimates models using data sampled bimodally from the far reaches of the choice space or beyond. However, regressions are most effective when thought of as *local linear approximations*. Thus, the result on sampling contrasts a sender's persuasive desire to fit a regression to "dispersed data" with the common perception that regressions are most effective when utilized as "local" linear approximations.

Accurate Sender's Gain Because the accurate sender's models are not misspecified, his data sampling strategy does not affect the alternative models R considers. However, the accurate sender *gains* from the existence of simple senders: he can make the receiver more optimistic than the truth. Suppose the accurate sender provides a model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Because the receiver believes that, with some probability, this model came from a simple type δ sender, she also entertains alternative models. The accurate sender, however, knows that f_S is the true model. His gain, the gap between the receiver's perceived and true maximal expected utility, is given by:

$$G(f_S) = \max_{x \in X} f_S(x) + \Pr(\theta = \delta | f_S) \Delta(f_S, x_\delta) \left(\sigma_\delta^2 - (x - x_\delta)^2\right) - \max_{x \in X} f_S(x).$$

The following proposition shows that the accurate sender strictly gains from the existence of simple senders.

Proposition 2. *Suppose the accurate sender provides a model f_S with curvature δ, that $\Pr(\theta = \delta | f_S) > 0$, that the type δ simple sender samples as in Theorem 1, and that $\Delta(f_S, x_\delta)$ has the same sign as $\mathbb{E}_\gamma^R[\gamma] - \delta$.*

1. *If $\delta < \mathbb{E}_\gamma^R[\gamma]$ and $\overline{\sigma} > \overline{x}$, then $G(f_S) > 0$.*
2. *If $\delta > \mathbb{E}_\gamma^R[\gamma]$, then $G(f_S) > 0$ whenever $\arg\max_{x \in X} f_S(x)$ lies more than $\underline{\sigma}$ away from x_δ^*.*

Intuitively, the receiver can always choose the value of x that maximizes $f_S(x)$, so the accurate sender gains whenever the alternative models lie above f_S at that point. If the simple type δ is low-curvature, the alternative models are steeper, and because he samples dispersed data at $x_\delta^* \pm \overline{\sigma}$, they lie above f_S everywhere between these two points, which includes the entire choice set. If the simple type δ is high-curvature, the alternative models are flatter, and because he samples local data at an endpoint of X, they lie above f_S everywhere except within $\underline{\sigma}$ of that endpoint.

In either case, the receiver is more optimistic than the truth. She weighs not only the true model (the sender's original model) but also models that promise higher expected utility at her optimal choice. It is precisely simple senders' restricted treatment of *curvature*, coupled with their *strategic sampling*, that generates this gain, which is larger the more likely the accurate sender is to be perceived as a simple sender. I probe this result further in Section 5, when I characterize which composition of simple senders advantages the accurate sender the most.

3.3 Multidimensional Choices

While the baseline model considered a single choice variable x, I also consider what happens when choices are multidimensional. For example, imagine that the policymaker can shape consumer welfare not just through a carbon tax but also through clean energy subsidies. I briefly show that many of the findings of the baseline model generalize. However, I also show that when the sender samples data about multiple variables, he always has an incentive to introduce nearly perfect *collinearity* between them. In particular, if there are two choice variables — x_1 and x_2 — and the sender can choose how *correlated* the two variables are when sampling the data, he will make them either as correlated as possible or as *anticorrelated* as possible.

Multidimensional Models Suppose now that the receiver's choices are given by $x = (x_1, \ldots, x_n) \in X^n \subseteq \mathbb{R}^n$. The models f_R the receiver entertains are now as follows. For each $f_R \in F_R$, we have

$$f_R(x) = \alpha_0 + \alpha^T x - x^T \gamma x,$$

where $\alpha_0 \sim P_0^R$ and $\alpha \sim P_1^R$ are continuous and uniform[9], and $\gamma \sim P_\gamma^R$ on a set Γ. Γ is a compact, convex set of positive-semidefinite matrices such that $0 \in \Gamma$ and such that there exists $\gamma \in \Gamma$ with $\gamma_{ii} > 0$ for all i.

F_R now *generalizes* the idea of a quadratic data-generating process to a multivariate setting using a set of concave *quadratic forms*. This set of forms is given by a set of positive-semidefinite matrices Γ. Accordingly, I assume the type space Θ of senders is given by $\{a\} \cup \Gamma$.

For simple senders, I assume that a type δ sender provides models of the form:

$$f_S(x) = \beta_0 + \beta^T x - x^T \delta x,$$

where $\delta \in \Gamma$ is the sender's type.

Finally, I describe how the sender samples data in multiple variables. Formally, let $\epsilon \in \mathbb{R}^n$ be a *spherically symmetric* random vector that admits a density with $\mathbb{E}[\epsilon] = 0$ and $\mathrm{Cov}(\epsilon) = I_n$. Let Σ_δ be a symmetric positive-definite matrix.

Then, the sender's choice of distribution is characterized by $(x_\delta, \Sigma_\delta)$. In this case, his data x are sampled with respect to the distribution $\mu_\delta(x_\delta, \Sigma_\delta)$, such that:

$$\mu_\delta(x_\delta, \Sigma_\delta) = x_\delta + \Sigma_\delta^{\frac{1}{2}} \epsilon.$$

I place the following assumptions on data sampling. As before, I assume $x_\delta \in X^n$. Additionally,

- $\sigma_i := \sqrt{\Sigma_{\delta,ii}} \in [\underline{\sigma}, \overline{\sigma}]$ and
- $\rho_{ij} := \frac{\Sigma_{\delta,ij}}{\sigma_i \sigma_j} \in [-\overline{\rho}, \overline{\rho}]$ for $\overline{\rho} \in (0,1)$, with $\overline{\rho} < \frac{1}{n-1}$ when $n \geq 3$, so that every such Σ_δ is positive definite.

[9] Each component of α is distributed uniformly on a bounded interval, as in the univariate case.

This assumption is a multidimensional generalization of the idea that the "variance" of a variable x_i must be between $\underline{\sigma}^2$ and $\overline{\sigma}^2$. However, the key addition with n choice variables is that the sender can choose the degree of *correlation* between different variables when he samples data.[10]

Learning with Multidimensional Choices The following result generalizes receiver learning in multiple variables. As before, ϕ denotes the set of additional models R considers, given a model f_S and sampling strategies μ_δ and μ_a.

Proposition 3. *Suppose $f_S = \beta_0 + \beta^T x - x^T \delta x$ best-fits the data and $\mu_\delta = \mu_\delta(x_\delta, \Sigma_\delta)$. Then, for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$, we have:*

$$f_R(x) = f_S(x) + \operatorname{tr}\big((\gamma - \delta)\Sigma_\delta\big) - (x - x_\delta)^T(\gamma - \delta)(x - x_\delta).$$

The proposition reads almost identically to Proposition 1, and the intuition is similar as well. However, in place of the original scalar "variance" term, we instead have $\operatorname{tr}\big((\gamma - \delta)\Sigma_\delta\big)$. To see the impact that this has on the set of models the receiver will entertain, consider the following example with $x \in \mathbb{R}^2$.

$$\gamma = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \qquad \delta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Sigma_\delta = \overline{\sigma}^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

Here, the receiver's model allows for a "cross effect" between x_1 and x_2, while the sender's model neglects it. In turn, we thus have:

$$\operatorname{tr}\big((\gamma - \delta)\Sigma_\delta\big) = \overline{\sigma}^2 \operatorname{tr}\left(\begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right) = \overline{\sigma}^2 \rho.$$

This trace term represents, as before, the *gap* between the sender's original model and the receiver's model at x_δ. The sender can *increase* this gap if he makes ρ as large as possible.

More formally, the sender in this case provides models that suffer from *neglecting interactions*. In the running example, the consultant whom the policymaker hires may effectively evaluate the effects of both a subsidy and tax *separately*, but not take into account the *interaction* of their effects.

The receiver is aware of the sender's misspecification, and she entertains additional models. However, the sender can give the receiver a more optimistic view of these models if he artificially creates a high degree of correlation between x_1 and x_2 when sampling data. In the example, if the consultant cannot figure out how to model these interactive effects, he compensates by skewing his data collection towards settings where both a carbon tax and a clean energy subsidy are active at the same time, rather than towards settings that cleanly isolate the effect of one or the other.

[10] I bound the correlation away from -1 and 1 to ensure that learning does not collapse at $\rho = 1$.

Results with Multidimensional Choices As before, given $f_S = \beta_0 + \beta^T x - x^T \delta x$ and fixing x_δ, we denote $\Delta(f_S, x_\delta)$ as:

$$\Delta(f_S, x_\delta) = \mathbb{E}^R_\gamma[\gamma | f_S, x_\delta, \theta = \delta] - \delta.$$

The intuitive mapping from the one-dimensional problem to the multidimensional one can be seen by looking at the diagonal entries of $\Delta(f_S, x_\delta)$. If the ith diagonal entry is *positive*, this means that S tends to *neglect tradeoffs* with respect to variable x_i. If it is negative, he tends to *exaggerate* tradeoffs. Thus, fixing x_δ, we say that a sender estimates a **low-curvature model in** x_i if $\mathbb{E}^\delta_{f_S}[\Delta_{ii}(f_S, x_\delta)] > 0$ and a **high-curvature model in** x_i if $\mathbb{E}^\delta_{f_S}[\Delta_{ii}(f_S, x_\delta)] < 0$.

However, what happens on the *non-diagonal* entries of Δ is also of interest. Suppose $\Delta_{ij}(f_S, x_\delta) + \Delta_{ji}(f_S, x_\delta) > 0$. This means that the receiver expects the coefficients on the interactions of x_i and x_j to be *larger* than what the sender provides. If $\Delta_{ij}(f_S, x_\delta) + \Delta_{ji}(f_S, x_\delta) < 0$, the sender *exaggerates* interactions between x_i and x_j. With these new definitions in hand, I can state the key multidimensional result.

Theorem 2. *Simple senders always maximize the collinearity of underlying datasets, i.e. make variables as highly correlated or highly anticorrelated as possible. In particular:*

- *if* $\mathbb{E}^\delta_{f_S}[\Delta_{ij}(f_S, x^*_\delta) + \Delta_{ji}(f_S, x^*_\delta)] > 0$, *then* $\rho^*_{ij} = \overline{\rho}$.
- *If* $\mathbb{E}^\delta_{f_S}[\Delta_{ij}(f_S, x^*_\delta) + \Delta_{ji}(f_S, x^*_\delta)] < 0$, *then* $\rho^*_{ij} = -\overline{\rho}$.

If γ *is diagonal for all* $\gamma \in \Gamma$, *the sender additionally sets* $\sigma^*_i = \overline{\sigma}$ *or* $\underline{\sigma}$. *In particular:*

- *if* $\mathbb{E}^\delta_{f_S}[\Delta_{ii}(f_S, 0)] > 0$, *then* $\sigma^*_i = \overline{\sigma}$.
- *If* $\mathbb{E}^\delta_{f_S}[\Delta_{ii}(f_S, 0)] < 0$, *then* $\sigma^*_i = \underline{\sigma}$.

First, Theorem 2 replicates the intuitions of the univariate result: when there are no interaction effects (i.e. γ is always diagonal), the simple sender collects data from the widest variance distribution or smallest variance distribution in variable i, as opposed to pursuing an intermediate amount of sampling, exactly as before. With interaction effects, a variable in which the sender estimates a high-curvature model may instead be sampled with an intermediate variance, since spreading it out also increases the gain from its correlation with other variables.

However, Theorem 2 also shows that the sender's dataset *maximizes collinearity* between variables. If he *neglects* an interaction effect between variables x_i and x_j (e.g. by always assuming it is $= 0$), he samples data so that they are as *correlated as possible*. If he *exaggerates* interactions, he samples data so that they are as *anticorrelated as possible.*

Concretely, suppose that the policymaker believes that the interaction of a carbon tax and clean energy subsidy may have particularly deleterious welfare effects at very high values, and that this is totally neglected by the consultant. The theorem states that this sender will sample data to favor settings where both a tax and subsidy are active simultaneously, rather than separately.

Recall that any additional models the receiver entertains all have the same sample average value of y. Because the tax and subsidy are highly correlated in those data, this sample average is depressed in large part by the (negative) interaction that the sender's model neglects. To rationalize the same sample average, the policymaker's additional models must hence assign larger separate effects to the tax and subsidy than the sender's model. But since the tax and subsidy are also choice variables, the receiver need not correlate them in her eventual choice, increasing her maximal expected utility. While this paper does not explicitly deal with finite samples, in practice, the introduction of this collinearity can create large standard errors and biased parameter estimates.

In this sense, when choices are multivariate, simple senders have incentives to sample data in an *unbalanced way*: rather than ensuring that the marginal effects of different variables are estimated independently of each other, there may be an arbitrary degree of collinearity between variables in the datasets that senders produce. Coupled with the previous results where senders collect data about irrelevant choices — either dispersed, extremal data or local data about a single extreme choice — high collinearity between variables only worsens the receiver's ability to learn.

4 Additional Results: Information and Forecasting

This section provides additional results from the baseline, univariate model. First, I characterize the *value of information* for the receiver for contracting a sender's services. I show that although this expression does not depend on whether a sender samples local or dispersed data, it potentially *does* depend on *where* the sender sampled data.

Second, I augment the receiver's baseline preferences with *forecast error*. As a result, the model captures settings where the receiver cares not just about maximizing y but also about predicting y accurately. I show that the receiver's choices are more accurate when she receives information from a sender who samples local data than from a sender who samples dispersed data. Consequently, the receiver may be better off hiring a high-curvature sender than a low-curvature sender, although I also show that neither is first best.

The Appendix deals with additional extensions. Appendix B.1 studies the model where simple senders can model an *interval* of γ values, rather than a singleton γ value. Appendix B.3 studies the model with non-quadratic utility and shows that the central intuitions on sender sampling generalize.

4.1 Receiver's Value of Information

While it is clear that model simplicity inhibits the receiver's ability to learn the true model of the world and, thus, to make the right choice, there is also a question of *how much* the receiver's ability to learn is distorted, conditional on receiving a simple model. Characterizing the value of information for the receiver from hiring a sender allows me to determine how decision quality

varies with sender behavior. [11]

The value of information from hiring a simple sender of type δ is given by:

$$V(\delta) := \mathbb{E}^R_{f_S \in F_\delta}\big[\max_x \mathbb{E}^R_{f_R|f_S}[f_R(x)|f_S, x^*_\delta]\big] - \max_x \mathbb{E}^R_{f_R}[f_R(x)] \tag{3}$$

In the absence of the sender, R would solely rely on her prior to maximize the expected value of y. This allows me to characterize the *value of information* for the receiver as follows.

Proposition 4. *Fix δ. The value of information V for the receiver from hiring a sender of type δ is given by:*

$$V(\delta) = \mathbb{E}_{f_S \in F_\delta}\left[\mathbb{E}^R_\gamma[\gamma|f_S, \theta = \delta, x^*_\delta]\left(x^*(f_S) - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right]$$

and does not depend on σ^*_δ. Moreover, for $x^*(f_S)$ interior, we have

$$V(\delta) = \mathbb{E}^R_{f_R}\left[\mathbb{E}^R_\gamma[\gamma|f'_R(x^*_\delta)]\left(x^*_\delta + \frac{f'_R(x^*_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x^*_\delta)]} - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right].$$

The key result is that V does not depend on σ^*_δ, i.e. the spread of the simple sender's data. The reason is that the set of alternative models that the receiver entertains are simply vertical shifts of each other. This means that the receiver is effectively receiving a noisy signal about the value of the constant term. Furthermore, they do not affect the receiver's eventual choice $x^*(f_S)$.

Because the receiver is an expected utility maximizer, these noisy signals about the constant term have no bearing on her expected utility. This notion of informational value ignores a key tension, which is that the *forecast accuracy* of the sender's will depend on the vertical spread between the DGPs she places positive probability on in equilibrium. I study this in the next subsection.

Nevertheless, the receiver's value of learning *does* depend on x^*_δ, which is the *mean* of the simple sender's sampling distribution. The reason is that, by choosing a value of x_δ, the sender reveals $f'_R(x_\delta) = \alpha_1 - 2\gamma x_\delta$. When $x_\delta = 0$, the sender is effectively revealing α_1 and no information about γ. As x_δ grows higher, the sender reveals more information about γ and less about α_1, which in turn affects the receiver's expected decision quality. We thus parametrize the value of information as:

$$V(\delta, x_\delta) := \mathbb{E}^R_{f_S \in F_\delta}\big[\max_x \mathbb{E}^R_{f_R|f_S}[f_R(x)|f_S, \theta = \delta, x_\delta]\big].$$

The following proposition characterizes how $V(\delta, x_\delta)$ varies with x_δ.

Proposition 5. *Suppose the receiver's choice $x^*(f_S)$ is interior. Then $x^R_\delta = \arg\max_{x_\delta \in [0,\overline{x}]} V(\delta, x_\delta)$ satisfies $x_\delta \geq \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}$. Moreover, if S is low-curvature and $\overline{\sigma}$ is sufficiently large or if S is high-curvature and $x^*_\delta = 0$, $x^R_\delta \geq x^*_\delta$.*

[11] It also represents the receiver's *willingness to pay* to hire the sender, prior to data sampling and model provision.

In particular, the proposition shows that, conditional on misspecification, the *location* around which the simple sender samples is not first-best for the receiver, and may in fact be slightly to the left. The formal intuition can be gleaned from the proof of Theorem 1. Recall that the gain generated by the alternative models the receiver considers is *decreasing* is approximately proportional to $|\mathbb{E}^R_\gamma[\gamma|f_S, \theta = \delta, x_\delta] - \delta|$. This gain is decreasing in x_δ. This gives simple senders a preference for x_δ values that tend to be systematically lower than what the receiver would choose if she were to estimate a simple model.

4.2 Forecast Accuracy

So far, I have assumed only that the receiver would like to maximize $\mathbb{E}[y|x]$. However, in many settings, the receiver may also directly (or indirectly) value *forecast accuracy*. In our running example, a policymaker may not just value not just picking the optimal carbon tax , but also being able to *accurately forecast* the effects of her policy choice. Being able to predict the outcome (or set of outcomes) her policy choice could generate may allow her to better defend her choice to the politicians and the press, as well as to be more confident in her decision. That is, the receiver values *knowing the true data-generating-process*.

To illustrate the impact of model simplicity and sampling on forecast-minded receivers, I consider a receiver whose expected utility is given by

$$(1 - \psi)\mathbb{E}^R[y|x] - \psi \mathrm{Var}^R(y|x) \tag{4}$$

for $\psi \in (0, 1)$. The first term, as before, corresponds to a sender's desire to maximize the expected value of y. The second term represents the ex-ante *accuracy* with which she can predict y, after making her choice but prior to the resolution of any uncertainty. If $\psi = 0$, she does not value her ability to forecast at all, and only cares about her expectation of y. As $\psi \to 1$, she places a greater and greater emphasis on forecasting.

Exogenous Forecast Error Before studying what happens when the receiver chooses x to maximize (4), I first characterize how the variance of y behaves with respect to σ_δ when the receiver simply maximizes the expected value of y. The following proposition shows that, fixing x_δ, the variance of outcomes is minimized at an *intermediate* value of σ_δ. That is, the way in which the simple type δ samples data to fit his model — either using heavily dispersed or highly localized datasets — never minimizes forecast error for the receiver.

Proposition 6. *Fix a sender type δ and x_δ. There exists σ_δ such that $W(\sigma_\delta)$ is strictly decreasing in σ_δ for $\sigma_\delta < \sigma^R_\delta$ and strictly increasing for $\sigma_\delta > \sigma^R_\delta$ at $x^*(f_S)$.*

The intuition for the result is as follows. Writing x^* as the value of x that maximizes the expected value of y, recall the previous formulation of the receiver's conditional expected utility:

$$f_S(x^*) + \Pr(\theta = \delta|f_S)\Delta(f_s, x_\delta)\left(\sigma^2 - (x^*(f_S) - x_\delta)^2\right)$$

The receiver's forecast error is driven by the latter set of terms multiplied by $\Delta(f_s, x_\delta)$. In particular, the variance of outcomes can be expressed as the difference between the simple sender's spread of sampling σ_δ^2 and the squared distance between $x^*(f_S)$ and x_δ (the optimal choice and the point around which data are sampled). When σ_δ^2 is small, the dispersion is driven by $(x^*(f_S) - x_\delta)^2$, which is always nonzero in expectation. This means the variance can be reduced by increasing σ_δ^2. Intuitively, because there is uncertainty about what the ex-ante optimal choice x^* will be, an intermediate level of σ_δ^2 allows the receiver to "hedge" and gather information in a slightly larger region where she expects the choice to be, rather than precisely at a single point. When σ^2 is very large, however, there is a great deal of vertical dispersion in the potential values of y that may materialize at x^*, and that dispersion can be reduced by decreasing σ_δ^2.

Thus, the *second-best* degree of sampling σ_δ^R that the receiver would choose to minimize outcome variance is necessarily intermediate: that is, neither a sender who samples with dispersed data $\sigma_\delta^* = \overline{\sigma}$ nor a sender who samples locally with $\sigma_\delta^* = \underline{\sigma}$ minimizes the variance of the receiver's outcomes. However, as $\overline{\sigma}$ grows arbitrarily large, the receiver *will* be better off under "local" sampling than "dispersed" sampling.

Endogenous Forecast Error I now study the setting where R jointly trades off her expected gain in y with forecast error. Conditional on f_S and fixing x_δ, her expected value is

$$V^F(f_S|\sigma_\delta) = \max_x (1-\psi)\,\mathbb{E}^R_{f_R|f_S}\big[f_R(x) \,\big|\, f_S, \theta = \delta, x_\delta, \sigma_\delta\big] - \psi\,\mathrm{Var}^R_{f_R|f_S}\big(f_R(x) \,\big|\, f_S, \theta = \delta, x_\delta, \sigma_\delta\big).$$

The sender's preferences are the same, which is to maximize the receiver's perceived expected value of y given her optimal choice. The key difference is that the receiver's optimal choice itself is affected by how much she values forecast error. The following proposition shows how the sender's choice of σ_δ is shaped by these considerations.

Proposition 7. *Suppose R chooses $x^*(f_S)$ to maximize (4), and let σ_δ^R be the choice of σ_δ that minimizes the expected variance of outcomes. Then:*

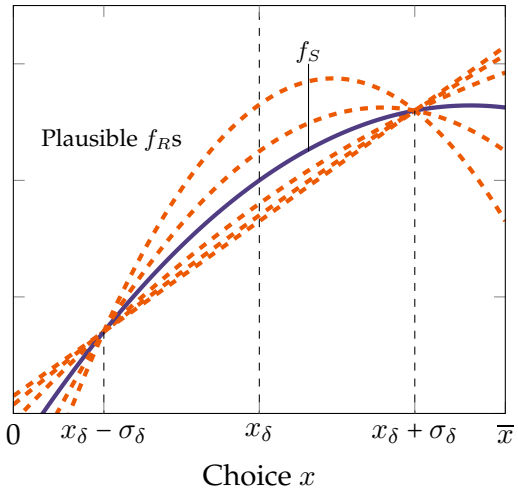
1. *A low-curvature sender chooses $\sigma_\delta^* > \sigma_\delta^R$ and a high-curvature sender chooses $\sigma_\delta^* < \sigma_\delta^R$.*
2. *As $\psi \to 1$, if $x_\delta \pm \sigma_\delta \in [0, \overline{x}]$, then $x^*(f_S) \to x_\delta - \sigma_\delta$ or $x_\delta + \sigma_\delta$.*

The first part of the result generalizes the intuition of Theorem 1. Namely, low-curvature senders sample data with a *larger spread* than what the receiver would choose, and high-curvature senders sample with *less spread*. The intuition is that the sender accounts for the fact that the receiver will shift her choices to take into account forecast error. For very high values of σ^2, a low-curvature sender trades off between increasing the receiver's expectation of y through more dispersed data sampling and increasing the variance of outcomes she faces. The opposite logic holds for high-curvature senders.

The second part of the result shows that if the receiver increasingly only values forecast error, her choice approaches $x_\delta \pm \sigma_\delta$. This is illustrated in Figure 6. Thus, R trades off between a choice

that maximizes her expected utility (at an interior point) and minimizing the variance of outcomes, which happens at $x_\delta \pm \sigma_\delta$. When σ_δ is very large, R is forced to choose arbitrarily extreme points $x_\delta \pm \sigma_\delta$ to minimize variance. When σ is very small, R chooses points close to x_δ.

Figure 6: Intersection of receiver's models with sender's models



5 The Supply of Simple Senders

This section studies the *supply* of simple senders that the receiver will receive models from in equilibrium. So far, I have taken the distribution of simple sender types as exogenous. However, Proposition 2 shows that accurate senders gain from being misperceived as simple senders, in that their existence allows the receiver to entertain more optimistic models than the truth. Moreover, this gain is different depending on the type δ of a simple sender. Thus, if an accurate sender could endogenously choose the *supply* of simple senders who operated alongisde him, what simple senders would he choose? Formally, I ask what *distribution* of simple senders would maximize the accurate sender's expected gain.

I thus augment the model with an additional stage: prior to being hired by the receiver, the accurate sender a first chooses a *distribution* T of simple sender types. After he chooses that distribution, nature then either selects the accurate sender or a simple sender and, in the latter case, draws the simple sender from T. The interpretation is that the accurate sender is able to *hire* additional senders who may provide simple models. Conditional on contracting the services of his organization, a receiver may receive information from either the accurate sender or one of these simple senders. For example, the accurate sender could be thought of as:

- a consultant or investment advisor who can employ other consultants/advisors, whose recommendations may take the form of simple rules of thumb;
- a politician who can recruit co-partisans to propagate simple talking points among voters;
- an AI company that can deploy simpler or more complex LLMs through its chatbot.

The key is that he is able to *shape* the receiver's beliefs about whether she is facing a simple or accurate model.

I show that the distribution of simple senders that the accurate sender would choose is characterized by *polarization* in how senders consider tradeoffs. Formally, for each $\delta \in [0, \overline{\gamma}]$, the conditional probability of facing a type δ sender is given by a density $t(\delta)$.[12] This density $t(\delta)$ takes the accurate sender's prior over δ and pushes mass away from intermediate values of δ towards the extremes of $[0, \overline{\gamma}]$. I.e., simple senders either estimate models with low δ values (close to 0) or high δ values (close to $\overline{\gamma}$), and never estimate models with intermediate δ values. Thus, the set of simple sender types is fully composed of senders whose models have (arbitrarily) low or high curvature. As the probability of seeing the accurate sender's model approaches 1, the supply of simple senders converges to a one- or two-point distribution concentrated on either $\delta = 0$, $\delta = \overline{\gamma}$, or both points. These respectively correspond to senders who only provide "linear regression" models or "worst-case scenario" models, as I studied in Section 3.

5.1 Motivation

To motivate the role of the accurate sender further, I return to the problem of the consultant and the policymaker. Imagine an "accurate" consultant employed by a consulting organization, who can think about tradeoffs from carbon taxation in *exactly* the way the policymaker would like. However, he can also choose to hire additional colleagues, who might think about tradeoffs in a simpler way than both himself and the policymaker. He can choose how many of these colleagues to hire, as well as how simply they might think about problems.

The consultant thus hires other consultants. The policymaker, in turn, is aware of the composition of the consultant's organization. When she contracts the organization's services, she is randomly assigned either the "accurate" consultant or one of the "simple" consultants. Relatedly, she may contract a team composed of both the accurate consultant as well as other simple consultants, but not know how much oversight the accurate consultant exercised. Based on the model she is ultimately provided, she in turn attempts to *infer* the sender's type. Thus, the problem boils down to: will the accurate consultant hire "simple" consultants to join him, what kind of simplicity will they adopt, and how many of those consultants will he hire?

The key interpretation of the design problem is thus that an accurate sender can endogenously shape the set of senders that the receiver considers, such as by actively hiring additional senders who model tradeoffs differently. Beyond hiring, I also highlight additional interpretations based on the motivating examples.

1. **Portfolio management.** Suppose the sender is a financial advisor who can estimate a highly accurate model of asset returns. The advisor may help train other managers or deploy "robo-advisors" that use simple rules of thumb.

[12] Appendix B.1 shows that even if the accurate sender can consider "interval" types, he would never choose an interval type in equilibrium.

2. **Political narratives.** Suppose the sender is a politician who can accurately provide a "narrative model" of a policy's effects on voter welfare. However, he can also use online media accounts to propagate "simple narratives" of a policy's effects. He could also align himself with extremist politicians who the public knows provide simple narratives.

3. **AI recommendations.** Suppose the sender is an AI platform that uses a sophisticated LLM to collect data and estimate a robust model that provides users accurate recommendations. The platform can also deploy simpler LLMs that apply a "fixed" way of thinking about a user's problem, and can determine how often users are shown each model under the hood.

The key mechanism through which this design problem will operate is through manipulating the receiver's *beliefs* about whether the model she sees came from an accurate or simple sender.

5.2 Designing the Supply of Senders

Design Problem Setup Formally, consider the following problem for the accurate (type a) sender. Conditional on a model f_S, with probability p_a, R believes she has received a model from a. With probability $1 - p_a$, she believes the model was simple.[13] We write $P^a_{f_S}$ for his prior over models f_S. His objective is:

$$\max_T p_a \int_{f_S \in F_R} \max_x \mathbb{E}_{q|f_S,T}[f_R(x)] dP^a_{f_S}(f_S).$$

As before, the type a sender wants to make the receiver optimistic about the expected value of her choice. We note that the impact of T is through the receiver's *posterior*. I take p_a as fixed; this could represent the type a sender's capacity to hire others (higher p_a means he has less latitude to hire), the exogenous choice of a platform/organization, or a market belief.[14]

The order of events is as follows. a first distribution of types T over $[0, \overline{\gamma}]$. We denote his optimal choice by T^*. The receiver contracts the services of a sender, and receives a model from a with probability p_a or from the simple sender with probability $1 - p_a$. Conditional on the latter event, a type δ sender is then drawn according to the distribution T. Finally, the deployed sender strategically samples data as before, supplies the model of best fit, and provides it to the receiver, who makes her decision.

Distribution of Simple Senders The distribution of simple senders that maximizes the accurate sender's ability to make the receiver optimistic is as follows.

Theorem 3. *There exists a function $\pi(\delta)$ such that T^* is given by a continuous distribution t with*

$$t(\delta) = p^R_\gamma(\delta)\pi(\delta). \qquad (T^*)$$

[13] Appendix B.2 allows a to also choose types that are intervals of curvature values, and shows that he would never do so.

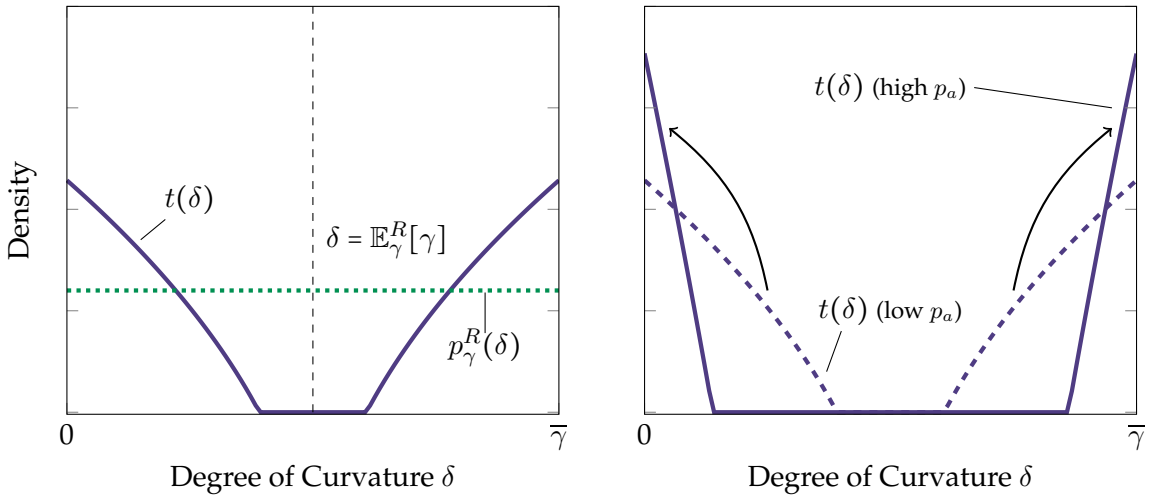
[14] p_a could also endogenously be chosen by the type a sender himself to trade off the probability with which his services will be contracted with receivers' perceptions of him.

In particular, there exist thresholds $0 \leq \underline{\delta}_0 < \mathbb{E}^R_\gamma[\gamma] < \overline{\delta}_0 \leq \overline{\gamma}$ *such that* $\pi(\delta)$ *is strictly decreasing below* $\underline{\delta}_0$*, strictly increasing above* $\overline{\delta}_0$*, and equal to* 0 *on* $[\underline{\delta}_0, \overline{\delta}_0]$*.*

Theorem 3 characterizes the *shape* of the distribution of δ values the type a sender utilizes. The solution is that the type a sender pushes mass to the *edges* of the interval $[0, \overline{\gamma}]$; formally, the distribution is a *warped* version of his and the receiver's prior belief about curvature values. In this sense, the type a sender *polarizes* simple senders towards models calibrated to (arbitrarily) low curvature (with δ close to 0) or high curvature (with δ close to $\overline{\gamma}$). In particular, $t(\mathbb{E}^R_\gamma[\gamma]) = 0$, i.e. $t(\delta) = 0$ when δ is close to the prior expectation of γ.

This idea is shown on the left panel of Figure 7 below. The dotted green line shows the prior distribution over δ, which is given by a continuous uniform distribution on $[0, \overline{\gamma}]$. The solid purple line shows that the types he creates are concentrated towards the edges at $\delta = 0$ (i.e. linear models) and $\delta = \overline{\gamma}$ (i.e. worst-case models).

Figure 7: Prior distribution $p^R_\gamma(\delta)$ of γ vs. optimal distribution of sender types $t(\delta)$



(a) Optimal distribution of sender types $t(\delta)$ vs. prior over curvature values $p^R_\gamma(\delta)$

(b) Increased polarization of $t(\delta)$ with increase in likelihood of seeing true model p_a

First, recall by Proposition 2 that the accurate sender benefits from being misperceived as a simple sender. In particular, this gain comes from the combination of the simple sender's strategic sampling with his assumption of a fixed curvature value when providing a model.

But while the accurate sender always gains, but his gain also depends on the true curvature value γ in the data. Thus, he must consider which γ values are ex-ante more plausible. His problem is thus to choose a distribution over types which, in this case, boils down to just being a distribution over the set of plausible simple types δ. In particular, he designates an amount of probability density to allocate to each $\delta \in [0, \overline{\gamma}]$, subject to the constraint that these densities must integrate to 1.

Suppose $\delta = \mathbb{E}^R_\gamma[\gamma]$, which is the *receiver*'s ex-ante expectation of γ. Looking back to Proposition 1, simple senders do not gain sampling from a higher or lower variance distribution in this case.

Relatedly, this means that the type a sender has no incentive to place mass on δ values near $\mathbb{E}^R_\gamma[\gamma]$. On the other hand, as δ gets farther and farther away from $\mathbb{E}^R_\gamma[\gamma]$, simple senders provide a larger and larger gain for the accurate sender. In fact, the gains to the sender are conditionally maximal if $\delta = 0$ or $\delta = \overline{\gamma}$. This is what drives the "polarization" of simple senders: the accurate sender, ex-ante, does not know what value of γ will materialize, but does know that the gain will be small for intermediate values of γ, and thus pushes more mass to the edge.

Variation with p_a We note that as p_a increases, the receiver is more likely to think that she saw the accurate sender model, and that as p_a decreases, this is less likely. In the example with the consultant and policymaker, the type a sender may be required to provide his services p_a of the time, but has latitude to choose the senders who the receiver contracts $1 - p_a$ of the time. As p_a increases, this latitude decreases, and the receiver is conditionally more and more sure that she received the true model. The following proposition shows that simple sender "polarization" is increasing in p_a, in the sense that as p_a increases, $t(\delta)$ places more and more mass on extreme types with very high or low δ values, and less (or none) on intermediate types.

Proposition 8. *The polarization of simple senders is increasing in p_a. That is, $\underline{\delta}_0$ is decreasing in p_a and $\overline{\delta}_0$ is increasing in p_a. As $p_a \to 1$, $t(\delta)$ converges to a distribution that either places all mass only on $\delta = 0$, $\delta = \overline{\gamma}$, or both. As $p_a \to 0$, $t(\delta)$ converges to a distribution that is strictly decreasing on $[0, \mathbb{E}^R_\gamma[\gamma])$, equal to 0 at $\mathbb{E}^R_\gamma[\gamma]$, and strictly increasing on $(\mathbb{E}^R_\gamma[\gamma], \overline{\gamma}]$.*

The proposition is visualized above in the right panel of Figure 7. The dashed purple line graphs $t(\delta)$ for a lower value of p_a. The solid line graphs $t(\delta)$ for a higher value of p_a, showcasing the "increased polarization" generated by an increase in p_a.

The intuition is as follows. The type a sender's gain from being mistaken for a simple sender is high at $\delta = 0$, decreases to 0 at $\delta = \mathbb{E}^R_\gamma[\gamma]$, and then rises again until $\delta = \overline{\gamma}$. That is, he conditionally gains the most when δ is very low or very high. When p_a is large, the ex-ante probability with which he will be seen as a simple sender is low, and he is thus best off concentrating simple senders close to $\delta = 0$ and $\delta = \overline{\gamma}$, where he expects the gain to be largest. However, when p_a is small, he is willing to distribute simple senders more evenly throughout the population although, in either case, he concentrates few senders near $\delta = \mathbb{E}^R_\gamma[\gamma]$.

I highlight two additional features of this result. First, polarization survives even as $p_a \to 0$ (so the receiver is less likely to think she sees the true model). Second, as $p_a \to 1$, the optimal distribution either converges to only deploying senders with type 0 or type $\overline{\gamma}$ and nothing in between. That is, the model returns precisely to the two-type example introduced in Section 2: in the limit, the only simple senders are those who *only provide* linear regression models or *only provide* "worst-case" models.

6 Conclusion

This paper theoretically studies how strategic data collection and model simplicity shape communication with models and data. A receiver R contracts a sender S to help her make a decision. The sender S first collects data and then estimates a model. However, the sender may be accurate or simple. If he is accurate, he correctly estimates the data-generating-process mapping choices to outcomes. If he is simple, he assumes the *curvature* of the relationship beteween choices and outcomes is calibrated to a fixed value. In the real world, these correspond to senders who calibrate their models to pre-defined parameters: economic consultants may calibrate models to elasticities or labor shares from the literature rather than organically estimating them in their setting; robo-advisors may apply fixed investing rules of thumb across clients; and politicians may weave narratives in which a policy's costs are fixed in advance.

S's goal is to make R as optimistic as possible, in the sense of maximizing her *perceived expected utility*, so that he is more likely to receive credit for the programs or initiatives that the receiver may adopt as a result of that optimism. The setting I study has a wide variety of applications to the economics of organizations, portfolio management, political economy, delegation in policymaking, and the use of AI-based decisionmaking.

I first characterize how R learns based on the model that S provides her. I show that R may sometimes believe that S provided an accurate model of her decisions, but that more often than not, R must entertain additional models. If S estimated his model on a small set of "local" choices, I show that the additional models R entertains are all tangent to S's original model at a point. If S estimated his model on a large range of choices, the additional models R entertains are all secants through S's original model.

I then show a close correspondence between what kinds of data S collects and his model *simplicity*. If S's model assumes low-curvature, i.e. *neglect* tradeoffs, the data underlying his model is dispersed bimodally across an arbitrarily wide set of choices, and may draw on information that is irrelevant to the receiver's eventual choice. Low-curvature models include linear regression models If S estimates high-curvature models that *exaggerate* tradeoffs, then he collects data locally about a single extreme receiver choice. These include "worst-case scenario" models that always assume the costs of choices are as high as possible.

In both cases, the receiver considers not only the sender's original model, but also models that deliver strictly higher expected utility. When the sender presents a low-curvature model, the expected utility delivered by these alternative models is strictly increasing in the spread of data collection. Thus, conditional on receiving a low-curvature model, the receiver's perceived expected utility is increasing in the *spread* of the underlying data. When the sender presents a high-curvature model, the expected utility delivered by these alternative models is strictly decreasing in the spread of data collection, which gives a high-curvature sender an incentive to sample data with minimal spread, i.e. locally in a small neighborhood of a point. In either case, this strategic data sampling increases the receiver's optimism about the potential impact of her optimal choice, in the sense of maximizing her perceived expected utility. When choices are multivariate, I show

that this mechanism generalizes, and the sender samples data in a way that maximizes collinearity between variables, and makes them as correlated (or anticorrelated) as possible.

Taken together, these points suggest that the quality of *data* may often be *low* when contracting external experts to present model-based recommendations. Moreover, this low quality of data emerges *precisely* as a result of the fact that senders estimate simple models. In both cases, the data embed information about choices the receiver is unlikely to end up making. The results on multivariate choices show further that the data senders collect may be particularly unbalanced and highly collinear. This collinearity makes it difficult to separate the levers that these different choices control.

I then show that an accurate sender gains from being misperceived as a simple sender, providing a channel for the existence of simple senders in equilibrium. That is, if a sender provides a true, accurate model of a receiver's choices, he always gains if the receiver thinks his model might have (artificially) low or high curvature. The intuition is that, in the latter cases, the receiver entertains not only a sender's original model but also additional models that lie above it. Thus, the sender who provides an accurate model can make the receiver have a more optimistic belief about her maximal expected utility than the true model. Accurate senders may therefore try to masquerade as simple ones: clients of a large consulting firm may not know whether a senior partner or junior analyst built their model, investors may be unable to tell whether an experienced manager or a robo-advisor suggested a rule, and users of an AI platform may not know whether a recommendation came from a small or a large model.

I use this insight on model simplicity to characterize the supply of senders that a receiver may receive information from in equilibrium. I study a design problem where an accurate sender can "hire" additional simple senders, and characterize what distribution of simple senders provides him the maximal *gain*, in the sense of making the receiver the most optimistic. I show that he would polarize simple senders into two distinct groups: those whose models have (arbitrarily) low curvature, close to "linear regression" models, and those whose models have (arbitrarily) high curvature, close to "worst-case" models. Thus, the market of senders is composed of three groups: some senders who always provide accurate models, some who provide very low-curvature models, and some who provide very high-curvature models. In the limit, a receiver who receives a simple model sees only linear regressions or worst-case models. These simple models tend to recommend either very extreme or very conservative actions: low-curvature models suggest making choices as extreme as possible, while high-curvature models may imply sticking to the status quo. While this form of polarization is common among political narratives, the paper suggests that it may occur more generally when an expert is hired to provide a model or *narrative* of the effects of different choices. A market for models may be saturated with models that recommend very extreme or conservative choices.

Simple models, strategic data sampling, and polarization all represent key frictions that may emerge when decisionmakers rely on expertise and modeling to shape their choices. Future work will investigate these frictions further in the settings of price-based markets and the polarization

of political platforms, as well as through empirical work that investigates delegation to agentic AI programs or the role of consultants' narratives in shaping organizational decisionmaking.

References

Aina, C. (2023). Tailored stories. *Unpublished manuscript 12*, 52.

Aina, C. and F. Schneider (2025). Weighting competing models.

Alonso, R. and O. Câmara (2016). Bayesian persuasion with heterogeneous priors. *Journal of Economic Theory 165*, 672–706.

Altonji, J. G., T. E. Elder, and C. R. Taber (2005). Selection on observed and unobserved variables: Assessing the effectiveness of catholic schools. *Journal of political economy 113*(1), 151–184.

Barron, K. and T. Fries (2024). Narrative persuasion. Technical report, WZB Discussion Paper.

Bloedel, A. W. and I. R. Segal (2018). Persuasion with rational inattention. *Available at SSRN 3164033*.

Bohren, J. A. and D. N. Hauser (2021). Learning with heterogeneous misspecified models: Characterization and robustness. *Econometrica 89*(6), 3025–3077.

Bohren, J. A. and D. N. Hauser (2025). Misspecified models in learning and games. *Annual Review of Economics 17*(1), 427–451.

Burkovskaya, A. and E. Starkov (2026). Causal persuasion.

Carroll, G. R. and J. B. Sørensen (2024). Strategy theory using analogy: Rationale, tools and examples. *Strategy Science 9*(4), 483–498.

Cheng, H. and A. Hsiaw (2022). Distrust in experts and the origins of disagreement. *Journal of economic theory 200*, 105401.

Colo, P. (2024). Communicating about confidence: Cheap talk with an ambiguity-averse receiver. *American Economic Journal: Microeconomics 16*(3), 43–75.

Dai, Y., D. Fudenberg, and H. Pei (2026). Bayesian persuasion with selective disclosure. *arXiv preprint arXiv:2601.05914*.

De Clippel, G. and X. Zhang (2022). Non-bayesian persuasion. *Journal of Political Economy 130*(10), 2594–2642.

Deimen, I. and D. Szalay (2019). Delegated expertise, authority, and communication. *American Economic Review 109*(4), 1349–1374.

Di Tillio, A., M. Ottaviani, and P. N. Sørensen (2021). Strategic sample selection. *Econometrica 89*(2), 911–953.

Eliaz, K., S. Galperti, and R. Spiegler (2025). False narratives and political mobilization. *Journal of the European Economic Association 23*(3), 983–1027.

Eliaz, K. and R. Spiegler (2020). A model of competing narratives. *American Economic Review 110*(12), 3786–3816.

Eliaz, K. and R. Spiegler (2026). News media as suppliers of narratives (and information). *Theoretical Economics 21*(3), 928–972.

Eliaz, K., R. Spiegler, and H. C. Thysen (2021). Strategic interpretations. *Journal of Economic Theory 192*, 105192.

Esponda, I. and D. Pouzo (2016). Berk–nash equilibrium: A framework for modeling agents with misspecified models. *Econometrica 84*(3), 1093–1130.

Felgenhauer, M. and E. Schulte (2014). Strategic private experimentation. *American Economic Journal: Microeconomics 6*(4), 74–105.

Flynn, J. P. and K. Sastry (2025). Contagious business cycles. *Available at SSRN 4140751*.

Fudenberg, D., G. Lanzani, and P. Strack (2021). Limit points of endogenous misspecified learning. *Econometrica 89*(3), 1065–1098.

Gentzkow, M. and E. Kamenica (2017). Bayesian persuasion with multiple senders and rich signal spaces. *Games and Economic Behavior 104*, 411–429.

Gleyze, S. and A. Pernoud (2022). The value of model misspecification in communication.

Han, Y., D. Huffman, and Y. Liu (2026). Minds, models and markets: How managerial cognition affects pricing strategies.

Heidhues, P., B. Kőszegi, and P. Strack (2018). Unrealistic expectations and misguided learning. *Econometrica 86*(4), 1159–1214.

Henry, E. and M. Ottaviani (2019). Research and the approval process: The organization of persuasion. *American Economic Review 109*(3), 911–955.

Huang, W. (2024). Learning about informativeness. *arXiv preprint arXiv:2406.05299*.

Jin, Z., J. Liu, Z. Lyu, M. Sachan, R. Mihalcea, M. Diab, D. Ha, et al. (2024). Can large language models infer causation from correlation? In *International Conference on Learning Representations*, Volume 2024, pp. 28663–28679.

Kamenica, E. and M. Gentzkow (2011). Bayesian persuasion. *American Economic Review 101*(6), 2590–2615.

Kishishita, D. (2020). (not) delegating decisions to experts: The effect of uncertainty. *Journal of Economic Theory 190*, 105117.

Kolotilin, A. and A. Zapechelnyuk (2025). Persuasion meets delegation. *Econometrica 93*(1), 195–228.

Landsman, Z. and J. Nešlehová (2008). Stein's lemma for elliptical random vectors. *Journal of Multivariate Analysis 99*(5), 912–927.

Levy, G., I. M. d. Barreda, and R. Razin (2022). Persuasion with correlation neglect: a full manipulation result. *American Economic Review: Insights 4*(1), 123–138.

Levy, G., R. Razin, and A. Young (2022). Misspecified politics and the recurrence of populism. *American Economic Review 112*(3), 928–962.

Li, Y. and C. Wang (2018). Delegation uncertainty. *Columbia Business School Research Paper* (15-54).

Lipnowski, E., L. Mathevet, and D. Wei (2020). Attention management. *American Economic Review: Insights 2*(1), 17–32.

Mailath, G. J. and L. Samuelson (2020). Learning under diverse world views: Model-based inference. *American Economic Review 110*(5), 1464–1501.

Min, W. (2026). Information acquisition with uncertain signal structure. *Available at SSRN 5118651*.

Montiel Olea, J. L., P. Ortoleva, M. M. Pai, and A. Prat (2022). Competing models. *The Quarterly Journal of Economics 137*(4), 2419–2457.

Nielsen (2024). Audience targeting solutions. `https://www.nielsen.com/solutions/media-planning/audience-segments/`. Accessed 2026-09-03; page last modified 2024-09-12.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of business & economic statistics 37*(2), 187–204.

Perez-Richet, E. and D. Prady (2012). Complicating to persuade?

Schwartzstein, J. and A. Sunderam (2021). Using models to persuade. *American Economic Review 111*(1), 276–323.

Shiller, R. J. (2017, April). Narrative economics. *American Economic Review 107*(4), 967–1004.

Spiegler, R. (2016). Bayesian networks and boundedly rational expectations. *The Quarterly Journal of Economics 131*(3), 1243–1290.

Theil, H. (1957). Specification errors and the estimation of economic relationships. *Revue de l'Institut International de Statistique*, 41–51.

Wang, Y. (2013). Bayesian persuasion with multiple receivers. *Available at SSRN 2625399*.

Zuccarelli, E. (2024, April). Building trust in AI means moving beyond black-box algorithms. Here's why. https://www.weforum.org/stories/emerging-technologies/building-trust-in-ai-means-moving-beyond-black-box-algorithms-heres-why/. Accessed 2026-09-03.

A Proofs of Results

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ best-fits the data and the simple sender's sampling strategy μ_δ has mean and variance x_δ and σ_δ^2. Then, for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$, we have:*

$$f_R(x) = f_S(x) + (\gamma - \delta)\big(\sigma_\delta^2 - (x - x_\delta)^2\big).$$

Proof. We rewrite the sender's model as a local expansion around x_δ: $f_S(x) = \tilde{\beta}_0 + \tilde{\beta}_1(x - x_\delta) - \delta(x - x_\delta)^2$, where $\tilde{\beta}_1 = \beta_1 - 2\delta x_\delta$. Because a sender's type is the curvature δ of his model, the value of δ uniquely pins down the sender's type.

Suppose $f_R \in \phi(f_S; \mu_\delta)$. Similarly expanding f_R around x_δ, we have:

$$f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1(x - x_\delta) - \gamma(x - x_\delta)^2.$$

Because μ_δ is symmetric, it is elliptical (in one dimension). By Stein's Lemma (Landsman and Nešlehová, 2008), equation (2) can be written as

$$\begin{aligned}\mathbb{E}_{\mu_\delta}\big[x\big(f_R(x) - f_S(x)\big)\big] &= \mathrm{Cov}_{\mu_\delta}(x, f_R(x) - f_S(x)) \\ &= \sigma^2 \mathbb{E}_{\mu_\delta}[f_R'(x) - f_S'(x)] \\ &= \sigma^2 \mathbb{E}_{\mu_\delta}[\tilde{\alpha}_1 - 2\gamma(x - x_\delta) - \tilde{\beta}_1 + 2\delta(x - x_\delta)] = 0\end{aligned}$$

Because the mean of μ_δ is x_δ, this yields the condition $\tilde{\alpha}_1 = \tilde{\beta}_1$, which is *independent* of σ^2. This means that $\alpha_1 - 2\gamma x_\delta = \beta_1 - 2\delta x_\delta$, i.e. $\alpha_1 = \beta_1 - 2(\delta - \gamma)x_\delta$. Hence, we have

$$f_R(x) = \alpha_0 + (\beta_1 - 2(\delta - \gamma)x_\delta)x - \gamma x^2,$$

with the value of γ and α_0 potentially indeterminate. However, fixing a plausible γ value, from (1) we have

$$\begin{aligned}\mathbb{E}_{\mu_\delta}[f_R(x)] = \alpha_0 + (\beta_1 - 2(\delta - \gamma)x_\delta)x_\delta - \gamma(x_\delta^2 + \sigma^2) = \beta_0 + \beta_1 x_\delta - \delta(x_\delta^2 + \sigma^2) = \mathbb{E}_{\mu_\delta}[f_S(x)] \\ \implies \alpha_0 - 2(\delta - \gamma)x_\delta^2 = \beta_0 - \delta x_\delta^2 - \delta\sigma^2 \\ \implies \alpha_0 = \beta_0 + (\gamma - \delta)(\sigma^2 - x_\delta^2).\end{aligned}$$

Putting the pieces together, we have

$$\begin{aligned}f_R(x) &= \beta_0 + (\gamma - \delta)(\sigma^2 - x_\delta^2) + (\beta_1 - 2(\delta - \gamma)x_\delta)x - \gamma x^2 \\ &= \beta_0 + \beta_1 x - \delta x^2 + \big(-(\gamma - \delta)x^2 + (\gamma - \delta)(\sigma^2 - x_\delta^2) + 2(\gamma - \delta)x_\delta\big) \\ &= \beta_0 + \beta_1 x - \delta x^2 + (\gamma - \delta)\big(\sigma^2 - (x - x_\delta)^2\big) \\ &= f_S(x) + (\gamma - \delta)\big(\sigma^2 - (x - x_\delta)^2\big).\end{aligned}$$

Because the receiver's prior over α_1 is flat, her expected utility (and objective) is given by:

$$\beta_0 + \beta_1 x - \delta x^2 + \Pr(\theta = \delta | f_S)\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\big(\sigma^2 - (x - x_\delta)^2\big)$$

which is maximized at an interior value when

$$f'_S(x_\delta) - 2\Pr(\theta = \delta | f_S)\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)(x - x_\delta) = 0$$
$$x = x_\delta + \frac{f'_S(x_\delta)}{2\big(\Pr(\theta = \delta | f_S)\delta + \Pr(\theta = a | f_S)\mathbb{E}^R_\gamma[\gamma]\big)}$$

□

Theorem 1. *Suppose S is a type δ simple sender.*

1. *If S is low-curvature, he collects* dispersed *data about extreme choices. μ^*_δ is a bimodal distribution that places mass $\frac{1}{2}$ on each of $x^*_\delta - \overline{\sigma}$ and $x^*_\delta + \overline{\sigma}$, where $x^*_\delta \in \big[0, \mathbb{E}^\delta_{f_S}[x^*(f_S)]\big)$.*
2. *If S is high-curvature, he collects* local *data about a single extreme choice. μ^*_δ is any distribution in M with variance $\underline{\sigma}^2$, and has mean x^*_δ equal to either 0 or $\overline{x}$.*

Proof. Following from Proposition 1, given a type δ sender's model, any $\gamma \in [0, \overline{\gamma}]$ value is possible from the receiver's perspective. Fixing x_δ, σ_δ, and a model f_S with curvature δ, the receiver chooses $x = x^*(f_S)$ to maximize:

$$f_S(x) + \Pr(\theta = \delta | f_S)\Delta(f_S, x_\delta)\big(\sigma^2_\delta - (x - x_\delta)^2\big),$$

where $\Delta(f_S, x_\delta) := \mathbb{E}^R_\gamma[\gamma | \gamma, f_S, x_\delta] - \delta$. In turn, the type δ sender chooses x_δ, σ_δ to maximize his expectation of the receiver's perceived expected utility

$$\max_{x_\delta, \sigma_\delta} \mathbb{E}^\delta_{f_S}\left[\max_{x \in [0, \overline{x}]} f_S(x) + \Pr(\theta = \delta | f_S)\Delta(f_S, x_\delta)\big(\sigma^2_\delta - (x - x_\delta)^2\big)\right] \tag{5}$$

We note that because the receiver has a uniform prior over α_0, her posterior places the same mass on all models that are the same "up to a constant"; hence, changing σ^2 does not change the relative mass she places on each model, fixing a plausible γ value. Thus, by the Envelope Theorem, (5) is increasing in σ^2 if $\Delta(f_S, x_\delta)$ is positive and decreasing if $\Delta(f_S, x_\delta)$ is negative. Working backwards, when the sender takes expectations over (5), he chooses σ as large as possible. Fixing x_δ, the distribution with the largest possible variance corresponds to a bimodal distribution which places equal mass on $x_\delta - \underline{\sigma}$ and $x_\delta + \overline{\sigma}$. Otherwise, if $\mathbb{E}_{f_S}[\Delta(f_S, x_\delta)] \geq 0$, we have $\sigma^*_\delta = \underline{\sigma}$.

Next, we characterize the equilibrium value of x_δ and hence $\mathbb{E}_{f_S}[\Delta(f_S, x_\delta)]$. Note that $\Delta(f_S, x_\delta)$ depends not just on f_S but also x_δ. In particular:

$$\mathbb{E}^R_\gamma[\gamma | f_S, x_\delta, \theta = \delta] = \frac{\int_0^{\overline{\gamma}} \gamma p^R_\gamma(\gamma) p_1(\beta_1 + 2(\gamma - \delta)x_\delta) d\gamma}{\int_0^{\overline{\gamma}} p^R_\gamma(\gamma) p_1(\beta_1 + 2(\gamma - \delta)x_\delta) d\gamma}.$$

Note that when $x_\delta = 0$, this expression is the receiver's prior expectation of γ, i.e. $\mathbb{E}_R^\gamma[\delta]$.

Next, p_1 is uniform on $[\underline{\alpha}_1, \overline{\alpha}_1]$, p_1 is $\neq 0$ only when $\gamma \in \left(\delta + \frac{\underline{\alpha}-\beta_1}{2x_\delta}, \delta + \frac{\overline{\alpha}-\beta_1}{2x_\delta}\right)$. Conditional on f_S, the receiver's expectation and the sender's prior expectation of that expectation can then be written as

$$\mathbb{E}_\gamma^R[\gamma|f_S, x_\delta, \theta = \delta] = \mathbb{E}_\gamma^R\Big[\gamma|\gamma \in \left(\delta + \frac{\underline{\alpha}-\beta_1}{2x_\delta}, \delta + \frac{\overline{\alpha}-\beta_1}{2x_\delta}\right)\Big]$$

$$\implies \mathbb{E}_{f_S}^\delta\Big[\mathbb{E}_\gamma^R[\gamma|f_S, x_\delta, \theta = \delta]\Big] = \int_{\underline{\alpha}_1}^{\overline{\alpha}_1} \frac{\mathbb{E}_\gamma^R\Big[\gamma|\gamma \in \left(\delta + \frac{\underline{\alpha}-\beta_1}{2x_\delta}, \delta + \frac{\overline{\alpha}-\beta_1}{2x_\delta}\right)\Big]}{\overline{\alpha}_1 - \underline{\alpha}_1} d\beta_1.$$

That is, the simple sender is integrating the expectation out with respect to a (uniform) prior over β_1, which is identical to the prior over α_1. Substituting $u = \frac{\beta_1 - \underline{\alpha}_1}{2x_\delta}$:

$$\mathbb{E}_{f_S}^\delta\Big[\mathbb{E}_\gamma^R[\gamma|f_S, x_\delta, \theta = \delta]\Big] = \frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1} \int_0^{\frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta}} \mathbb{E}_\gamma^R\Big[\gamma|\gamma \in \Big[\delta - u, \delta - u + \frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta}\Big] \cap [0, \overline{\gamma}]\Big]du$$

Recall from above that when $x_\delta = 0$, the expectation is just the receiver's prior. Thus, we can write

$$\mathbb{E}_{f_S}^\delta\Big[\mathbb{E}_\gamma^R[\gamma|f_S, x_\delta, \theta = \delta]\Big] = \mathbb{E}_\gamma^R[\gamma] + \left(\mathbb{E}_{f_S}^\delta\Big[\mathbb{E}_\gamma^R[\gamma|f_S, x_\delta, \theta = \delta]\Big] - \mathbb{E}_\gamma^R[\gamma]\right)$$

Breaking the expectation into three regions gives:

$$\begin{aligned}
&\frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1}\Bigg(\int_0^\delta \mathbb{E}_\gamma^R\Big[\gamma|\gamma \in [\delta - u, \overline{\gamma}]\Big]du + \int_\delta^{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta} - (\overline{\gamma}-\delta)} \mathbb{E}_\gamma^R[\gamma]\, du \\
&+ \int_{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta} - (\overline{\gamma}-\delta)}^{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta}} \mathbb{E}_\gamma^R\Big[\gamma\Big|\gamma \in \Big[0, \delta - u + \frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta}\Big]\Big]du\Bigg) - \mathbb{E}_\gamma^R[\gamma] \\
=&\frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1}\Bigg(\int_0^\delta \mathbb{E}_\gamma^R\Big[\gamma|\gamma \in [\delta - u, \overline{\gamma}]\Big]du + \Big(\frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta} - \overline{\gamma}\Big)\mathbb{E}_\gamma^R[\gamma] \\
&+ \int_{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta} - (\overline{\gamma}-\delta)}^{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta}} \mathbb{E}_\gamma^R\Big[\gamma\Big|\gamma \in \Big[0, \delta - u + \frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta}\Big]\Big]du - \frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta}\mathbb{E}_\gamma^R[\gamma]\Bigg) \\
=&\frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1}\Bigg(\int_0^\delta \mathbb{E}_\gamma^R\Big[\gamma|\gamma \in [\delta - u, \overline{\gamma}]\Big]du - \delta\,\mathbb{E}_\gamma^R[\gamma] + \int_{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta} - (\overline{\gamma}-\delta)}^{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta}} \mathbb{E}_\gamma^R\Big[\gamma\Big|\gamma \in \Big[0, \delta - u + \frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta}\Big]\Big]du - (\overline{\gamma} - \delta)\,\mathbb{E}_\gamma^R[\gamma]\Bigg) \\
=&\frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1}\Bigg(\int_0^\delta \Big(\mathbb{E}_\gamma^R\Big[\gamma|\gamma \in [\delta - u, \overline{\gamma}]\Big] - \mathbb{E}_\gamma^R[\gamma]\Big)du + \int_{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta} - (\overline{\gamma}-\delta)}^{\frac{\overline{\alpha}_1-\underline{\alpha}_1}{2x_\delta}} \Big(\mathbb{E}_\gamma^R\Big[\gamma\Big|\gamma \in \Big[0, \delta - u + \frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta}\Big]\Big] - \mathbb{E}_\gamma^R[\gamma]\Big)du\Bigg)
\end{aligned}$$

Changing variables again with $v = -u + \frac{\overline{\alpha}_1 - \underline{\alpha}_1}{2x_\delta} + \delta$ in the last integral term gives

$$\begin{aligned}
&\frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1}\left(\int_0^\delta \mathbb{E}_\gamma^R\big[\gamma|\gamma \in \big[\delta - u, \overline{\gamma}\big]\big] - \mathbb{E}_\gamma^R[\gamma]du + \int_\delta^{\overline{\gamma}} \mathbb{E}_\gamma^R\big[\gamma|\gamma \in \big[0, v\big]\big] - \mathbb{E}_\gamma^R[\gamma]dv \right) \\
&= \frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1}\left(\int_0^\delta \mathbb{E}_\gamma^R\big[\gamma|\gamma \geq u\big] - \mathbb{E}_\gamma^R[\gamma]du + \int_\delta^{\overline{\gamma}} \mathbb{E}_\gamma^R\big[\gamma|\gamma \leq v\big] - \mathbb{E}_\gamma^R[\gamma]dv \right) \\
&= \frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1}\left(\int_0^\delta \mathbb{E}_\gamma^R\big[\gamma|\gamma \geq u\big]du + \int_\delta^{\overline{\gamma}} \mathbb{E}_\gamma^R\big[\gamma|\gamma \leq v\big]dv - \overline{\gamma}\mathbb{E}_\gamma^R[\gamma] \right) := \frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1} K(\delta),
\end{aligned}$$

where $K(\delta)$ does not depend on x_δ fixing δ. Thus, fixing δ, $\mathbb{E}_{f_S}^\delta\big[\mathbb{E}_\gamma^R[\gamma|f_S, x_\delta, \theta = \delta]$ is linear in x_δ. $K(\delta)$ is strictly increasing in δ, since its derivative is

$$\mathbb{E}_\gamma^R[\gamma|\gamma \geq \delta] - \mathbb{E}_\gamma^R[\gamma|\gamma \leq \delta].$$

Moreover, because γ is symmetric, we have that $\overline{\gamma} - \gamma$ and γ have the same distribution, i.e.

$$\mathbb{E}_\gamma^R\big[\gamma|\gamma \geq u\big] = \mathbb{E}_\gamma^R\big[\overline{\gamma} - \gamma|\overline{\gamma} - \gamma \geq u\big] = \overline{\gamma} - \mathbb{E}_\gamma^R\big[\gamma|\gamma \leq \overline{\gamma} - u\big].$$

But then we have

$$\begin{aligned}
K(\mathbb{E}_\gamma^R[\gamma]) &= \int_0^{\mathbb{E}_\gamma^R[\gamma]} \mathbb{E}_\gamma^R\big[\gamma|\gamma \geq u\big]du + \int_{\mathbb{E}_\gamma^R[\gamma]}^{\overline{\gamma}} \mathbb{E}_\gamma^R\big[\gamma|\gamma \leq v\big]dv - \overline{\gamma}\mathbb{E}_\gamma^R[\gamma] \\
&= \int_0^{\mathbb{E}_\gamma^R[\gamma]} \overline{\gamma} - \mathbb{E}_\gamma^R\big[\gamma|\gamma \leq \overline{\gamma} - u\big]du + \int_{\mathbb{E}_\gamma^R[\gamma]}^{\overline{\gamma}} \mathbb{E}_\gamma^R\big[\gamma|\gamma \leq v\big]dv - \overline{\gamma}\mathbb{E}_\gamma^R[\gamma] \\
&= \int_0^{\mathbb{E}_\gamma^R[\gamma]} -\mathbb{E}_\gamma^R\big[\gamma|\gamma \leq \overline{\gamma} - u\big]du + \int_{\mathbb{E}_\gamma^R[\gamma]}^{\overline{\gamma}} \mathbb{E}_\gamma^R\big[\gamma|\gamma \leq v\big]dv + \overline{\gamma}\mathbb{E}_\gamma^R[\gamma] - \overline{\gamma}\mathbb{E}_\gamma^R[\gamma] = 0.
\end{aligned}$$

Thus, for $\delta < \mathbb{E}_\gamma^R[\gamma]$, $\mathbb{E}_{f_S}^\delta\big[\mathbb{E}_\gamma^R[\gamma|f_S, x_\delta, \theta = \delta]$ is strictly decreasing and linear in x_δ. For $\delta > \mathbb{E}_\gamma^R[\gamma]$, $\mathbb{E}_{f_S}^\delta\big[\mathbb{E}_\gamma^R[\gamma|f_S, x_\delta, \theta = \delta]$ is strictly increasing and linear in x_δ. When $\delta = \mathbb{E}_\gamma^R[\gamma]$, it is equal to 0 for all x_δ.

Thus, we have:

$$\mathbb{E}_{f_S}^\delta[\Delta(f_S, x_\delta)(\sigma_\delta^2 - (x_\delta - x^*(f_S))^2] = \left(\mathbb{E}_\gamma^R[\gamma] - \delta + \frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1} K(\delta)\right)\big(\sigma_\delta^2 - \mathbb{E}_{f_S}^\delta[(x_\delta - x^*(f_S))^2]\big).$$

Differentiating with respect to x_δ and using an Envelope Argument ($x^*(f_S)$ maximizes the receiver's objective) yields that the sender's objective is increasing in x_δ if and only if

$$\frac{2}{\overline{\alpha}_1 - \underline{\alpha}_1} K(\delta)\big(\sigma_\delta^2 - \mathbb{E}_{f_S}^\delta[(x_\delta - x^*(f_S))^2]\big) - 2\left(\mathbb{E}_\gamma^R[\gamma] - \delta + \frac{2x_\delta}{\overline{\alpha}_1 - \underline{\alpha}_1} K(\delta)\right)\mathbb{E}_{f_S}^\delta[(x_\delta - x^*(f_S))].$$

Because σ_δ can be arbitrarily small or large, we have that $\big(\sigma_\delta^2 - \mathbb{E}_{f_S}^\delta[(x_\delta - x^*(f_S))^2]\big) > 0$. We thus deal with the following cases.

First, suppose $\delta < \mathbb{E}_\gamma^R[\gamma]$. Note that because σ_δ is sufficiently large, setting $x_\delta = 0$ always deliv-

ers a strictly positive gain. Indeed, the first set of terms in the derivative is negative (since $K(\delta)$ is negative) and pushes the optimal choice of x^*_δ towards 0 linearly. The latter set of terms pushes towards $\mathbb{E}^\delta_{f_S}[x^*(f_S)]$: since $\mathbb{E}^\delta_{f_S}\big[\mathbb{E}^R_\gamma[\gamma|f_S, x_\delta, \theta = \delta]\big] - \delta > 0$, it is positive when $x_\delta < \mathbb{E}^\delta_{f_S}[x^*(f_S)]$ and negative when $x_\delta > \mathbb{E}^\delta_{f_S}[x^*(f_S)]$. Thus, we have $x^*_\delta \in [0, \mathbb{E}^\delta_{f_S}[x^*(f_S)]$.

Second, suppose $\delta > \mathbb{E}^R_\gamma[\gamma]$. Then $K(\delta) > 0$ and $\sigma^2_\delta - \mathbb{E}^\delta_{f_S}[(x_\delta - x^*(f_S))^2] \le 0$. The first set of terms thus pushes to 0 linearly. The latter set of terms represent a convex function and push to either 0 or $\overline{x}$. Thus, we have either $x^*_\delta = 0$ or $\overline{x}$.

Finally, putting the pieces together, whenever $\delta < \mathbb{E}^R_\gamma[\gamma]$, a type δ sender always has a positive gain from setting $x^*_\delta \ge 0$. Thus, these senders set $\sigma^*_\delta = \overline{\sigma}$ via a bimodal distribution with symmetric mass on $x^*_\delta \pm \overline{\sigma}$. For $\delta > \mathbb{E}^R_\gamma[\gamma]$, these senders set $\sigma^*_\delta = \underline{\sigma}$.

□

Proposition 2. *Suppose the accurate sender provides a model f_S with curvature δ, that $\Pr(\theta = \delta|f_S) > 0$, that the type δ simple sender samples as in Theorem 1, and that $\Delta(f_S, x_\delta)$ has the same sign as $\mathbb{E}^R_\gamma[\gamma] - \delta$.*

1. *If $\delta < \mathbb{E}^R_\gamma[\gamma]$ and $\overline{\sigma} > \overline{x}$, then $G(f_S) > 0$.*
2. *If $\delta > \mathbb{E}^R_\gamma[\gamma]$, then $G(f_S) > 0$ whenever $\arg\max_{x\in X} f_S(x)$ lies more than $\underline{\sigma}$ away from x^*_δ.*

Proof. Note that for each f_S, we have

$$G(f_S) = \Pr(\theta = \delta|f_S) \cdot \begin{cases} (\mathbb{E}^R_\gamma[\gamma] - \delta)\big(\overline{\sigma}^2 - (x^*(f_S) - \mathbb{E}^\delta_{f_S}[x^*(f_S)])^2\big) & \delta < \mathbb{E}^R_\gamma[\gamma] \\ (\mathbb{E}^R_\gamma[\gamma] - \delta)\big(\underline{\sigma}^2 - (x^*(f_S) - x^*_\delta)^2\big) & \delta \ge \mathbb{E}^R_\gamma[\gamma] \end{cases}$$

where x^*_δ in the latter case is either 0 or $\overline{x}$. In the first case, note that the $\mathbb{E}^R_\gamma[\gamma] - \delta$ term is strictly positive. Because we assumed $\overline{\sigma}$ is large, the latter set of terms is also positive. In the second case, $\mathbb{E}^R_\gamma[\gamma] - \delta$ is ngative. However, because $\underline{\sigma}$ is arbitrarily small, $\underline{\sigma}^2 - (x^*(f_S) - x^*_\delta)^2 < 0$. Thus, their product is always strictly positive. This gives the result. □

Proposition 3. *Suppose $f_S = \beta_0 + \beta^T x - x^T \delta x$ best-fits the data and $\mu_\delta = \mu_\delta(x_\delta, \Sigma_\delta)$. Then, for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$, we have:*

$$f_R(x) = f_S(x) + \operatorname{tr}\big((\gamma - \delta)\Sigma_\delta\big) - (x - x_\delta)^T(\gamma - \delta)(x - x_\delta).$$

Proof. As above, we necessarily have the following moment conditions for all i:

$$\mathbb{E}_{\mu_\delta}\big[x_i\big(f_R(x) - f_S(x)\big)\big] = 0 \qquad \mathbb{E}_{\mu_\delta}\big[f_R(x) - f_S(x)\big] = 0$$

and we may or may not have the correct moment condition with respect to the entries of γ. Crucially, however, because μ_δ is elliptical, by Stein's Lemma for elliptical random vectors (Landsman

and Nešlehová, 2008), we can simplify the first equation to

$$\begin{aligned}
\mathbb{E}_{\mu_\delta}\big[x_i\big(f_R(x) - f_S(x)\big)\big] = 0 \implies & \mathbb{E}_{\mu_\delta}\big[\frac{\partial}{\partial x_i} f_R(x)\big] = \mathbb{E}_{\mu_\delta}\big[\frac{\partial}{\partial x_i} f_S(x)\big] \qquad \forall i \\
\implies & \alpha - (\gamma + \gamma')\mathbb{E}_{\mu_\delta}[x] = \beta - (\delta + \delta')\mathbb{E}_{\mu_\delta}[x] \\
\implies & \alpha = \beta + \big((\gamma + \gamma') - (\delta + \delta')\big)x_\delta \\
\implies & \alpha' x = \beta' x + x_\delta'\big((\gamma + \gamma') - (\delta + \delta')\big)x
\end{aligned}$$

which is invariant to Σ_δ. Next, working with the second equation, we have

$$\mathbb{E}_{\mu_\delta}\big[f_R(x)\big] = \alpha_0 + \alpha x_\delta - \big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big)'\gamma\big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big) = \beta_0 + \beta x_\delta - \big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big)'\delta\big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big) = \mathbb{E}_{\mu_\delta}\big[f_S(x)\big].$$

Note that

$$\big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big)'\gamma\big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big) = x_\delta'\gamma x_\delta + 2x_\delta'\gamma\Sigma_\delta^{\frac{1}{2}}\epsilon + \Sigma_\delta^{\frac{1}{2}}\epsilon'\gamma\Sigma_\delta^{\frac{1}{2}}\epsilon$$

and taking the expectation with respect to ϵ thus gives

$$x_\delta'\gamma x_\delta + \mathrm{tr}(\gamma\Sigma_\delta).$$

Utilizing the same logic with δ and plugging in the α equation gives

$$\begin{aligned}
\alpha_0 &= \beta_0 + (\beta - \alpha)x_\delta + x_\delta'(\gamma - \delta)x_\delta + \mathrm{tr}((\gamma - \delta)\Sigma_\delta) \\
&= \beta_0 - x_\delta'\big((\gamma + \gamma') - (\delta + \delta')\big)x_\delta + \mathrm{tr}((\gamma - \delta)\Sigma_\delta) \\
&= \beta_0 - x_\delta'(\gamma - \delta)x_\delta + \mathrm{tr}((\gamma - \delta)\Sigma_\delta)
\end{aligned}$$

Hence, we renormalize f_R, plug in the values of α_0, α, and reorganize the quadratic terms to get:

$$\begin{aligned}
f_R(x) &= \alpha_0 + \alpha' x - x'\gamma x \\
&= \beta_0 - x_\delta'(\gamma - \delta)x_\delta + \mathrm{tr}((\gamma - \delta)\Sigma_\delta) + \beta' x + x_\delta'\big((\gamma + \gamma') - (\delta + \delta')\big)x - x'\gamma x \\
&= \beta_0 + \beta' x + \mathrm{tr}((\gamma - \delta)\Sigma_\delta) - (x - x_\delta)'(\gamma - \delta)(x - x_\delta).
\end{aligned}$$

□

Theorem 2. *Simple senders always maximize the collinearity of underlying datasets, i.e. make variables as highly correlated or highly anticorrelated as possible. In particular:*

- *if* $\mathbb{E}_{f_S}^{\delta}[\Delta_{ij}(f_S, x_\delta^*) + \Delta_{ji}(f_S, x_\delta^*)] > 0$, *then* $\rho_{ij}^* = \overline{\rho}$.
- *If* $\mathbb{E}_{f_S}^{\delta}[\Delta_{ij}(f_S, x_\delta^*) + \Delta_{ji}(f_S, x_\delta^*)] < 0$, *then* $\rho_{ij}^* = -\overline{\rho}$.

If γ *is diagonal for all* $\gamma \in \Gamma$, *the sender additionally sets* $\sigma_i^* = \overline{\sigma}$ *or* $\underline{\sigma}$. *In particular:*

- *if* $\mathbb{E}_{f_S}^{\delta}[\Delta_{ii}(f_S, 0)] > 0$, *then* $\sigma_i^* = \overline{\sigma}$.

- *If* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ii}(f_S, 0)] < 0$, *then* $\sigma_i^* = \underline{\sigma}$.

Proof. The proof of this result is similar to that of Theorem 1. The problem for a type δ sender is given by:

$$\max_{\Sigma_\delta, x_\delta} \mathbb{E}^{\delta}_{f_S}\left[\max_{x\in[0,\overline{x}]} \beta_0 + \beta' x + \Pr(\theta = \delta|f_S)\Big(\operatorname{tr}((\gamma-\delta)\Sigma_\delta) - (x-x_\delta)'(\gamma-\delta)(x-x_\delta)\Big)\right]$$

subject to the constraints that Σ_δ is positive definite and variances/correlations are in their bounded intervals:

$$\Big(\prod_{i=1}^{j}\sigma_i^2\Big)\det\begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1j} \\ \rho_{12} & 1 & \cdots & \rho_{2j} \\ \vdots & & \ddots & \vdots \\ \rho_{1j} & \rho_{2j} & \cdots & 1 \end{pmatrix} > 0 \quad \text{for } j = 2,\dots,n, \quad \sigma_i \in [\underline{\sigma}, \overline{\sigma}], \quad \rho_{ij} \in [-\overline{\rho}, \overline{\rho}].$$

We first work with Σ_δ, fixing x_δ, and note that the program with respect to Σ_δ depends only on $\gamma - \delta$ and $\Pr(\theta = a|f_S)$. Recalling the definition of $\Delta(f_s, x_\delta) = \mathbb{E}[\gamma|f_S, \theta \neq a] - \delta$, the maximizer is equivalent to the maximizer of

$$\max_{\Sigma_\delta} \mathbb{E}^{\delta}_{f_S}\big[\operatorname{tr}(\Delta(f_s, x_\delta)\Sigma_\delta)\big] = \max_{\Sigma_\delta} \mathbb{E}^{\delta}_{f_S}\Big[\sum_{i=1}^{n}\Delta_{ii}(f_S, x_\delta)\sigma_i^2 + \sum_{i<j}(\Delta_{ij}+\Delta_{ji})\rho_{ij}\sigma_i\sigma_j\Big]$$

subject to the same constraints. This program is linear in the objective, meaning the solution is given by an extreme point of the feasible set. In particular, fixing σ_i and σ_j, the objective is linear in ρ_{ij}. Thus, we have

$$\mathbb{E}^{\delta}_{f_S}[\Delta_{ij} + \Delta_{ji}] > 0 \implies \rho_{ij} = \overline{\rho}$$
$$\mathbb{E}^{\delta}_{f_S}[\Delta_{ij} + \Delta_{ji}] < 0 \implies \rho_{ij} = -\overline{\rho},$$

noting that because $|\rho| < 1$, we have:

$$\det\begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1j} \\ \rho_{12} & 1 & \cdots & \rho_{2j} \\ \vdots & & \ddots & \vdots \\ \rho_{1j} & \rho_{2j} & \cdots & 1 \end{pmatrix} > 0$$

We further note that the value of ρ does not depend on σ_i or σ_j, since they are always positive. With the values of ρ_{ij} in hand, we then note that because $\sigma_i > 0$ for all i, we necessarily have that $\prod_{i=1}^{j}\sigma_i^2 > 0$, meaning the positive definiteness constraint is always satisfied strictly. Finally, the optimal choice of σ_i is given by taking the program above, plugging in the values of ρ_{ij}, and solving the linear program, which yields either $\sigma_i = \overline{\sigma}_i$ or $\underline{\sigma}_i$ as maximizers.

The program is linear in the entries of Σ_δ. In particular, the sign of the coefficient on Σ_{ij} is

given by the sign of $\Delta_{ij}(f_S, x_\delta) + \Delta_{ji}(f_S, x_\delta)$. □

Proposition 4. *Fix δ. The value of information V for the receiver from hiring a sender of type δ is given by:*

$$V(\delta) = \mathbb{E}_{f_S \in F_\delta}\left[\mathbb{E}^R_\gamma[\gamma|f_S, \theta = \delta, x^*_\delta]\left(x^*(f_S) - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right]$$

and does not depend on σ^*_δ. Moreover, for $x^*(f_S)$ interior, we have

$$V(\delta) = \mathbb{E}^R_{f_R}\left[\mathbb{E}^R_\gamma[\gamma|f'_R(x^*_\delta)]\left(x^*_\delta + \frac{f'_R(x^*_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x^*_\delta)]} - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right].$$

Proof. Suppose x^* maximizes $\mathbb{E}^R_{f_R|f_S}[f_R(x)|f_S]$ and x_0 maximizes $\mathbb{E}^R_{f_R}[f_R(x)]$. Conditional on f_S from a type δ sender, we have:

$$\begin{aligned}
&\mathbb{E}^R_{f_R|f_S}[f_R(x^*)] - \mathbb{E}^R_{f_R|f_S}[f_R(x_0)] \\
&= f_S(x^*) + \Delta(f_S, x^*_\delta)\big(\sigma^2_\delta - (x^* - x^*_\delta)^2\big) - f_S(x_0) - \Delta(f_S, x^*_\delta)\big(\sigma^2_\delta - (x_0 - x^*_\delta)^2\big) \\
&= \beta_1(x^* - x_0) - \delta\big(x^{*2} - x_0^2\big) + \Delta(f_S, x^*_\delta)\big((x_0 - x^*_\delta)^2 - (x^* - x^*_\delta)^2\big) \\
&= \beta_1(x^* - x_0) - \delta\big(x^{*2} - x_0^2\big) + \Delta(f_S, x^*_\delta)\big(2x^*_\delta(x^* - x_0) - (x^{*2} - x_0^2)\big) \\
&= \big(\beta_1 + 2\Delta(f_S, x^*_\delta)x^*_\delta\big)(x^* - x_0) - \big(\delta + \Delta(f_S, x^*_\delta)\big)\big(x^{*2} - x_0^2\big) \\
&= (x^* - x_0)\Big[\beta_1 + 2\Delta(f_S, x^*_\delta)x^*_\delta - \big(\delta + \Delta(f_S, x^*_\delta)\big)(x^* + x_0)\Big] \\
&= (x^* - x_0)\Big[2\big(\delta + \Delta(f_S, x^*_\delta)\big)x^* - \big(\delta + \Delta(f_S, x^*_\delta)\big)(x^* + x_0)\Big] \\
&= \big(\delta + \Delta(f_S, x^*_\delta)\big)(x^* - x_0)^2 \\
&= \mathbb{E}^R_\gamma[\gamma|f_S, \theta = \delta, x^*_\delta]\,(x^*(f_S) - x_0)^2 \\
&= \mathbb{E}^R_\gamma[\gamma|f_S, \theta = \delta, x^*_\delta]\left(x^*(f_S) - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2 \\
&= \mathbb{E}^R_\gamma[\gamma|f_S, \theta = \delta, x^*_\delta]\left(x^*(f_S) - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2.
\end{aligned}$$

Since $f'_S(x^*_\delta) = f'_R(x^*_\delta)$, for interior solutions, we have:

$$\mathbb{E}^R_\gamma[\gamma|f_S, \theta = \delta, x^*_\delta]\left(x^*(f_S) - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2 = \mathbb{E}^R_\gamma[\gamma|f'_R(x^*_\delta)]\left(x^*_\delta + \frac{f'_R(x^*_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x^*_\delta)]} - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2.$$

Integrating over f_R gives:

$$V(\delta) = \mathbb{E}^R_{f_R}\left[\mathbb{E}^R_\gamma[\gamma|f'_R(x^*_\delta)]\left(x^*_\delta + \frac{f'_R(x^*_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x^*_\delta)]} - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right].$$

□

Proposition 5. *Suppose the receiver's choice $x^*(f_S)$ is interior. Then $x_\delta^R = \arg\max_{x_\delta \in [0,\overline{x}]} V(\delta, x_\delta)$ satisfies $x_\delta \geq \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}$. Moreover, if S is low-curvature and $\overline{\sigma}$ is sufficiently large or if S is high-curvature and $x_\delta^* = 0$, $x_\delta^R \geq x_\delta^*$.*

Proof. Recalling that $f_R''(x_\delta) = -2\gamma$ and using the previous proposition, we note that

$$\begin{aligned}\frac{\partial V(\delta)}{\partial x_\delta} = \mathbb{E}_{f_R}^R\Bigg[&\left(\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)^2 \\ &+ 2\,\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right) \\ &\cdot\left(1 - \frac{\gamma}{\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]^2}\,\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\Bigg]\end{aligned}$$

In particular, we note that by Bayes' Rule:

$$\mathbb{E}_{f_R}^R\big[\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\big] = \mathbb{E}_\gamma^R[\gamma] \implies \mathbb{E}_{f_R}^R\big[\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\big] = 0.$$

Using this, we simplify the above expression to

$$\begin{aligned}&\mathbb{E}_{f_R}^R\Bigg[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right) \\ &\quad\cdot\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]} - \frac{f_R'(x_\delta)}{\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]}\right) \\ &\quad\underbrace{+\,2\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)\cdot\left(\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)] - \gamma\right)}_{=0 \text{ conditioning on } f_R'}\Bigg] \\ =&\mathbb{E}_{f_R}^R\left[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)\left(x_\delta - \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)\right] \\ =&\mathbb{E}_{f_R}^R\left[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\left(\left(x_\delta - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)^2 - \left(\frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]}\right)^2\right)\right] \\ =&-\mathbb{E}_{f_R}^R\left[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\left(\frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]}\right)^2\right].\end{aligned}$$

Since α_1 is uniform on $[\underline{\alpha}_1, \overline{\alpha}_1]$ with $\overline{\alpha}_1 - \underline{\alpha}_1 \geq 2\overline{x}\,\overline{\gamma}$, and $f_R'(x_\delta) = \alpha_1 - 2\gamma x_\delta$, we have

$$\mathbb{E}_\gamma^R[\gamma | f_R'(x_\delta)] = \mathbb{E}_\gamma^R\left[\gamma \;\middle|\; \gamma \in \left[\frac{\underline{\alpha}_1 - f_R'(x_\delta)}{2x_\delta},\ \frac{\overline{\alpha}_1 - f_R'(x_\delta)}{2x_\delta}\right] \cap [0, \overline{\gamma}]\right].$$

Using $f_R''(x_\delta) = -2\gamma$, the lower limit has derivative $\frac{\alpha_1 - \underline{\alpha}_1}{2x_\delta^2} \geq 0$ and the upper limit has derivative $-\frac{\overline{\alpha}_1 - \alpha_1}{2x_\delta^2} \leq 0$ with respect to x_δ. At most one limit lies inside $[0, \overline{\gamma}]$: the lower limit does if $\alpha_1 - \underline{\alpha}_1 < 2\gamma x_\delta$, and the upper limit does if $\overline{\alpha}_1 - \alpha_1 < 2(\overline{\gamma} - \gamma)x_\delta$. Otherwise the posterior equals the prior and its derivative is zero. Hence

$$\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma | f_R'(x_\delta)] = \begin{cases} \dfrac{p_\gamma^R(t)\big(\mathbb{E}_\gamma^R[\gamma|\gamma \geq t] - t\big)}{\Pr(\gamma \geq t)} \cdot \dfrac{\alpha_1 - \underline{\alpha}_1}{2x_\delta^2}, & t = \frac{\underline{\alpha}_1 - f_R'(x_\delta)}{2x_\delta} \in (0, \overline{\gamma}), \\ -\dfrac{p_\gamma^R(t)\big(t - \mathbb{E}_\gamma^R[\gamma|\gamma \leq t]\big)}{\Pr(\gamma \leq t)} \cdot \dfrac{\overline{\alpha}_1 - \alpha_1}{2x_\delta^2}, & t = \frac{\overline{\alpha}_1 - f_R'(x_\delta)}{2x_\delta} \in (0, \overline{\gamma}), \\ 0 & \text{otherwise,} \end{cases}$$

and in the two edge cases $f_R'(x_\delta) = \underline{\alpha}_1 - 2x_\delta t$ and $f_R'(x_\delta) = \overline{\alpha}_1 - 2x_\delta t$, respectively.

Writing $\mathbb{E}_{f_R}^R$ as an integral over α_1 and γ and using the arguments in Theorem 1 gives

$$\begin{aligned}\frac{\partial V(\delta)}{\partial x_\delta} = -\frac{1}{\overline{\alpha}_1 - \underline{\alpha}_1}\int_0^{\overline{\gamma}} \Bigg[&\frac{2p_\gamma^R(t)\big(\mathbb{E}_\gamma^R[\gamma|\gamma \geq t] - t\big)}{\Pr(\gamma \geq t)}\left(\frac{\underline{\alpha}_1 - 2x_\delta t}{2\mathbb{E}_\gamma^R[\gamma|\gamma \geq t]}\right)^2 \int_t^{\overline{\gamma}} (\gamma - t)\, p_\gamma^R(\gamma)\, d\gamma \\ &- \frac{2p_\gamma^R(t)\big(t - \mathbb{E}_\gamma^R[\gamma|\gamma \leq t]\big)}{\Pr(\gamma \leq t)}\left(\frac{\overline{\alpha}_1 - 2x_\delta t}{2\mathbb{E}_\gamma^R[\gamma|\gamma \leq t]}\right)^2 \int_0^t (t - \gamma)\, p_\gamma^R(\gamma)\, d\gamma \Bigg] dt.\end{aligned}$$

(to-proofread) Since $\int_t^{\overline{\gamma}} (\gamma - t)\, p_\gamma^R(\gamma)\, d\gamma = \Pr(\gamma \geq t)\big(\mathbb{E}_\gamma^R[\gamma|\gamma \geq t] - t\big)$ and $\int_0^t (t - \gamma)\, p_\gamma^R(\gamma)\, d\gamma = \Pr(\gamma \leq t)\big(t - \mathbb{E}_\gamma^R[\gamma|\gamma \leq t]\big)$, this simplifies to

$$\frac{\partial V(\delta)}{\partial x_\delta} = \frac{2}{\overline{\alpha}_1 - \underline{\alpha}_1}\int_0^{\overline{\gamma}} p_\gamma^R(t)\left[\big(t - \mathbb{E}_\gamma^R[\gamma|\gamma \leq t]\big)^2\left(\frac{\overline{\alpha}_1 - 2x_\delta t}{2\mathbb{E}_\gamma^R[\gamma|\gamma \leq t]}\right)^2 - \big(\mathbb{E}_\gamma^R[\gamma|\gamma \geq t] - t\big)^2\left(\frac{\underline{\alpha}_1 - 2x_\delta t}{2\mathbb{E}_\gamma^R[\gamma|\gamma \geq t]}\right)^2\right] dt.$$

Next, we use the symmetry of p_γ^R, as in the proof of Theorem 1: γ and $\overline{\gamma} - \gamma$ have the same distribution, so $p_\gamma^R(t) = p_\gamma^R(\overline{\gamma} - t)$, $\mathbb{E}_\gamma^R[\gamma] = \frac{\overline{\gamma}}{2}$, and

$$\mathbb{E}_\gamma^R[\gamma|\gamma \geq t] = \overline{\gamma} - \mathbb{E}_\gamma^R[\gamma|\gamma \leq \overline{\gamma} - t], \qquad \mathbb{E}_\gamma^R[\gamma|\gamma \geq t] - t = (\overline{\gamma} - t) - \mathbb{E}_\gamma^R[\gamma|\gamma \leq \overline{\gamma} - t].$$

Substituting these into the second term and changing variables from t to $\overline{\gamma} - t$ there gives

$$\frac{\partial V(\delta)}{\partial x_\delta} = \frac{1}{2(\overline{\alpha}_1 - \underline{\alpha}_1)}\int_0^{\overline{\gamma}} p_\gamma^R(t)\big(t - \mathbb{E}_\gamma^R[\gamma|\gamma \leq t]\big)^2\left[\left(\frac{\overline{\alpha}_1 - 2x_\delta t}{\mathbb{E}_\gamma^R[\gamma|\gamma \leq t]}\right)^2 - \left(\frac{\underline{\alpha}_1 - 2x_\delta(\overline{\gamma} - t)}{\overline{\gamma} - \mathbb{E}_\gamma^R[\gamma|\gamma \leq t]}\right)^2\right] dt.$$

The term outside the brackets is nonnegative, and strictly positive on a set of positive measure.

Moreover, $\mathbb{E}^R_\gamma[\gamma|\gamma \le t] \le \mathbb{E}^R_\gamma[\gamma] = \frac{\overline{\gamma}}{2}$, so $\overline{\gamma} - \mathbb{E}^R_\gamma[\gamma|\gamma \le t] \ge \mathbb{E}^R_\gamma[\gamma|\gamma \le t]$, and the term in brackets is at least

$$\frac{\left(\overline{\alpha}_1 - 2x_\delta t\right)^2 - \left(\underline{\alpha}_1 - 2x_\delta(\overline{\gamma} - t)\right)^2}{\mathbb{E}^R_\gamma[\gamma|\gamma \le t]^2} = \frac{\left(\overline{\alpha}_1 - \underline{\alpha}_1 + 2x_\delta\overline{\gamma} - 4x_\delta t\right) \cdot 2\left(\mathbb{E}^R_1[\alpha_1] - x_\delta\overline{\gamma}\right)}{\mathbb{E}^R_\gamma[\gamma|\gamma \le t]^2},$$

using $\overline{\alpha}_1 + \underline{\alpha}_1 = 2\mathbb{E}^R_1[\alpha_1]$. The first factor in the numerator is at least $\overline{\alpha}_1 - \underline{\alpha}_1 - 2x_\delta\overline{\gamma} \ge 0$, with strict inequality for $t < \overline{\gamma}$. Since $\mathbb{E}^R_\gamma[\gamma] = \frac{\overline{\gamma}}{2}$, the second factor is positive if and only if $x_\delta < \frac{\mathbb{E}^R_1[\alpha_1]}{\overline{\gamma}} = \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}$. Hence $\frac{\partial V(\delta)}{\partial x_\delta} > 0$ for all $x_\delta < \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}$, so $V(\delta, x_\delta)$ is strictly increasing on $\left[0, \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right]$, and every $x_\delta \in \arg\max_{x_\delta \in [0,\overline{x}]} V(\delta, x_\delta)$ satisfies $x_\delta \ge \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}$.

Finally, if S is high-curvature and $x^*_\delta = 0$, then $x^*_\delta < \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}$ immediately. If S is low-curvature, the proof of Theorem 1 shows that x^*_δ is bounded above by a term of order $\frac{1}{\overline{\sigma}^2}$, so $x^*_\delta < \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}$ for $\overline{\sigma}$ sufficiently large. In either case, every maximizer of $V(\delta, \cdot)$ exceeds x^*_δ. □

Proposition 6. *Fix a sender type δ and x_δ. There exists σ_δ such that $W(\sigma_\delta)$ is strictly decreasing in σ_δ for $\sigma_\delta < \sigma^R_\delta$ and strictly increasing for $\sigma_\delta > \sigma^R_\delta$ at $x^*(f_S)$.*

Proof. Given f_S and the other parameters, we have:

$$\begin{aligned}\mathrm{Var}^R_{f_R|f_S}\left(f_R(x^*(f_S))\right) &= \mathrm{Var}^R_{f_R|f_S}\left(f_S(x^*(f_S)) + (\gamma - \delta)(\sigma^2_\delta - (x^*(f_S) - x_\delta)^2)\right) \\ &= \left(\sigma^2_\delta - (x^*(f_S) - x_\delta)^2\right)^2 \mathrm{Var}^R_\gamma(\gamma|f_S, \theta = \delta, x_\delta),\end{aligned}$$

where we note that information about γ depends on both f_S as well as x_δ. Because the sender's model fixes curvature at a single value, R remains uncertain about γ after seeing f_S, and this variance is > 0. Hence, our problem is to choose σ_δ to minimize

$$\mathbb{E}^R_{f_S|\delta}\left[\left(\sigma^2_\delta - (x^*(f_S) - x_\delta)^2\right)^2 \mathrm{Var}^R_\gamma(\gamma|f_S, \theta = \delta, x_\delta)\right].$$

Since neither $x^*(f_S)$ nor $\mathrm{Var}^R_\gamma(\gamma|f_S, \theta = \delta, x_\delta)$ depends on σ_δ, this function is quadratic in σ^2_δ and minimized at

$$\sigma^2_\delta = \frac{\mathbb{E}^R_{f_S|\delta}\left[(x^*(f_S) - x_\delta)^2 \mathrm{Var}^R_\gamma(\gamma|f_S, \theta = \delta, x_\delta)\right]}{\mathbb{E}^R_{f_S|\delta}\left[\mathrm{Var}^R_\gamma(\gamma|f_S, \theta = \delta, x_\delta)\right]}.$$

The terms in the numerator represent the ex-ante expectation of the squared difference between x_δ and the eventual choice $x^*(f_S)$ multiplied by the variance of γ. The term in the denominator is the expected variance of γ. Because there is uncertainty about $x^*(f_S)$, the numerator is strictly positive, and because $x^*(f_S), x_\delta \in [0, \overline{x}]$, this ratio is at most $\overline{x}^2 < \overline{\sigma}^2$, which gives the result. □

Proposition 7. *Suppose R chooses $x^*(f_S)$ to maximize (4), and let σ^R_δ be the choice of σ_δ that minimizes the expected variance of outcomes. Then:*

1. *A low-curvature sender chooses* $\sigma_\delta^* > \sigma_\delta^R$ *and a high-curvature sender chooses* $\sigma_\delta^* < \sigma_\delta^R$.

2. *As* $\psi \to 1$, *if* $x_\delta \pm \sigma_\delta \in [0, \overline{x}]$, *then* $x^*(f_S) \to x_\delta - \sigma_\delta$ *or* $x_\delta + \sigma_\delta$.

Proof. Given f_S and δ and using the previous result, R's problem is to maximize

$$\max_x (1-\psi)\big(f_S(x) + \Delta(f_S, x_\delta)(\sigma_\delta^2 - (x - x_\delta)^2)\big) - \psi\big(\sigma_\delta^2 - (x - x_\delta)^2\big)^2 \mathrm{Var}_\gamma^R(\gamma | f_S, \theta = \delta, x_\delta),$$

which S then integrates out over the plausible values of f_S. Denote this maximizer by $x^\dagger(f_S)$. Since neither $\Delta(f_S, x_\delta)$ nor $\mathrm{Var}_\gamma^R(\gamma | f_S, \theta = \delta, x_\delta)$ depends on σ_δ, by the Envelope Theorem the derivative with respect to σ_δ^2 is

$$\mathbb{E}_{f_S|\delta}^R\big[(1-\psi)\Delta(f_S, x_\delta) - 2\psi\big(\sigma_\delta^2 - (x^\dagger(f_S) - x_\delta)^2\big)\mathrm{Var}_\gamma^R(\gamma | f_S, \theta = \delta, x_\delta)\big].$$

Recall from above that the expectation of $\big(\sigma_\delta^2 - (x^\dagger(f_S) - x_\delta)^2\big)\mathrm{Var}_\gamma^R(\gamma | f_S, \theta = \delta, x_\delta)$ is negative below σ_δ^R and positive above. For a low-curvature sender, $\mathbb{E}_{f_S|\delta}^R[\Delta(f_S, x_\delta)] > 0$, meaning the derivative is positive at and below σ_δ^R, and thus that his choice of σ_δ is higher. The opposite holds for a high-curvature sender.

The second result follows by noting that as $\psi \to 1$, R is simply choosing $x^\dagger(f_S)$ to minimize variance. When $x_\delta \pm \sigma_\delta \in [0, \overline{x}]$, this variance can be set to zero by choosing $x^\dagger(f_S) = x_\delta \pm \sigma_\delta$, which gives the result. □

Theorem 3. *There exists a function* $\pi(\delta)$ *such that* T^* *is given by a continuous distribution* t *with*

$$t(\delta) = p_\gamma^R(\delta)\pi(\delta). \tag{T^*}$$

In particular, there exist thresholds $0 \le \underline{\delta}_0 < \mathbb{E}_\gamma^R[\gamma] < \overline{\delta}_0 \le \overline{\gamma}$ *such that* $\pi(\delta)$ *is strictly decreasing below* $\underline{\delta}_0$, *strictly increasing above* $\overline{\delta}_0$, *and equal to* 0 *on* $[\underline{\delta}_0, \overline{\delta}_0]$.

Proof. Consider a sender model $f_S = \beta_0 + \beta_1 x - \delta x^2$. First, we show that $\Theta = \Gamma$. For each $\delta \in \Gamma$, a can create a type δ. This type samples as in Theorem 1, so by Proposition 2, $\max_x f_S(x) \le \max_x f_S(x) + \Pr(\theta = \delta | f_S)\Delta(f_S, x_\delta)\big(\sigma_\delta^2 - (x - x_\delta)^2\big)$. Thus, conditional on seeing a model of the form $\beta_0 + \beta_1 x - \delta x^2$, R believes that $f_S(x)$ is either the true model or the model from the type δ sender. a's utility conditional on providing this model increases from the existence of the type δ sender whenever $\delta \neq \mathbb{E}_\gamma^R[\gamma]$, and the gain is increasing in the posterior probability that R places on the type δ sender. Hence, $\Theta = \Gamma$.

We now determine the distribution of types T. Fix a model $\beta_0 + \beta_1 x - \delta x^2$. First, for a distribution $T \in \mathcal{P}([0, \overline{\gamma}])$ (the set of measures over $[0, \overline{\gamma}]$), we write as $q(T, \delta)$ the posterior probability that this model came from a type δ sender.

$$q(T, \delta) = \begin{cases} \frac{(1-p_a)t(\delta)}{(1-p_a)t(\delta) + p_a p_\gamma^R(\delta)} & T \text{ admits density } t(\delta) \\ 1 & T(\{\delta\}) \text{ has a mass point} \end{cases}$$

By Theorem 1, high-curvature types sample at a corner, and low-curvature types sample at $x^*_\delta \to 0$ as $\overline{\sigma}$ grows. At $x_\delta = 0$, the slope a simple sender reveals is $f'_R(0) = \alpha_1$, which is uninformative about γ, so $\mathbb{E}^R_\gamma[\gamma|f_S, x_\delta, \theta = \delta] = \mathbb{E}^R_\gamma[\gamma]$ and $\Delta(f_S, x_\delta) = \mathbb{E}^R_\gamma[\gamma] - \delta$. We therefore evaluate the simple type's posterior at $\mathbb{E}^R_\gamma[\gamma]$; this is exact at $x^*_\delta = 0$, and otherwise holds up to terms of order $\frac{\overline{x}\,\overline{\gamma}}{\overline{\alpha}_1 - \underline{\alpha}_1}$, as shown in the proof of Theorem 1. R's posterior mean of γ is then

$$\mathbb{E}^R_\gamma[\gamma|f_S] = q(T,\delta)\mathbb{E}^R_\gamma[\gamma] + \big(1 - q(T,\delta)\big)\delta.$$

Given a model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$, the receiver's maximal expected value is

$$f_S(x^*) + q(T,\delta)\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\big(\sigma^2_\delta - (x^* - x_\delta)^2\big)$$

where x^* is a function of β_1, δ, and $\mathbb{E}^R_\gamma[\gamma|f_S]$, and σ_δ and x_δ are determined as in Theorem 1.

Our objective can thus be rewritten as:

$$\max_{T \in \mathcal{P}([0,\overline{\gamma}])} \int_{\mathbb{R}} \int_{\mathbb{R}} \int_0^{\overline{\gamma}} f_S(x^*) + q(T,\delta)\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\big(\sigma^2_\delta - (x^* - x_\delta)^2\big) dP^a_\delta(\delta) dP^a_1(\beta_1) dP^a_0(\beta_0),$$

Note that because P^a_δ is continuous, there are no gains to placing a positive mass on any singleton δ value. Hence, our problem boils down to

$$\max_{t(\delta)} \int_{\mathbb{R}} \int_{\mathbb{R}} \int_0^{\overline{\gamma}} f_S(x^*) + q(T,\delta)\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\big(\sigma^2_\delta - (x^* - x_\delta)^2\big) dP^a_\delta(\delta) dP^a_1(\beta_1) dP^a_0(\beta_0)$$
$$\text{s.t. } t(\delta) \geq 0 \text{ and } \int_0^{\overline{\gamma}} t(\delta) d\delta = 1.$$

Using a calculus of variations approach, we can write a Lagrangian as:

$$\mathcal{L}(t,\lambda) = \int_{\mathbb{R}} \int_{\mathbb{R}} \int_0^{\overline{\gamma}} \Big[f_S(x^*) + q(T,\delta)\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\big(\sigma^2_\delta - (x^* - x_\delta)^2\big)\Big] p^R_\gamma(\delta)\, d\delta\, dP^a_1(\beta_1) dP^a_0(\beta_0) - \lambda\left(\int_0^{\overline{\gamma}} t(\delta)\, d\delta - 1 \right).$$

Using the Envelope Theorem (since x^* maximizes the receiver's objective), we obtain the following first-order condition.

$$\mathbb{E}^a_{0,1}\left[\frac{(1-p_a)\big((1-p_a)t(\delta) + p_a p^R_\gamma(\delta)\big) - (1-p_a)^2 t(\delta)}{((1-p_a)t(\delta) + p_a p^R_\gamma(\delta))^2} \big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\big(\sigma^2_\delta - (x^* - x_\delta)^2\big) p^R_\gamma(\delta) \right] - \lambda$$
$$= \frac{(1-p_a)p_a p^R_\gamma(\delta)}{((1-p_a)t(\delta) + p_a p^R_\gamma(\delta))^2} p^R_\gamma(\delta)\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\big(\sigma^2_\delta - \mathbb{E}^a_1\big[(x^* - x_\delta)^2\big]\big) - \lambda,$$

where we note that x^* depends on β_1 and δ. Now, define

$$P = p_a p^R_\gamma(\delta) \qquad R = (1-p_a)t(\delta).$$

Recall that $x^* = x_\delta + \frac{\beta_1 - 2\delta x_\delta}{2\mathbb{E}^R_\gamma[\gamma|f_S]}$ and $\mathbb{E}^R_\gamma[\gamma|f_S] = \frac{P\delta + R\mathbb{E}^R_\gamma[\gamma]}{P+R}$. We then have

$$\begin{aligned}\frac{(1-p_a)P^2}{p_a(P+R)^2}\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma_\delta^2 - \mathbb{E}^a_1\big[(x^*-x_\delta)^2\big]\big) &= \frac{(1-p_a)P^2}{p_a(P+R)^2}\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma_\delta^2 - \mathbb{E}^a_1\big[\big(\frac{\beta_1-2\delta x_\delta}{2\mathbb{E}^R_\gamma[\gamma|f_S]}\big)^2\big]\big)\\ &= \frac{(1-p_a)P^2}{p_a(P+R)^2}\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma_\delta^2 - \frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{\frac{4(P\delta+R\mathbb{E}^R_\gamma[\gamma])^2}{(P+R)^2}}\big)\\ &= \frac{(1-p_a)}{p_a}P^2\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(P+R)^2} - \frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{4(P\delta+R\mathbb{E}^R_\gamma[\gamma])^2}\big).\end{aligned}$$

Next, define $\pi(\delta) := \frac{t(\delta)}{p^R_\gamma(\delta)}$ and $c = \frac{1-p_a}{p_a}$. In particular, $\frac{R}{P} = c\pi(\delta)$. This definition gives:

$$\begin{aligned}&\frac{(1-p_a)}{p_a}P^2\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(P+R)^2} - \frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{4(P\delta+R\mathbb{E}^R_\gamma[\gamma])^2}\big)\\ &= c\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{4(\delta+c\pi(\delta)\mathbb{E}^R_\gamma[\gamma])^2}\big).\end{aligned}$$

Finally, we verify the second-order condition; because $\pi(\delta) = \frac{t(\delta)}{p^R_\gamma(\delta)}$, differentiating with respect to t yields:

$$c^2 p^R_\gamma(\delta)\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(-\frac{2\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}^R_\gamma[\gamma]\frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{2(\delta+c\pi(\delta)\mathbb{E}^R_\gamma[\gamma])^3}\big).$$

For $\delta < \mathbb{E}^R_\gamma[\gamma]$, we note that $\sigma_\delta = \overline{\sigma}$, meaning $-\frac{2\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}^R_\gamma[\gamma]\frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{2(\delta+c\pi(\delta)\mathbb{E}^R_\gamma[\gamma])^3} < 0$ for $\overline{\sigma}$ sufficiently large. For $\delta > \mathbb{E}^R_\gamma[\gamma]$, $\sigma_\delta = \underline{\sigma} \to 0$, meaning $-\frac{2\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}^R_\gamma[\gamma]\frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{2(\delta+c\pi(\delta)\mathbb{E}^R_\gamma[\gamma])^3} > 0$, and thus that the whole expression is negative. Hence, the original objective is concave in t, meaning the first-order condition identifying a solution is given by

$$c\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{4(\delta+c\pi(\delta)\mathbb{E}^R_\gamma[\gamma])^2}\big) = \lambda.$$

Note that the left-hand side is strictly positive except at $\delta = \mathbb{E}^R_\gamma[\gamma]$, when it is equal to 0. Indeed, the second factor equals $\big(\sigma_\delta^2 - \mathbb{E}^a_1[(x^*-x_\delta)^2]\big)/(1+c\pi(\delta))^2$. For $\delta < \mathbb{E}^R_\gamma[\gamma]$, $\sigma_\delta = \overline{\sigma} \geq \overline{x} \geq |x^* - x_\delta|$, so it is positive, as is $\mathbb{E}^R_\gamma[\gamma] - \delta$. For $\delta > \mathbb{E}^R_\gamma[\gamma]$, $\sigma_\delta = \underline{\sigma} \to 0$, so it is negative, as is $\mathbb{E}^R_\gamma[\gamma] - \delta$. We define:

$$\xi(\delta,\pi) := c\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(1+c\pi)^2} - \frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{4(\delta+c\pi\mathbb{E}^R_\gamma[\gamma])^2}\big).$$

By the Implicit Function Theorem, we have $\pi'(\delta) = -\frac{\xi_\delta}{\xi_\pi}$. But note that $\xi_\pi < 0$ (since the objective was concave in t, and $t = p^R_\gamma(\delta)\cdot\pi$), meaning that the sign of π' is just the sign of ξ_δ.

Now, consider $\delta < \mathbb{E}^R_\gamma[\gamma]$. By Theorem 1, these senders sample so that $\sigma_\delta = \overline{\sigma}$ and $x^*_\delta \to 0$ as $\overline{\sigma}$

grows, so that $\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta)^2] \to \mathbb{E}_1^a[\beta_1^2]$ and

$$\xi(\delta,\pi) = c\big(\mathbb{E}_\gamma^R[\gamma] - \delta\big)\Big(\frac{\overline{\sigma}^2}{(1+c\pi)^2} - \frac{\mathbb{E}_1^a[\beta_1^2]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}\Big).$$

The derivative of this expression with respect to δ is < 0 if and only if

$$\frac{\overline{\sigma}^2}{(1+c\pi)^2} > \frac{\mathbb{E}_1^a[\beta_1^2]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}\left(1 + \frac{2(\mathbb{E}_\gamma^R[\gamma] - \delta)}{\delta + c\pi\mathbb{E}_\gamma^R[\gamma]}\right),$$

which holds for $\overline{\sigma}$ sufficiently large. Thus, $\xi(\delta,\pi)$ is positive and decreasing in δ for $\delta < \mathbb{E}_\gamma^R[\gamma]$, and equals 0 at $\delta = \mathbb{E}_\gamma^R[\gamma]$, so $\pi(\delta)$ is decreasing in δ for $\delta < \mathbb{E}_\gamma^R[\gamma]$.

Now, we consider $\delta > \mathbb{E}_\gamma^R[\gamma]$. In this case, we have $\sigma_\delta = \underline{\sigma} \to 0$, and by Theorem 1, $x_\delta^* \in \{0, \overline{x}\}$: high-curvature senders sample at whichever edge of X they expect f_S to be steepest, i.e. $x_\delta^* = \arg\max_{x\in\{0,\overline{x}\}} \mathbb{E}_1^a[(\beta_1 - 2\delta x)^2]$. In this case, we have

$$\xi(\delta,\pi) \approx c\big(\delta - \mathbb{E}_\gamma^R[\gamma]\big)\frac{\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta^*)^2]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}.$$

By the Envelope Theorem, $\frac{\partial}{\partial\delta}\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta^*)^2] = -4x_\delta^*\,\mathbb{E}_1^a[\beta_1 - 2\delta x_\delta^*]$, which is zero at $x_\delta^* = 0$ and nonnegative at $x_\delta^* = \overline{x}$, since $\overline{x}$ is chosen over 0 only if $\mathbb{E}_1^a[\beta_1] \le \delta\overline{x}$. Hence, the derivative of ξ with respect to δ has the sign of at least

$$\big(\delta + c\pi\mathbb{E}_\gamma^R[\gamma]\big) - 2\big(\delta - \mathbb{E}_\gamma^R[\gamma]\big) = (2 + c\pi)\mathbb{E}_\gamma^R[\gamma] - \delta,$$

which is positive for all $\delta < \overline{\gamma}$, since symmetry of p_γ^R gives $\overline{\gamma} = 2\mathbb{E}_\gamma^R[\gamma]$. Thus, $\xi(\delta,\pi)$ is positive and increasing in δ for $\delta > \mathbb{E}_\gamma^R[\gamma]$, so $\pi(\delta)$ is increasing in δ for $\delta > \mathbb{E}_\gamma^R[\gamma]$. In sum, the marginal gain $\xi(\delta,\pi)$ from a type δ sender is decreasing to 0 at $\delta = \mathbb{E}_\gamma^R[\gamma]$ and increasing thereafter.

Thus, whenever $\pi(\delta) > 0$, it is strictly decreasing for $\delta < \mathbb{E}_\gamma^R[\gamma]$ and strictly increasing for $\delta > \mathbb{E}_\gamma^R[\gamma]$. To see that $\pi(\delta) = 0$ on an interior interval, we simply note that for every $\lambda > 0$, there exists an interval of δ values containing $\mathbb{E}_\gamma^R[\gamma]$ such that

$$c\big(\mathbb{E}_\gamma^R[\gamma] - \delta\big)\Big(\frac{\sigma_\delta^2}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^2}\Big) < \lambda$$

at $\pi = 0$ (and, by concavity, for all $\pi \ge 0$). □

Proposition 8. *The polarization of simple senders is increasing in p_a. That is, $\underline{\delta}_0$ is decreasing in p_a and $\overline{\delta}_0$ is increasing in p_a. As $p_a \to 1$, $t(\delta)$ converges to a distribution that either places all mass only on $\delta = 0$, $\delta = \overline{\gamma}$, or both. As $p_a \to 0$, $t(\delta)$ converges to a distribution that is strictly decreasing on $[0, \mathbb{E}_\gamma^R[\gamma])$, equal to 0 at $\mathbb{E}_\gamma^R[\gamma]$, and strictly increasing on $(\mathbb{E}_\gamma^R[\gamma], \overline{\gamma}]$.*

Proof. We recall our initial first-order condition implicitly defining $\pi(\delta)$

$$c\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\Big(\frac{\sigma^2_\delta}{(1+c\pi(\delta))^2}-\frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{4(\delta+c\pi(\delta)\mathbb{E}^R_\gamma[\gamma])^2}\Big)=\lambda,$$

where the left-hand side is decreasing in π, decreasing in δ for $\delta<\mathbb{E}^R_\gamma[\gamma]$, and increasing in δ for $\delta>\mathbb{E}^R_\gamma[\gamma]$. We also recall that $c:=\frac{1-p_a}{p_a}$, and that as $c\to 0$, $p_a\to 1$, and that as $c\to\infty$, $p_a\to 0$.

Recall also that $\overline{\delta}_0$ and $\underline{\delta}_0$ are the points at which $\pi(\delta)=0$ and are, in turn, functions of c. By the constraint that $t(\delta)$ is a density and by writing $\pi(\delta,c)$ as a function of c, we have

$$\int_0^{\overline{\gamma}} t(\delta)d\delta=\int_0^{\underline{\delta}_0(c)}\pi(\delta,c)p^R_\gamma(\delta)d\delta+\int_{\overline{\delta}_0(c)}^{\overline{\gamma}}\pi(\delta,c)p^R_\gamma(\delta)d\delta=1$$

$$\pi(\overline{\delta}_0(c),c)p^R_\gamma(\overline{\delta}_0(c))\overline{\delta}_0'(c)-\pi(\underline{\delta}_0(c),c)p^R_\gamma(\underline{\delta}_0(c))\underline{\delta}_0'(c)+\int_0^{\underline{\delta}_0(c)}\pi_c(\delta,c)p^R_\gamma(\delta)+\int_{\overline{\delta}_0(c)}^{\overline{\gamma}}\pi_c(\delta,c)p^R_\gamma(\delta)$$
$$=\int_0^{\overline{\gamma}}\pi_c(\delta,c)p^R_\gamma(\delta)d\delta=0,$$

where we use the fact that $\pi(\overline{\delta}_0(c),c)=\pi(\underline{\delta}_0(c),c)=0$.

Next, note that, fixing δ, the first-order condition can be rearranged as

$$\Xi(c\pi,\delta):=\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\Big(\frac{\sigma^2_\delta}{(1+c\pi)^2}-\frac{\mathbb{E}^a_1[(\beta_1-2\delta x_\delta)^2]}{4(\delta+c\pi\mathbb{E}^R_\gamma[\gamma])^2}\Big)=\frac{\lambda(c)}{c}$$

The FOC then gives

$$\Xi_{(c\pi)}(c\pi_c+\pi)=\frac{\lambda'(c)}{c}-\frac{\lambda(c)}{c^2}\implies c\pi_c=\frac{\frac{\lambda'(c)}{c}-\frac{\lambda(c)}{c^2}}{\Xi_{(c\pi)}}-\pi$$

But then, using the boundary condition, we have

$$c\int_0^{\overline{\gamma}}\pi_c(\delta,c)p^R_\gamma(\delta)d\delta=\int_0^{\overline{\gamma}}\frac{\frac{\lambda'(c)}{c}-\frac{\lambda(c)}{c^2}}{\Xi_{(c\pi)}}p^R_\gamma(\delta)-\pi(\delta,c)p^R_\gamma(\delta)d\delta=0$$
$$\implies\left(\lambda'(c)-\frac{\lambda(c)}{c}\right)\int_0^{\overline{\gamma}}\frac{1}{\Xi_{(c\pi)}(c\pi,\delta)}p^R_\gamma(\delta)d\delta=c$$
$$\implies\frac{\lambda'(c)}{c}-\frac{\lambda(c)}{c^2}=\frac{1}{\int_0^{\overline{\gamma}}\frac{1}{\Xi_{(c\pi)}(c\pi,\delta)}p^R_\gamma(\delta)d\delta}<0$$

since $\Xi_{(c\pi)}<0$ (i.e. Ξ is decreasing in $c\pi$, from the proof of the previous Theorem).

Now, consider $\Xi(0,\underline{\delta}_0(c))$, which implicitly defines $\underline{\delta}_0(c)$ as

$$\Xi(0,\underline{\delta}_0(c)):=\big(\mathbb{E}^R_\gamma[\gamma]-\underline{\delta}_0(c)\big)\Big(\sigma^2_\delta-\frac{\mathbb{E}^a_1[(\beta_1-2\underline{\delta}_0(c)x_\delta)^2]}{4(\underline{\delta}_0(c))^2}\Big)=\frac{\lambda(c)}{c}.$$

In this case, we have

$$\Xi_\delta(0, \underline{\delta}_0(c))\underline{\delta}_0'(c) = \frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2} < 0.$$

But recall from the proof of the previous result that $\Xi_\delta < 0$ for $\delta < \mathbb{E}^R_\gamma[\gamma]$. This means that $\underline{\delta}_0(c)$ is increasing in c, i.e. is increasing in $\frac{1-p_a}{p_a}$. Hence, as p_a increases, $\underline{\delta}_0$ decreases. An analogous proof shows that as p_a increases, $\overline{\delta}_0$ increases. That is, the "polarization" induced by π is increasing in p_a.

Finally, we investigate the limiting cases as $p_a \to 0$ ($c \to \infty$) and $p_a \to 1$ ($c \to 0$). We begin with $c \to \infty$, and claim that $\underline{\delta}_0 \to \mathbb{E}^R_\gamma[\gamma]$. First, by our work above $\frac{\lambda(c)}{c}$ is decreasing in c, meaning it converges to a limit. We claim $\lim_{c\to\infty} \frac{\lambda(c)}{c} = 0$. Suppose by contradiction that this did not hold, and instead it converged to some $k > 0$. This would mean that there exists $\overline{c\pi} > 0$ such that $c\pi(\delta) \leq \overline{c\pi}$ for all c and $\delta \in [0, \overline{\gamma}] - \lim_{c\to\infty}[\underline{\delta}_0(c), \overline{\delta}_0(c)]$. But then, for all c we would have

$$\int_0^{\overline{\gamma}} \pi(\delta, c) p^R_\gamma(\delta) d\delta \leq \int_0^{\overline{\gamma}} \overline{c\pi} p^R_\gamma(\delta) d\delta = \frac{\overline{c\pi}}{c},$$

which approaches 0 as $c \to \infty$, contradicting that $\int_0^{\overline{\gamma}} \pi(\delta, c) p^R_\gamma(\delta) d\delta = 1$. Hence, we have that $\lim_{c\to\infty} \frac{\lambda(c)}{c} = 0$, meaning $\lim_{c\to\infty} \underline{\delta}_0(c) = \lim_{c\to\infty} \overline{\delta}_0(c) = \mathbb{E}^R_\gamma[\gamma]$. Moreover, we note that even for c arbitrarily large, $\pi(\mathbb{E}^R_\gamma[\gamma]) = 0$, with $\pi'(\delta) < 0$ below $\mathbb{E}^R_\gamma[\gamma]$ and $\pi'(\delta) > 0$ above $\mathbb{E}^R_\gamma[\gamma]$ since $\Xi(c\pi, \delta)$ is decreasing in δ and increasing in δ on each range, respectively.

Finally, we investigate the case where $c \to 0$. We claim that as $c \to 0$, $\underline{\delta}_0 \to 0$ and $\overline{\delta}_0 \to \overline{\gamma}$ (and hence t converges to a discrete distribution that either concentrates mass on 0 or $\overline{\gamma}$ or both). In particular, note that for all c, we have

$$c = \int_0^{\overline{\gamma}} c\pi(\delta, c) p^R_\gamma(\delta) d\delta,$$

which means that as $c \to 0$, we necessarily must have $c\pi(\delta, c) \to 0$ as well. This can happen only if

$$\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\Big(\sigma^2_\delta - \frac{\mathbb{E}^a_1[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta)^2}\Big) \leq \lim_{c\to 0} \frac{\lambda(c)}{c},$$

i.e. either the multiplier constraint must hold with equality or fail to hold (generating a corner solution at $\pi(\delta, c) = 0$). Since this must hold pointwise for each δ for c sufficiently small, we thus have that $\underline{\delta}_0 \to 0$ and $\overline{\delta}_0 \to \overline{\gamma}$, giving the result. □

B Extensions

This appendix relaxes two assumptions from the main text. Sections B.1 allow sender types to be an *interval*. Appendix B.3 allows tradeoffs to be non-quadratic.

B.1 Interval Type Spaces

In the main text, a type δ sender's model fixes curvature at a single value δ. However, in reality, senders and receivers alike may consider multiple tradeoffs in different ways.

- Senders could estimate *some* tradeoffs and assume the worst-case or neglect others. In the example, the consultant may thoroughly estimate how carbon taxes affect energy prices, but assume a worst-case scenario for their effects on political backlash.
- Senders could *calibrate* the value of a tradeoff at a fixed parameter, without estimating it on the real data. For example, the consultant may calibrate the effect of carbon taxes on energy prices using an elasticity from the literature, rather than organically estimating it on data.

Accordingly, I now assume that a sender's type θ is a *subset* of Γ, with the following structure.

For each $\theta \in \Theta$, $\theta \subseteq \Gamma$ is a closed, nondegenerate interval. For each $\theta \neq \theta' \in \Theta$, $\theta \cap \theta' = \varnothing$. In turn, I now assume a type θ sender provides models in the set

$$F_\theta := \{\beta_0 + \beta_1 x - \delta x^2 : (\beta_0, \beta_1) \in \mathbb{R}^2, \delta \in \theta\}.$$

Thus, each simple type corresponds to an interval within $[0, \overline{\gamma}]$. I initially assume type intervals do not intersect for expositional purposes.[15] We will denote $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$. Accordingly, type spaces have the following interpretations.

- If $\overline{\delta}_\theta < \overline{\gamma}$, then a type θ S is *neglecting at least some tradeoffs*. Intuitively, these senders do not allow for the case where γ is close to $\overline{\gamma}$.
- If $\underline{\delta}_\theta > 0$, then a type θ S is *exaggerating at least some tradeoffs*. These senders do not allow for the case where γ is close to 0.

In addition to priors over β_0 and β_1, a type θ sender now has a prior over δ given by $\delta \sim P^\theta_\delta$ with support on $\theta \subseteq \Gamma$. We denote by $P^\theta_{f_S}$ a type θ sender's prior over models f_S.

Learning with Interval Types The characterization of $\phi(f_S; \mu_\theta)$ — the set of models R considers in addition to f_S — is the same as above. However, we note that if θ is non-degenerate, we may also have a moment condition corresponding to δ:

$$\mathbb{E}_{\mu_\theta}\left[x^2\left(f_R(x) - \beta_0 - \beta_1 x + \delta x^2\right)\right] \tag{6}$$

This equation may be zero or nonzero, depending on S's type. With interval types, $\Delta(f_s, x_\delta)$ is defined as in the main text. In particular, if $\delta \in \theta^\circ$, we necessarily have that (6) is equal to 0, and that f_S is the true model. In this case, $\Delta(f_S, x_\delta) = 0$. However, if $\delta \in \partial\theta$ (the boundary of θ), then the receiver may entertain additional models.

[15] The case where they do is of particular interest, and I study it in Appendix **??**.

The statement of Proposition 1 is identical in the case with interval types, and the expression for R's expected utility in Section 3 continues to hold. This generalizes the intuitions in the singleton type model. If $\delta = \overline{\delta}_\theta < \overline{\gamma}$, then $\Delta(f_s, x_\delta) > 0$: R considers models that *only* have higher curvature than the one S has provided because he has neglected some tradeoffs. If $\delta = \underline{\delta}_\theta > 0$, then $\Delta(f_s, x_\delta) < 0$: R considers models that *only* have lower curvature than the one S has provided because he has exaggerated some tradeoffs.

Strategic Sampling with Interval Types Let x^*_θ denote the mean of the underlying data that a type θ sender samples and σ^{*2}_θ its variance.

Proposition 9. *Suppose S is a type θ simple sender.*

1. *If $\mathbb{E}^\theta_{f_S}[\Delta(f_S, x^*_\theta)] > 0$, he collects* dispersed *data about extreme choices. μ^*_θ is a bimodal distribution that places mass $\frac{1}{2}$ on each of $x^*_\theta - \overline{\sigma}$ and $x^*_\theta + \overline{\sigma}$.*
2. *If $\mathbb{E}^\theta_{f_S}[\Delta(f_S, x^*_\theta)] < 0$, μ^*_θ is a distribution with variance $\underline{\sigma}^2$.*

The proposition characterizes the spread of the data underlying a simple sender's models and the logic and proof are identical to the singleton case.

The multidimensional results of Section 3 extend in the same way when each type θ is a closed and convex subset of Γ, distinct types are disjoint, a type θ sender has a prior P^θ_δ over $\delta \in \theta$, and, if γ is not an extreme point of Γ, $P^\theta_\delta(\partial\theta \cap \Gamma^\circ) > 0$.

B.2 Designing Interval Type Spaces

In the design problem of Section 5, suppose that a can choose any type space $\Theta \subseteq \mathcal{P}([0, \overline{\gamma}])$ such that for all $\theta \in \Theta$, θ is a closed interval.

Proposition 10. *The optimal type space contains no non-degenerate intervals, and is the collection of singletons on $[0, \overline{\gamma}] = \Gamma$: $\Theta = \big\{\{\delta\} : \delta \in \Gamma\big\}$.*

Proof. Suppose by contradiction that there existed a nondegenerate interval of type $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$. Suppose $\theta^\circ \neq \varnothing$ and consider $\delta \in \theta^\circ$. Conditional on seeing this model, the receiver believes the model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ is the true model.

Now, for each $\delta \in \theta^\circ$, as before, a can create a type $\{\delta\}$. This type strategically samples by setting σ such that $\max_x f_S(x) \leq \max_x f_S(x) + (\mathbb{E}[\gamma|\delta])(\sigma^2 - (x - x_\theta)^2)$. Thus, conditional on seeing a model of the form $\beta_0 + \beta_1 x - \delta x^2$, R believes that $f_S(x)$ is either the true model or the model from the $\{\delta\}$ type. Let $q(\delta|\beta_0, \beta_1, T)$ be the conditional probability of the type $\{\delta\}$ sender relative to the original type θ sender. Let $\overline{\gamma}_{\{\delta\}}$ be the receiver's expectation of γ if $\delta \geq \gamma$ given f_S and $\underline{\gamma}_{\{\delta\}}$ likewise

if $\delta < \gamma$. Then, this generates a difference on the order of

$$\int_{\underline{\delta}_\theta}^{\overline{\delta}_\theta} \max_x f_S(x) + q(\delta|\beta_0,\beta_1,T)\big(\overline{\gamma}_{\{\delta\}} - \delta\big)\big(\overline{\sigma}^2 - (x-x_\theta)^2\big) - \max_x f_S(x) dP_\delta^a(\delta \le \mathbb{E}_\gamma^R[\gamma]|\delta \in \theta^\circ)$$
$$+ \int_{\underline{\delta}_\theta}^{\overline{\delta}_\theta} \max_x f_S(x) + q(\delta|\beta_0,\beta_1,T)\big(\underline{\gamma}_{\{\delta\}} - \delta\big)\big(\underline{\sigma}^2 - (x-x_\theta)^2\big) - \max_x f_S(x) dP_\delta^a(\delta > \mathbb{E}_\gamma^R[\gamma]|\delta \in \theta^\circ)$$

The first line considers any simple senders who sample dispersed data. The second line considers any simple senders who sample local data. In both cases, the gain is positive and is increasing in $q(\delta|\beta_0,\beta_1,T)$; i.e., the gain is higher the more likely R is to think that $f_S(x)$ came from type $\{\delta\}$.

We are left to consider the boundary types of θ: $\overline{\delta}_\theta$ and $\underline{\delta}_\theta$. These may generate higher conditional expected utility than a singleton type, but because p_γ^R is continuous, they occur with measure 0. By contrast, replacing the non-degenerate interval with a continuum of types generates a non-measure-zero positive gain. □

B.3 Non-Quadratic Tradeoffs

While the main model utilizes quadratic utility to express tradeoffs, in this subsection, I probe the robustness of many of the main results to *non-quadratic* utility. In particular, I show an analogue of Theorem 1: if $\overline{\sigma}$ is sufficiently large and S is low-curvature, sampling dispersed data dominates sampling local data. If S is high-curvature, sampling local data dominates dispersed data.

General Loss Function The following two assumptions consider two generalizations of functional forms for the receiver.

Assumption 1. *The receiver's models and a type θ sender's models are given by:*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma\ell(x - x_\theta) \tag{7}$$
$$f_S(x) = \beta_0 + \beta_1 x - \delta\ell(x - x_\theta) \tag{8}$$

where $\ell : \mathbb{R} \to \mathbb{R}^+$ is strictly convex and differentiable. Moreover, $\ell(0) = 0$, $\ell'(0) = 0$, and ℓ is symmetric around $x = 0$.

One way to interpret this functional form is to think of a sender's or receiver's model as a *Taylor expansion* around a point x_θ. This Taylor expansion includes a constant component, a linear term, and a concave $\mathcal{O}(2)$ term centered at x_θ. While the convex loss $\ell(x - x_\theta)$ is symmetric around x_θ, the assumption does allow for f_R or f_S themselves to be asymmetric. This assumption subsumes the case where $\ell(x) = x^2$.

The case where this convex loss itself may be locally asymmetric is also of interest. Thus, I also consider the following alternative assumption.

Assumption 2. *The receiver's models and a type θ sender's models are given by:*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma m(x) \tag{9}$$

$$f_S(x) = \beta_0 + \beta_1 x - \delta m(x) \tag{10}$$

where $m : \mathbb{R} \to \mathbb{R}^+$ is strictly convex and differentiable, and $m'(x)$ is onto $\mathbb{R}$. Moreover, there exists x such that $m(x) = m'(x) = 0$.

This formulation of models allows for a convex loss function m to take many different shapes. With these two assumptions in hand, I prove a more general version of Theorem 1.

Theorem 4. *Suppose Assumption 1 holds. If a type δ simple sender is low-curvature, he collects dispersed data with $\sigma_\delta^{*2} = \overline{\sigma}_\delta^2$. If a type δ simple sender is high-curvature, he collects local data with $\sigma_\delta^{*2} = \underline{\sigma}_\delta^2$. Alternatively, suppose Assumption 2 holds and $\overline{\sigma}$ is sufficiently large. If a type δ simple sender is low-curvature, he prefers to collect dispersed data with $\sigma_\delta^2 = \overline{\sigma}_\delta^2$ to local data with $\sigma_\delta^2 = \underline{\sigma}_\delta^2$. If a type δ simple sender is high-curvature, he prefers to collect local data with $\sigma_\delta^2 = \underline{\sigma}_\delta^2$ to dispersed data with $\sigma_\delta^2 = \overline{\sigma}_\delta^2$.*

First, we note that the first part of Theorem 4 is identical to Theorem 1. The key reason is that the set of plausible f_Rs must obey the following moment conditions.

$$\mathbb{E}_{\mu_\delta}\big[f_R(x) - \beta_0 - \beta_1 x + \delta \ell(x - x_\delta)\big] = 0 \tag{11}$$

$$\mathbb{E}_{\mu_\delta}\big[x\big(f_R(x) - \beta_0 - \beta_1 x + \delta \ell(x - x_\delta)\big)\big] = 0 \tag{12}$$

The proof shows that (12) can be boiled down to the condition that the derivative of all plausible f_Rs must be the same across the data sample μ_δ. Since μ_δ and $\ell(x - x_\delta)$ are both symmetric around x_δ, we have that for μ'_δ with $x_\delta = x'_\delta$ but $\sigma'_\delta \neq \sigma_\delta$

$$\mathbb{E}_{\mu_\delta}[f'_R(x)] = \alpha_1 - \gamma \mathbb{E}_{\mu_\delta}[\ell'(x - x_\delta)] = \alpha_1 - \gamma \mathbb{E}_{\mu'_\delta}[\ell'(x - x_\delta)] \qquad \forall \sigma, \sigma' \in [\underline{\sigma}, \overline{\sigma}]$$

Thus, equation (12) is invariant to σ. This means that, as in the quadratic case, as σ changes, the set of models that R considers as a function of σ shifts up or down, allowing us to back out the same logic as in the original case.

However, if models are not characterized by a symmetric loss $\ell(x - x_\delta)$ around x_δ, it is more difficult to obtain a result. Theorem 4 simply states that, fixing x_δ, S prefers sampling dispersed data to local data if he is low-curvature (and the opposite holds if he is high-curvature). The reason is that, if the convex portion of a receiver's models is *not* symmetric around x_δ, the analogue of (12) no longer holds. I.e., the expression

$$\mathbb{E}_{\mu_\delta}[f'_R(x)] = \alpha_1 - \gamma \mathbb{E}_{\mu_\delta}[m(x)]$$

generally changes as the variance of a distribution μ_δ increases, fixing its mean. As a result, the proof allows α_1 to change to "absorb" any changes to $\mathbb{E}_{\mu_\delta}[m(x)]$ with respect to σ. For each $f_R(x)$

with a given γ value, changing σ thus not only changes α_0 (as in the benchmark and symmetric loss models) but also α_1. It is here where we utilize the condition in Assumption 2 that assigns the same prior mass to different α_1 values in the receiver's prior, which ensures that these models receive the same posterior weight.

Consequently, the value of the receiver's objective at her optimal choice x^* does not have an intuitive expression, and depends on nonintuitive features of the third derivative of $m(\cdot)$. However, the proof shows that if the sender is low-curvature, the value of each $f_R(x)$ the receiver entertains at x_δ is *increasing* in σ. Hence, the value of each f_R at x_δ for sufficiently high $\overline{\sigma}$ must necessarily be higher than the receiver's optimum x^* at $\underline{\sigma}$. The intuition for a high-curvature sender, as before, is flipped.

Proof of Theorem 4 We prove this result via the following lemmata. The first part of the theorem (assuming Assumption 1) follows immediately from the first lemma after applying the same reasoning as in Theorem 1.

Lemma 1. *Suppose Assumption 1 holds and fix f_S. Then, for all $f_R \in \phi(f_S; \delta, \mu_\delta)$, there exists $L(\sigma)$ strictly increasing in σ with $L(0) = 0$ such that:*

$$f_R(x) = f_S(x) + (\gamma - \delta)\big[L(\sigma) - \ell(x - x_\delta)\big].$$

Proof. Suppose $f_R \in \phi(f_S; \delta, \mu_\delta)$ for $f_S = \beta_0 + \beta_1 x - \delta\ell(x - x_\delta)$. This means that we have

$$f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1 x - \gamma\ell(x - x_\delta).$$

Applying Stein's Lemma as above, the moment condition with respect to $\tilde{\alpha}_1$ can be written as

$$\begin{aligned}\mathbb{E}_{\mu_\delta}\big[x\big(f_R(x) - f_S(x)\big)\big] &= \mathrm{Cov}_{\mu_\delta}(x, f_R(x) - f_S(x)) \\ &= \sigma^2\mathbb{E}_{\mu_\delta}[f_R'(x) - f_S'(x)] \\ &= \sigma^2\mathbb{E}_{\mu_\delta}[\tilde{\alpha}_1 - \gamma\ell'(x - x_\delta) - \beta_1 + \delta\ell'(x - x_\delta)] = 0.\end{aligned}$$

Because $\ell(x - x_\delta)$ is symmetric around x_δ, the terms with ℓ' are equal to 0. Thus, we have $\tilde{\alpha}_1 = \beta_1$ for all σ and are left to solve for $\tilde{\alpha}_0$. But for each γ value, as above, we have

$$\begin{aligned}\mathbb{E}_{\mu_\delta}[f_R(x)] = \mathbb{E}_{\mu_\delta}[f_S(x)] \implies \tilde{\alpha}_0 - \gamma\mathbb{E}_{\mu_\delta,\sigma}[\ell(x - x_\delta)] &= \beta_0 - \delta\mathbb{E}_{\mu_\delta,\sigma}[\ell(x - x_\delta)] \\ \implies \tilde{\alpha}_0 &= \beta_0 + (\gamma - \delta)\mathbb{E}_{\mu_\delta,\sigma}[\ell(x - x_\delta)].\end{aligned}$$

Defining $L(\sigma) = \mathbb{E}_{\mu_\delta,\sigma}[\ell(x - x_\delta)]$, we note that because $\ell(x - x_\delta)$ is convex, $L(\sigma)$ is increasing in σ. As $\sigma \to 0$, $L(\sigma) \to 0$. This gives the result. □

Lemma 2. *Suppose Assumption 2 holds and fix f_S. Then, for each $f_R \in \phi(f_S; \delta, \mu_\delta)$, $f_R(x_\delta)$ is increasing (decreasing) in σ if $\gamma > \delta$ ($\gamma < \delta$).*

Proof. As above, we write $f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1 x - \gamma m(x)$. Applying Stein's Lemma again to the slope moment condition, we have $\mathbb{E}_{\mu_\delta}[f_R'(x)] = \mathbb{E}_{\mu_\delta}[f_S'(x)]$. Using this condition, we implicitly define $\alpha_1(\sigma)$ as

$$\alpha_1(\sigma) = \beta_1 + (\gamma - \delta)\mathbb{E}_{\mu_\delta}[m'(x)]$$

and note that its sign with respect to σ is ambiguous. Similarly, we define $\alpha_0(\sigma)$ implicitly as

$$\alpha_0(\sigma) = \beta_0 + \big(\beta_1 - \alpha_1(\sigma)\big)x_\delta + (\gamma - \delta)\mathbb{E}_{\mu_\delta,\sigma}[m(x)]$$

and note in particular that, by definition, we have

$$\mathbb{E}_{\mu_\delta,\sigma}[f_S(x)] = \mathbb{E}_{\mu_\delta,\sigma}[f_R(x)] = \mathbb{E}_{\mu_\delta,\sigma}\big[\alpha_0(\sigma)x + \alpha_1(\sigma) - \gamma m(x)\big]$$

But then, by the Envelope Theorem, we have that

$$\begin{aligned}
\frac{\partial}{\partial\sigma} f_R(x_\delta|\sigma) &= \alpha_0'(\sigma) + \alpha_1'(\sigma)x_\delta - \gamma m(x_\delta) \\
&= -\alpha_1'(\sigma)x_\delta + (\gamma - \delta)\frac{\partial}{\partial\sigma}\mathbb{E}_{\mu_\delta,\sigma}[m(x)] + \alpha_1'(\sigma)x_\delta \\
&= (\gamma - \delta)\frac{\partial}{\partial\sigma}\mathbb{E}_{\mu_\delta,\sigma}[m(x)]
\end{aligned}$$

Since m is convex, this is weakly positive if and only if $\gamma \geq \delta$. Moreover, as $\sigma \to \infty$, $\mathbb{E}_{\mu_\delta,\sigma}[m(x)]$ is bounded away from 0, meaning that $f_R(x_\delta)$ approaches ∞ as $\sigma \to \infty$.

We note that because the receiver has a uniform prior over α_0 and α_1, her posterior places the same mass on all models that are the same "up to a constant and linear term." This means that for $\gamma > \delta$, there exists σ such that $f_R(x_\delta|\sigma) > f_R(x^*|\underline{\sigma})$, where x^* maximizes $f_R(x|\underline{\sigma})$. Hence, S is better sampling dispersed data than local data when $\gamma > \delta$. The same logic shows that S prefers local data when $\gamma < \delta$. Applying the same logic as in the proof of Theorem 1 gives the result. □